Shree Swami Atmanand Saraswati Institute of Technology

affiliated with

Gujrat Technological University, Ahmedabad

A

PROJECT REPORT SUBMITTED

ON

Online Handicraft System

Prepared as a part of the requirements for the subject of

Project (1360505)

Submitted By

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Shree Swami Atmanand Saraswati Institute of Technology
affiliated with
Gujrat Technological University, Ahmedabad

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **Online Handicraft System** has been carried out by **Mehta Jagnath Kamleshbhai** bearing enrollment no.: **220761305041** under the guidance of **Prof. Leena N. Chaudhari** as part of Project (1360505) 3rd Year (6th Semester) of Integrated M.Sc. (IT) in the academic year 2024-2025 and their work is satisfactory.

Date: / / 2025

Place: SSASIT, Surat

Prof. Leena N. Chaudhari
Internal Guide

Prof. Drasti J. Chauhan
Head of M.Sc (IT) Department





SHREE SWAMI ATMANAND SARASWATI INSTITUTE OF TECHNOLOGY

Self-Declaration

I hereby declare that the project report submitted along with the project entitled **Online Handicraft System** submitted as part of Project (1360505) 3rd Year (6th Semester) of Integrated M.Sc. (IT) to Gujarat Technological University, Ahmedabad, is a bonafide record of original project work carried out by me and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Mehta Jagnath
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Acknowledgments

The success of any task relies on the efforts made by person but it cannot be achieved without cooperation of other persons which are being helpful. So, i would like to thank **SHREE SWAMI ATMANAND SARASWATI INSTITUTE OF TECHNOLOGY, SURAT**. for giving me the opportunity of doing this Project. I would like to take this opportunity to express our sincere thanks to all those people without whose support and co-operation, it would have been difficult to complete this project.

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Online Handicraft System

1. PROJCET DESCRIPTION

➢ Introduction

The Online Handicraft System is a digital platform that connects artisans and customers, enabling the buying and selling of handmade products. It serves as a marketplace where sellers can showcase their handicrafts, and buyers can easily browse and purchase unique, handcrafted items.

1) Existing System

The current market for handicrafts relies on traditional methods, which come with several challenges:

- **Limited Online Presence** – Many handicraft sellers rely on physical stores or small-scale social media sales.
- **Manual Product Search** – Buyers must visit multiple websites or offline markets to find specific handicrafts.
- **Lack of Tracking System** – Customers often struggle to track orders and get real-time updates on purchases.

➔ Modules in the Online Handicraft System

- **Admin Module** – Manages product listings, sellers, and user activities.
- **User Module** – Allows buyers to browse, search, and purchase products.

2) Need for the New System

The Online Handicraft System addresses the shortcomings of the traditional market by providing:

- **A seamless interface between buyers and sellers** – Users can search for new or old handicrafts effortlessly.
- **Improved communication** – Sellers and buyers can interact directly.
- **Faster transactions** – Reduces time spent searching and purchasing products.
- **User-friendly environment** – Ensures a smooth shopping experience.
- **Increased profitability** – Expands market reach and enhances sales.
- **Special deals and offers** – Provides discounts, festival sales, and promotions.
- **Applicability to real-time businesses** – Any artisan or handicraft business can benefit from this platform.
- **Easy appointment booking** – Users can schedule meetings with artisans for custom orders.

3) Objectives of the New System

The primary goals of the Online Handicraft System include:

1. **High Reachability**
   - The system ensures a global reach, attracting more customers and increasing sales.

2. **Customer Satisfaction**
   - Offering quality handicrafts, secure payment options, fast delivery, and round-the-clock customer support.
3. **Broader Audience Reach**
   - The internet enables global access, allowing buyers from different locations to purchase unique handcrafted items.
4. **Brand Growth**
   - As more people interact with the platform, brand recognition and trust increase.

4) Problem Definition

- The platform provides an efficient way for artisans to sell handicrafts online.
- Customers expect an intuitive and easy-to-navigate platform to find products quickly.
- Traditional shopping takes time, whereas the system streamlines the buying process.
- The system manages sales, orders, payments, and user activities efficiently.
- Reduces conflicts by maintaining proper transaction records.
- Saves time and improves order processing.

5) Core Components

➔ User Categories

The Online Handicraft System has three types of users:

1. **Admin**
   - Manages product categories, user profiles, and orders.
   - Handles product inquiries, reviews, and inventory.

2. **Visitor**
   - Can browse product details and view the available services.
   - Can register to become a user.

3. **User (Buyer & Seller)**
   - Registers on the platform.
   - Manages profile and transactions.
   - Searches and purchases handicrafts.
   - Adds products to cart or Wishlist.
   - Cancels orders if necessary.
   - Tracks orders and receives notifications.
   - Provides reviews and feedback.

Online Handicraft System

1.1 PROJECT PROFILE

Project Attribute	Details
Project Title	Online Handicraft Website
Project Category	Web-Based E-Commerce Application
Time Duration	4 Months
Front-End	HTML, CSS, JavaScript
Back-End	PHP, MySQL
Database	MySQL
Payment Integration	Razorpay
Security Measures	Data Encryption, Secure Authentication
College Guide	Prof. Leena N. Chaudhari
Team Members	Jagnath K. Mehta

1.2 PROJECT DEFINATION

The Online Handicraft Website is a web-based e-commerce system designed to facilitate the buying and selling of handmade products. It serves as a digital marketplace where artisans can showcase and sell their handcrafted goods, while customers can explore, purchase, and track their orders conveniently.

➔ Key Aspects of the System:

1. **Purpose:**
   - To provide a platform for artisans to list and sell their handmade products.
   - To offer customers an easy and efficient way to browse and purchase handicrafts online.
   - To promote traditional and regional handicrafts by expanding their reach to a global audience.

2. **Core Functionalities:**
   - **User Management:** Secure registration, login, and profile management for customers, artisans, and administrators.
   - **Product Management:** Artisans can add, update, and categorize products with images, descriptions, and pricing.
   - **Shopping Cart & Checkout:** Customers can add products to their cart, proceed with secure payment, and track their orders.
   - **Admin Panel:** System administrators can manage users, products, orders, and promotional offers.
   - **Customer Engagement**: Wishlist, reviews & ratings, newsletter subscription, and order tracking.

3. **Technology Stack:**
   - **Frontend:** HTML, CSS, JavaScript, Bootstrap for a responsive UI.
   - **Backend:** PHP for dynamic content handling and MySQL for database management.
   - **Security Measures:** Secure authentication, encrypted payments, and data protection.

4. **Target Users:**
   - **Artisans/Sellers:** Register, upload products, manage orders, and track sales.
   - **Customers:** Browse, purchase, review, and track their orders.
   - **Administrators**: Oversee system operations, manage users, and ensure smooth transactions.

5. **Business Impact & Benefits:**
   - Expands market reach for artisans, providing them with a digital sales channel.
   - Offers customers access to a variety of unique handmade products in a single place.
   - Ensures a streamlined and automated process for order management, reducing manual effort.

This Online Handicraft Website aims to empower artisans, enhance customer convenience, and preserve cultural heritage by leveraging digital technology in the e-commerce domain.

1.3 Advantages and Limitations of the Proposed System

➔ **Advantages**

1. **Fast Buying and Selling** – Users can quickly find and purchase handicrafts.
2. **24/7 Availability** – Buyers can place orders anytime.
3. **Lower Operational Costs** – Sellers can list products without maintaining physical stores.
4. **Easy Product Comparison** – Customers can browse products from different artisans in one place.
5. **Daily Discounts & Updates** – Users receive notifications about special offers.

➔ **Limitations**

1. **Internet Requirement** – The system is dependent on internet connectivity for access and transactions.
2. **Logistics Challenges** – Shipping delays and damages can impact user experience.
3. **Product Authenticity Concerns** – Customers may hesitate to buy due to concerns about originality and quality.
4. **High Competition** – Many online platforms may offer similar handicrafts, making it harder to attract buyers.

2. ENVIRONMENT DESCRIPTION

The development and deployment of the Online Handicraft Website require a well-defined hardware and software environment to ensure smooth performance, scalability, and security. The following sections outline the essential hardware and software specifications, along with the development methodology and tools used for the project.

2.1 HARDWARE AND SOFTWARE SPECIFICATION

A. Hardware Requirements

The project requires a suitable hardware setup for development, testing, and deployment.

Component	Minimum Requirement	Recommended Requirement
Processor	Intel Core i3	Intel Core i5/i7
RAM	4 GB	8 GB or higher
Storage	100 GB HDD	256 GB SSD or higher
Operating System	Windows 10	Latest version of Windows
Internet Connection	2 Mbps	High-speed broadband connection
Peripherals	Keyboard, Mouse, Monitor	-

B. Software Requirements

The following software tools and technologies are used for development, testing, and deployment.

Software Component	Technology/Tool Used
Operating System	Windows 10/11
Backend Development	PHP (Version 7.4 or later)
Frontend Development	HTML5, CSS3, JavaScript
Database	MySQL (Version 5.7 or later)
Web Server	Apache (XAMPP/WAMP/LAMP)
IDE/Code Editor	Notepad++, Visual Studio Code
Payment Gateway	Razorpay

2.2 DEVELOPMENT DESCRIPTION

The development of the Online Handicraft Website follows a structured approach, ensuring modularity, scalability, and efficiency.

A. Development Approach

The project is developed using the Agile Development Methodology**,** which allows iterative improvements, faster feedback implementation, and flexibility in feature enhancement. The key phases include:

1. **Requirement Analysis** – Understanding user needs, defining features, and setting up the project scope.
2. **System Design** – Creating wireframes, database schema, and system architecture.
3. **Frontend & Backend Development** – Developing user interfaces, implementing business logic, and integrating APIs.
4. **Database Management** – Designing and optimizing the MySQL database structure.
5. **Testing & Debugging** – Conducting unit testing, integration testing, and security testing.
6. **Deployment & Maintenance** – Hosting the website on a live server and providing continuous updates.

B. Modules & Development Tools

Module	Technology Used
User Authentication	PHP Sessions, MySQL
Product Management	PHP, MySQL, JavaScript
Shopping Cart	PHP, JavaScript
Payment Integration	Razorpay
Admin Dashboard	PHP, MySQL
Search & Filters	JavaScript, MySQL Queries
Order Management	PHP, MySQL, API Calls

3. SYSTEM DESCRIPTION

3.1 INTRODUCTION

The Online Handicraft System is a comprehensive e-commerce platform designed to connect artisans with customers, enabling the seamless buying and selling of handcrafted products. This system serves as a dedicated digital marketplace that provides artisans with a global platform to showcase their work while offering customers a diverse range of handmade goods.

Developed using PHP and MySQL, the platform ensures robust backend processing and secure database management. The frontend, built with HTML, CSS, JavaScript, and Bootstrap, delivers an intuitive and responsive user experience. Additionally, secure payment gateways (e.g., Razorpay) provide safe transaction processing.

By automating the product listing, order management, and payment processes, the system reduces manual intervention, improves business efficiency, and fosters direct engagement between artisans and buyers.

3.2 SCOPE OF THE SYSTEM

The Online Handicraft System is designed to facilitate an end-to-end e-commerce experience, addressing the needs of artisans, customers, and administrators.

➔ Key Areas of the System:

- ✓ **User Management:** Secure registration, authentication, and role-based access control.
- ✓ **Product Management:** Artisans can add, update, and categorize products with images, descriptions, and pricing.
- ✓ **Shopping Cart & Checkout:** Customers can add items to the cart, proceed to checkout, and complete transactions securely.
- ✓ **Order Processing:** Automated order placement, tracking, and status updates.
- ✓ **Admin Control Panel:** Centralized dashboard for managing users, products, and orders.
- ✓ **Customer Engagement:** Wishlist, reviews, order tracking, newsletters, and notifications.
- ✓ **Security & Privacy:** Secure authentication, SSL encryption, and fraud prevention mechanisms.
- ✓ **Scalability:** Designed to support future **expansion**, including mobile apps and multi-vendor functionality.

The Online Handicraft System provides a seamless and efficient shopping experience, ensuring that both artisans and customers can interact smoothly while maintaining a high level of security and usability.

3.3 OBJECTIVEOF THE SYSTEM

The primary objective of the Online Handicraft System is to empower artisans by offering a digital marketplace that enables them to sell their handmade products directly to consumers. The system also enhances the customer shopping experience by providing a convenient, secure, and feature-rich online store.

➔ Specific Objectives:

- ✓ **Facilitate Global Market Access** – Allow artisans to sell their handcrafted products worldwide.
- ✓ **Automate Business Processes** – Reduce manual effort in order management, inventory tracking, and payment processing.
- ✓ **Enhance User Experience** – Provide an intuitive and engaging shopping platform.
- ✓ **Ensure Secure Transactions** – Implement secure payment gateways, SSL encryption, and fraud detection mechanisms.
- ✓ **Preserve & Promote Handicrafts** – Support local artisans by creating a sustainable and profitable business environment.
- ✓ **Optimize Platform Scalability** – Ensure that the system can accommodate an increasing number of users, products, and transactions without performance degradation.

By achieving these objectives, the Online Handicraft System will empower small businesses, promote traditional handicrafts, and offer an efficient digital commerce experience.

3.4 FUNCTIONALITY OF SYSTEM

The Online Handicraft System comprises multiple functional modules, each designed to streamline operations and improve user interaction.

A. User Management

Role	Key Responsibilities
Admin	Manages users, products, orders, and system settings. Generates business reports and oversees platform security.
Artisans (Sellers	Uploads products, manages inventory, processes orders, and monitors sales performance.
Customers (Buyers)	Browses products, makes purchases, tracks orders, and interacts with artisans through reviews and inquiries.

B. Product Management

- Artisans can:
  - ✓ Add, edit, and delete product listings.
  - ✓ Upload high-quality images and detailed product descriptions.
  - ✓ Categorize products for easier customer navigation.
- Customers can:
  - ✓ View and search for products based on categories, keywords, and filters.

C. Shopping Cart & Checkout

- ✓ Customers can add/remove products from the cart before proceeding to checkout.
- ✓ Secure payment processing through PayPal, Stripe, or Razorpay.
- ✓ Order confirmation via email and real-time notifications.

D. Order & Payment Management

- ✓ Automated order processing with real-time tracking.
- ✓ Order history and status updates for customers.
- ✓ Refund and cancellation policies to enhance customer trust.

E. Customer Engagement Features

- ✓ Wishlist feature for saving favourite products.
- ✓ Review and rating system for ensuring product quality.
- ✓ Newsletter subscription for exclusive offers and promotions.

F. Admin Dashboard

- ✓ Manages users, products, orders, and customer inquiries.
- ✓ Generates business analytics and performance reports.
- ✓ Configures security settings, promotional campaigns, and platform-wide updates.

By integrating these functionalities, the system ensures a smooth, efficient, and scalable e-commerce experience.

3.5 EXPECTED OUTCOME

The successful implementation of the Online Handicraft System is expected to deliver significant benefits to artisans, customers, and administrators.

A. For Artisans (Sellers):

- ✓ **Expanded Market Reach:** Ability to sell handcrafted products globally.
- ✓ **Business Growth:** Increased visibility and profitability through digital marketing.
- ✓ **Operational Efficiency:** Simplified order and inventory management.
- ✓ **Brand Development:** Enhanced artisan recognition through a professional online presence.

B. For Customers (Buyers):

- ✓ **Access to Unique Handicrafts:** A diverse selection of handmade products in one place.
- ✓ **User-Friendly Shopping Experience:** Intuitive browsing, secure checkout, and real-time tracking.
- ✓ **Product Assurance:** Reviews, ratings, and seller credibility verification.
- ✓ **Convenient Payment Options:** Secure multi-gateway transactions with refunds and order tracking.

C. For Admin:

- ✓ **Complete Control Over Platform Operations:** Efficient user, product, and order management.
- ✓ **Data-Driven Decision Making:** Analytics and insights for business growth strategies.
- ✓ **Fraud Prevention & Security:** Monitoring of transactions, user activity, and system vulnerabilities.

D. Business Impact:

- ✓ **Encourages Entrepreneurship:** Provides a cost-effective online store for artisans.
- ✓ **Boosts Economic Growth:** Supports handicraft industries and small-scale businesses.
- ✓ **Preserves Traditional Craftsmanship:** Promotes cultural heritage through e-commerce.

4. PROJECT PLANNING AND SCHEDULING

4.1 FEASIBILITY STUDY

A feasibility study is conducted to assess the practicality of the Online Handicraft Website by evaluating technical, economic, operational, and scheduling aspects. This study helps in identifying potential risks and ensuring the project's success.

A. Technical Feasibility

Technical feasibility examines whether the project can be developed using the available technology and infrastructure.

- Evaluation of Programming Languages, Frameworks, and Hosting Requirements
  - The website will be developed using PHP, MySQL, JavaScript, HTML, and CSS.
  - Bootstrap will be used for responsive design, and AJAX for interactive user experience.
  - Hosting will be managed on AWS, Hostinger, or DigitalOcean for scalability.
- Scalability Assessment
  - The platform should be able to handle a growing number of users and products.
  - Optimized database queries and caching mechanisms will be implemented for performance.
- Security Implementation
  - SSL encryption, secure login authentication, and payment gateway security.
  - Protection against SQL injection, XSS attacks, and data breaches.

B. Economic Feasibility

Economic feasibility determines whether the project is financially viable and how it can sustain operations in the long run.

- Cost-Benefit Analysis
  - Initial Costs: Web development, hosting, domain purchase, and security implementation.
  - Operational Costs: Server maintenance, updates, and customer support.
  - Potential Revenue Streams: Subscription fees for artisans, commission per sale, and advertising revenue.
- Return on Investment (ROI) Estimation
  - Expected revenue from handicraft sales, premium memberships, and featured listings.
  - Profitability projection over one to three years.

C. Operational Feasibility

Operational feasibility ensures that the system will be usable, efficient, and beneficial to artisans and customers.

- ♦ System Usability
  - The platform will have a user-friendly interface for smooth navigation.
  - Mobile compatibility to cater to customers using smartphones and tablets.
- ♦ Automation of Business Processes
  - Automated order management, inventory tracking, and customer notifications.
  - Integration of secure payment gateways (PayPal, Stripe, Razorpay).
- ♦ User Accessibility
  - Multi-language support for a diverse customer base.
  - Easy onboarding process for artisans with guided tutorials.

D. Schedule Feasibility

Schedule feasibility ensures that the project is completed on time without unnecessary delays.

- ♦ Timeline Evaluation
  - The project follows a 4-month development plan divided into key phases.
  - Agile methodology will be used for incremental progress.
- ♦ Risk Mitigation Strategies
  - Contingency plans for development delays, security threats, and system failures.
  - Regular progress reviews to ensure timely delivery.

4.2 SOFTWARE ENGINEERING MODEL

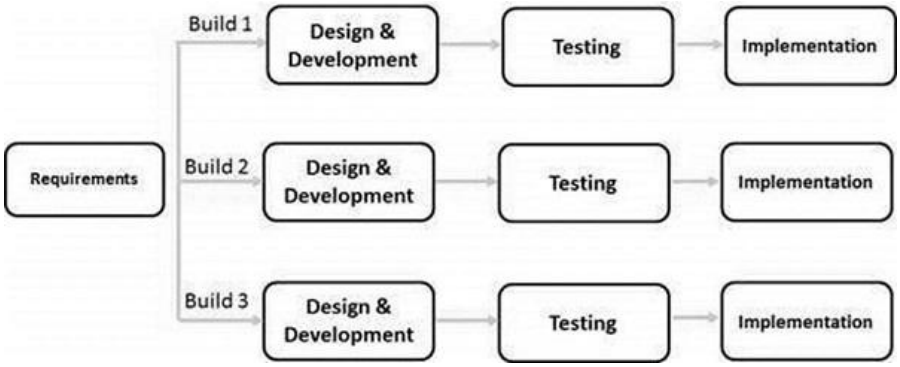
A. Overview of the Agile Model

The Agile Model is particularly beneficial for developing an Online Handicraft System as it enables quick adaptations to customer preferences, market trends, and seller requirements. Agile focuses on incremental delivery, continuous feedback, and collaborative development, ensuring a high-quality and user-friendly platform.

➔ **Iterative Development Process**

- Features of the Online Handicraft System, such as product listings, order management, and secure payments, are developed in small increments called sprints.
- Each sprint involves design, development, testing, and review cycles, allowing for continuous improvement.
- Regular updates ensure the platform stays competitive by incorporating new trends in handicraft sales, user experience, and security enhancements.

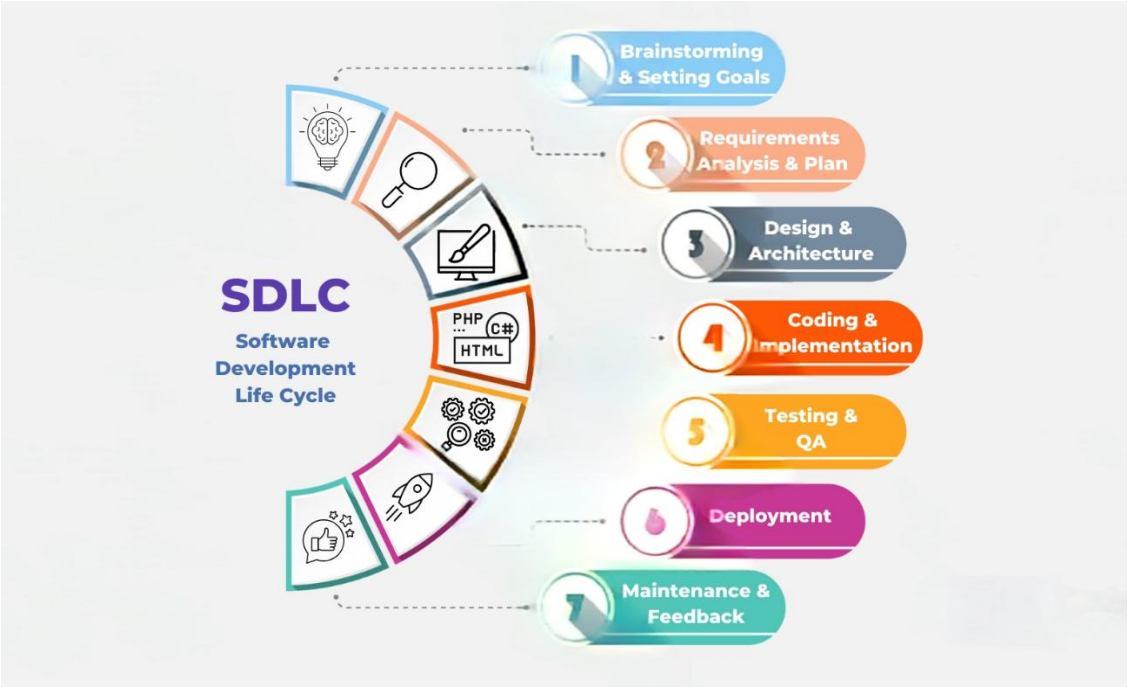
- **Diagram:**



➔ **Sprint Planning and Feedback Loops**

- The system is **continuously refined** based on feedback from artisans, buyers, and administrators.
- Regular **testing and integration** ensure seamless shopping experiences and secure transactions.
- Any usability, security, or performance issues are identified early and resolved efficiently.

B. Phases of Development



1. Requirement Gathering

- Identify functional requirements such as product listing, cart management, secure payments, and seller dashboards.

- Gather non-functional requirements like system scalability, security, and mobile responsiveness.
- Conduct surveys with artisans, analyze competitor platforms (e.g., Etsy, Amazon Handmade), and interview potential customers to understand expectations.

2. Planning

- Define key milestones, such as MVP (Minimum Viable Product) launch, feature enhancements, and security upgrades.
- Allocate resources for frontend design, backend development, payment integration, and security measures.
- Develop a roadmap for future updates, including AI-based recommendations, live chat support, and multi-language support.

3. Design

- Create UI/UX wireframes focusing on ease of use for both buyers and sellers.
- Develop database schemas to store product details, seller information, and transaction histories efficiently.
- Design secure system architecture to prevent fraud and ensure smooth operations.

4. Development

- Implement frontend components using Javascript for an interactive user experience.
- Develop backend services using PHP with a MySQL database to handle product listings, orders, and user management.
- Integrate payment gateways such as PayPal, Stripe, or Razorpay for seamless transactions.
- Implement AI-driven recommendations for personalized shopping experiences.

5. Testing

- Conduct unit testing to verify each module's functionality.
- Perform integration testing to ensure seamless communication between components like payment gateways and order processing.
- Apply security validation to prevent cyber threats, fraud, and data breaches.
- Optimize system performance for fast loading times and high scalability.

6. Deployment & Maintenance

- Deploy the platform on cloud services like AWS, Azure, or Google Cloud for scalability.
- Continuously monitor system performance to prevent downtime and technical glitches.
- Roll out regular updates, adding new features such as augmented reality (AR) for virtual product previews and enhanced security measures.
- Provide ongoing maintenance to ensure seller satisfaction, buyer trust, and market competitiveness.

4.3 REQUIREMENT ANALYSIS AND DATA GATHERING

Requirement analysis is a crucial phase in the development of an Online Handicraft System to ensure that the platform aligns with user needs, market trends, and business objectives. This process involves gathering and analyzing data from various sources to define the functional and non-functional requirements essential for a seamless user experience.

A. Requirement Gathering Methods

Effective requirement gathering ensures that the system meets user expectations, market demands, and technical feasibility. The following methods are employed to collect accurate and relevant information:

1. **User Interviews**
   - Conducted with artisans, buyers, and administrators to identify key challenges and expectations.
   - Helps understand the pain points of traditional selling methods and how digital transformation can benefit artisans.
   - Provides insights into customer preferences, such as personalized recommendations, secure payments, and real-time order tracking.

2. **Market Research**
   - Involves a comparative study of leading handicraft marketplaces such as Etsy, Craftsvilla, and Amazon Handmade.
   - Helps identify trending features, pricing models, and user experience improvements.
   - Assists in differentiating the platform with unique value propositions, such as sustainability-focused products or AI-powered recommendations.

3. **Surveys & Questionnaires**
   - Distributed among potential users to gather feedback on preferred features, usability expectations, and security concerns.
   - Helps prioritize functionalities such as multi-language support, diverse payment options, and interactive product previews.
   - Enables developers to incorporate data-driven decisions into the development process.

B. Functional Requirements

The Online Handicraft System consists of various modules designed to facilitate a smooth and secure experience for artisans, customers, and administrators.

Module	Description
User Management	Enables registration, authentication, and profile management for buyers and sellers.
Product Management	Allows artisans to add, update, categorize, and manage their products efficiently.
Shopping & Payments	Provides secure checkout, order tracking, notifications, and multiple payment gateway integration**.**
Admin Dashboard	Centralized system for administrators to manage users, orders, reports, security, and platform policies**.**

C. Non-Functional Requirements

Apart from functional aspects, non-functional requirements play a crucial role in ensuring a high-performing, secure, and scalable platform.

1. **Performance**
   - Optimized database queries and caching mechanisms to ensure fast page load times and smooth navigation.
   - Efficient content delivery network (CDN) integration to enhance global accessibility.
   - Use of lazy loading and optimized images for better performance.
2. **Security**
   - SSL encryption for secure data transmission and user privacy protection.
   - Multi-layered authentication (OTP, two-factor authentication) to prevent unauthorized access.
   - Fraud detection algorithms to monitor and prevent suspicious transactions.
3. **Scalability**
   - Cloud-based auto-scaling infrastructure (AWS, Azure, Google Cloud) to handle high traffic and large product databases.
   - Modular architecture to support multiple vendors, locations, and languages.
   - Load balancing to maintain system stability under heavy user demand.
4. **Cross-Platform Compatibility**
   - Ensuring a responsive design for seamless usability on mobile, tablet, and desktop devices.
   - Progressive Web App (PWA) development to enable offline access and improved mobile experience.
   - Support for multiple operating systems (Windows, macOS, Android, iOS) for broader accessibility.

4.4 TIME LINE CHART

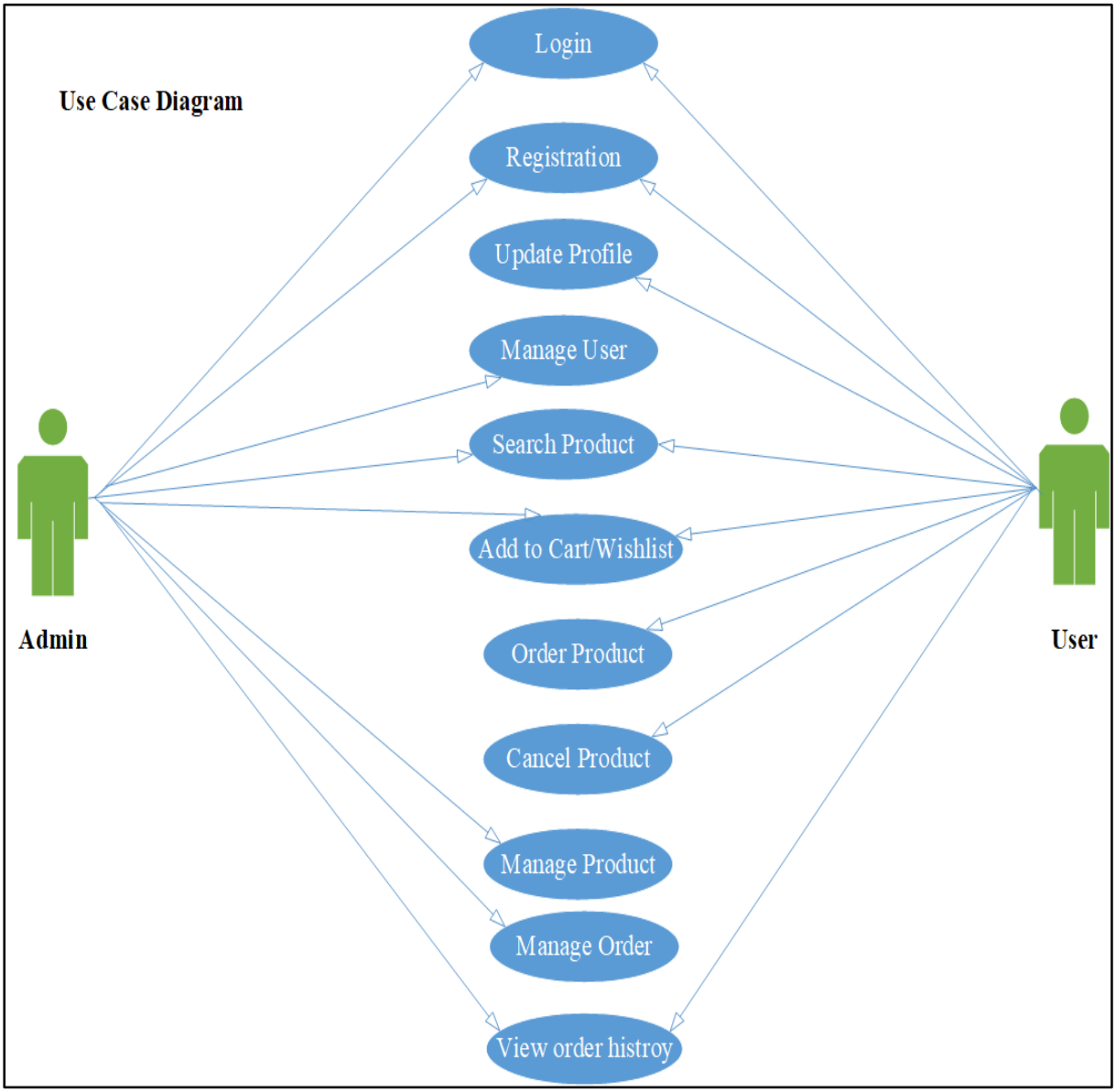
Tasks	Jan 2025	Feb 2025	Mar 2025	April 2025
Requirement Analysis				
o Requirement Gathering				
o Analyse gathered information				
o Determine scop of the system				
o Analysis Complete				
Planning and Risk analysis				
o Analyse Data for Possible Risks				
o Identify Technical Risks				
o Determine Different Modules				
o Planning Completed				
Coding / Implementation				
o **Sprint 1:** User Authentication & Authorization				
o **Sprint 2:** UI / UX Development				
o **Sprint 3:** Product Management				
o **Sprint 4:** Shopping Cart & Checkout				
o **Sprint 5:** Payment Gateway Integration				
o **Sprint 6:** Order Management				
o **Sprint 7:** Search & Filters				
o **Sprint 8:** Admin Panel				
o Coding Completed				
Testing				
o Validate & Check Accuracy of System				
o Testing Completed				

➢ Schedule Table

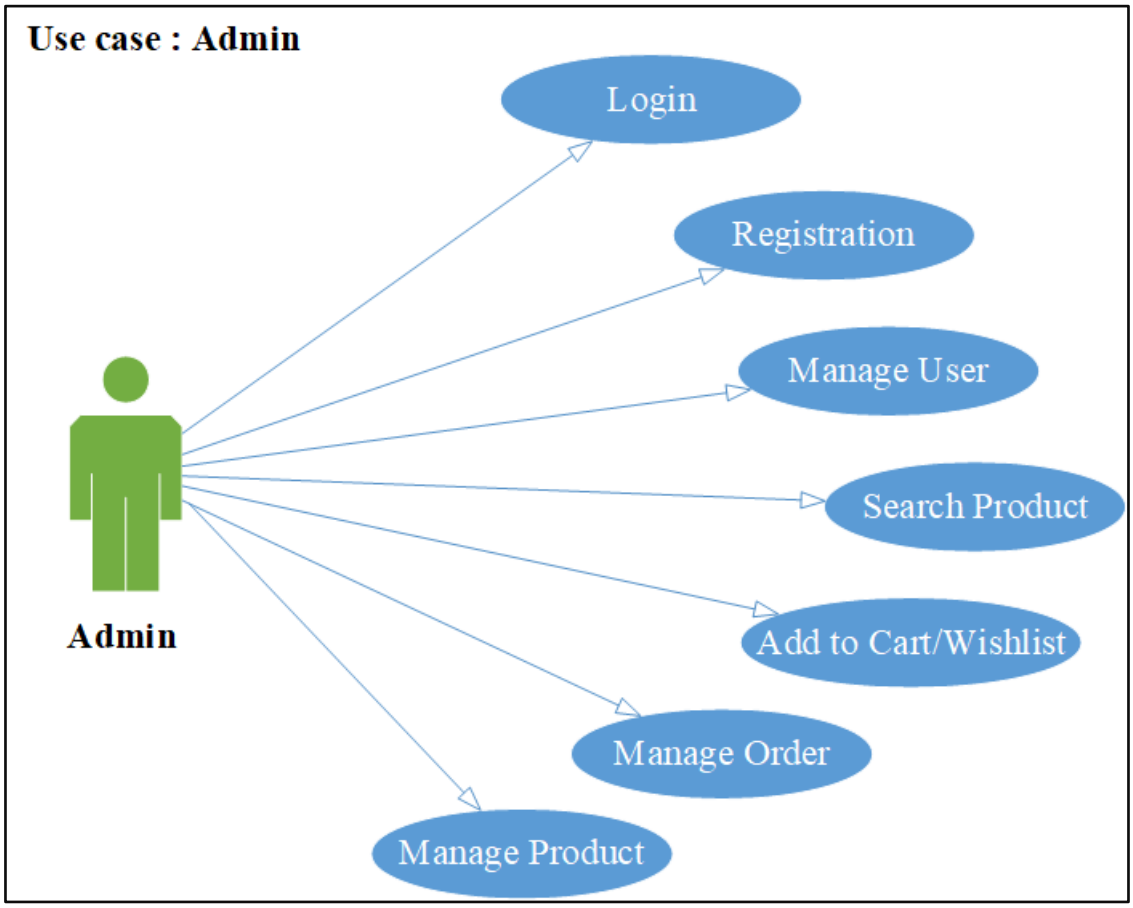
Task	Start Date	End Date	Duration (days)
Requirement Gathering	05-01-25	11-01-25	6
Analyse Gathered Information	12-01-25	16-01-25	5
Determine Scop of the System	17-01-25	19-01-25	3
Analyse Data for Possible Risks	20-01-25	25-01-25	6
Identify Technical Risks	26-01-25	30-01-25	5
Determine Different Module	31-01-25	05-02-25	6
User Authentication & Authorization	06-02-25	14-02-25	9
UI / UX Development	15-02-25	22-02-25	8
Product Management	23-02-25	01-03-25	7
Shopping Cart & Checkout	02-03-25	12-03-25	11
Payment Gateway Integration	13-03-25	22-03-25	10
Order Management	23-03-25	03-04-25	12
Search & Filters	04-04-25	11-04-25	8
Admin Panel	12-04-25	13-04-25	2
Validate & Check Accuracy of System	14-04-25	15-04-25	2

5. SYSTEM ANALYSIS & DESIGN

5.1 USE CASE DIAGRAM

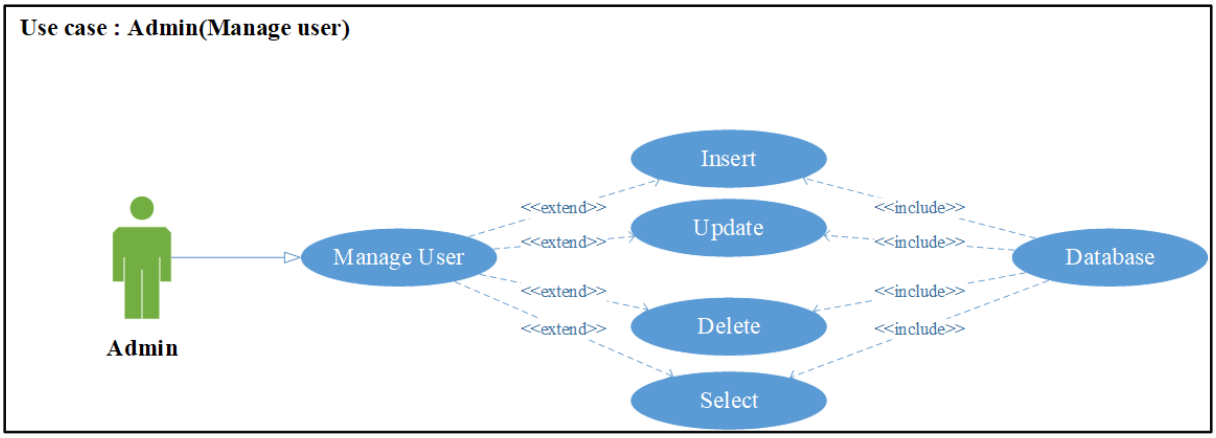


5.1.1 Admin Module:

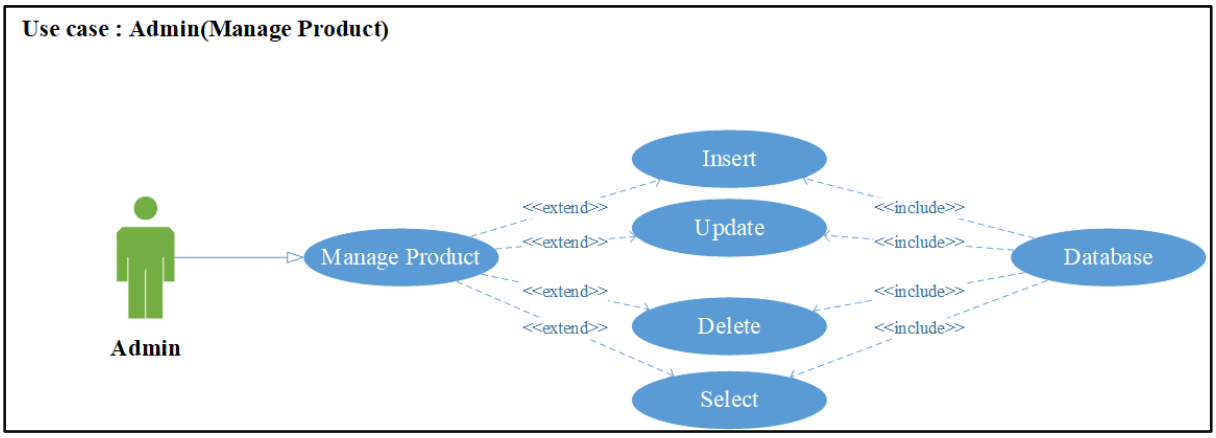


5.1.2 Micro level for Admin:

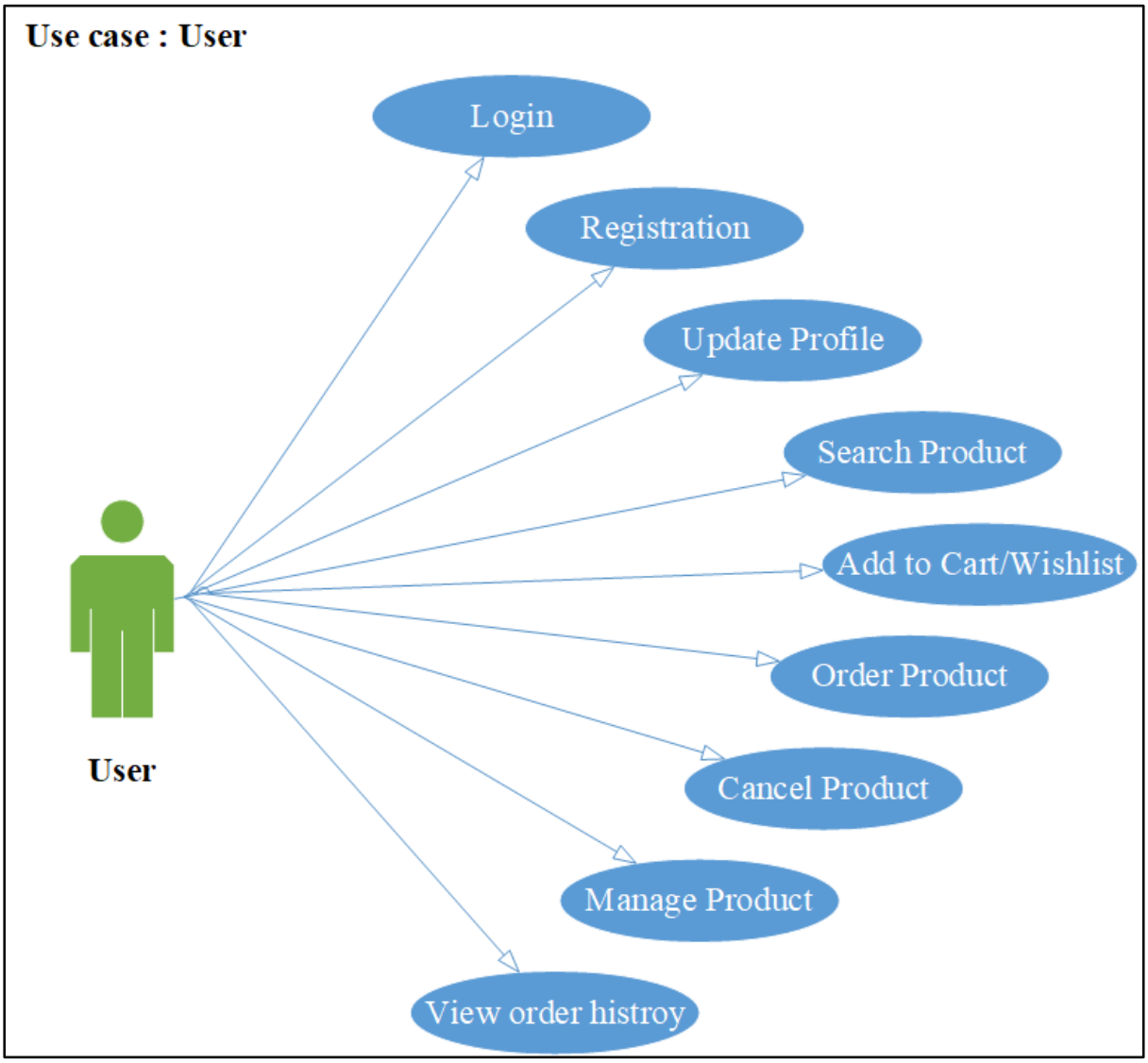
➔ **Manage user:**



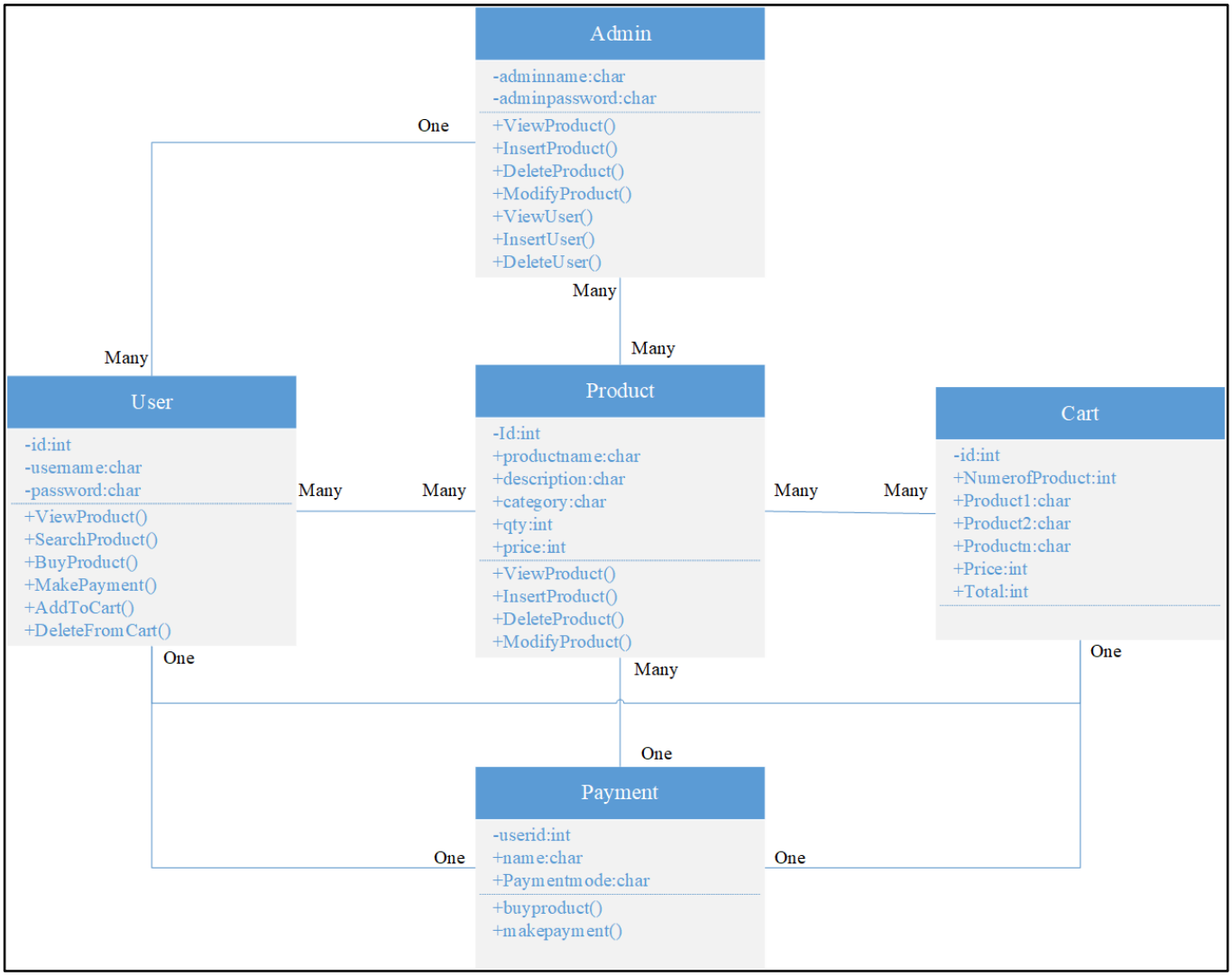
➔ **Manage Product:**



5.1.3 User Module:



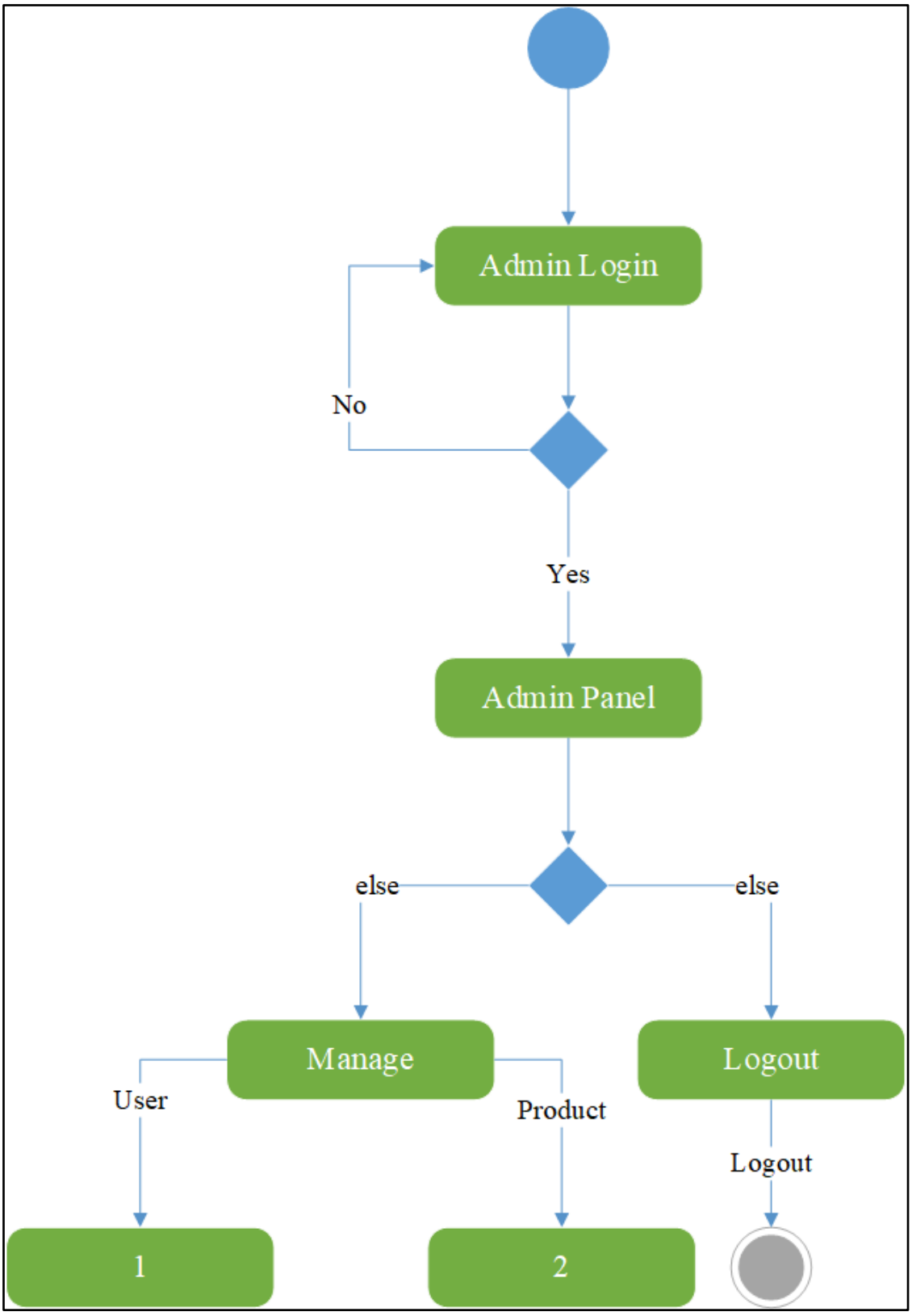
5.2 CLASS DIAGRAM



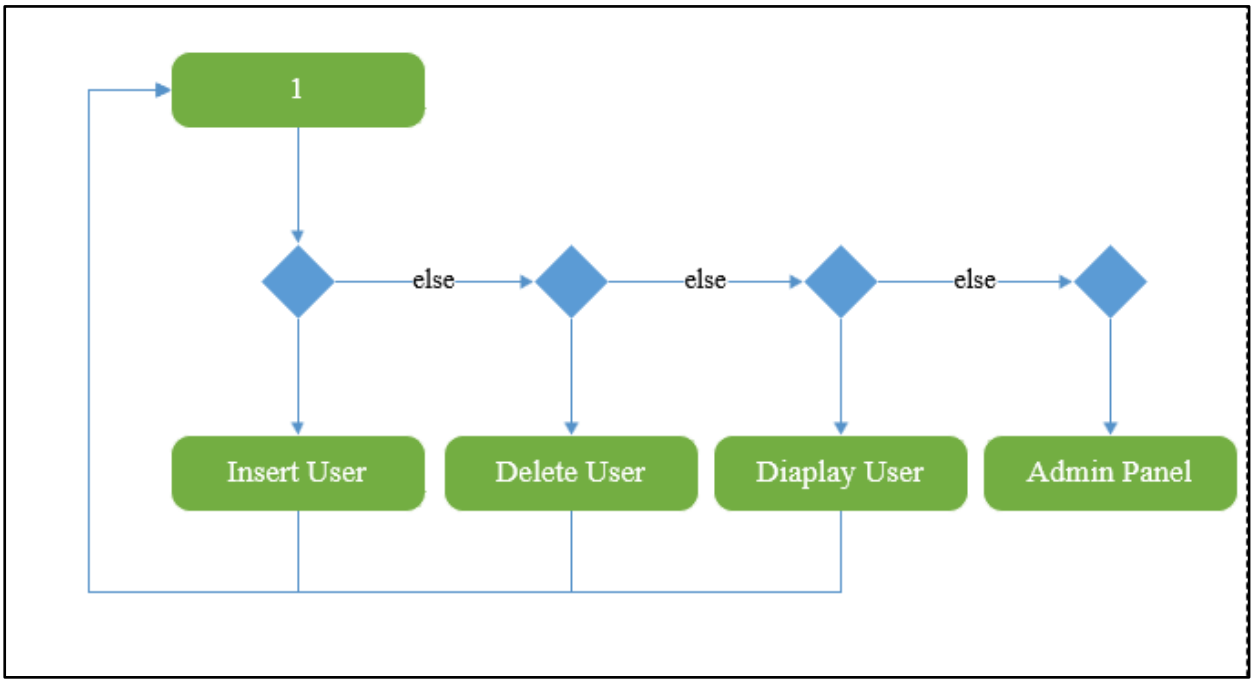
5.3 ACTIVITY DIAGRAM

5.3.1 Admin Module

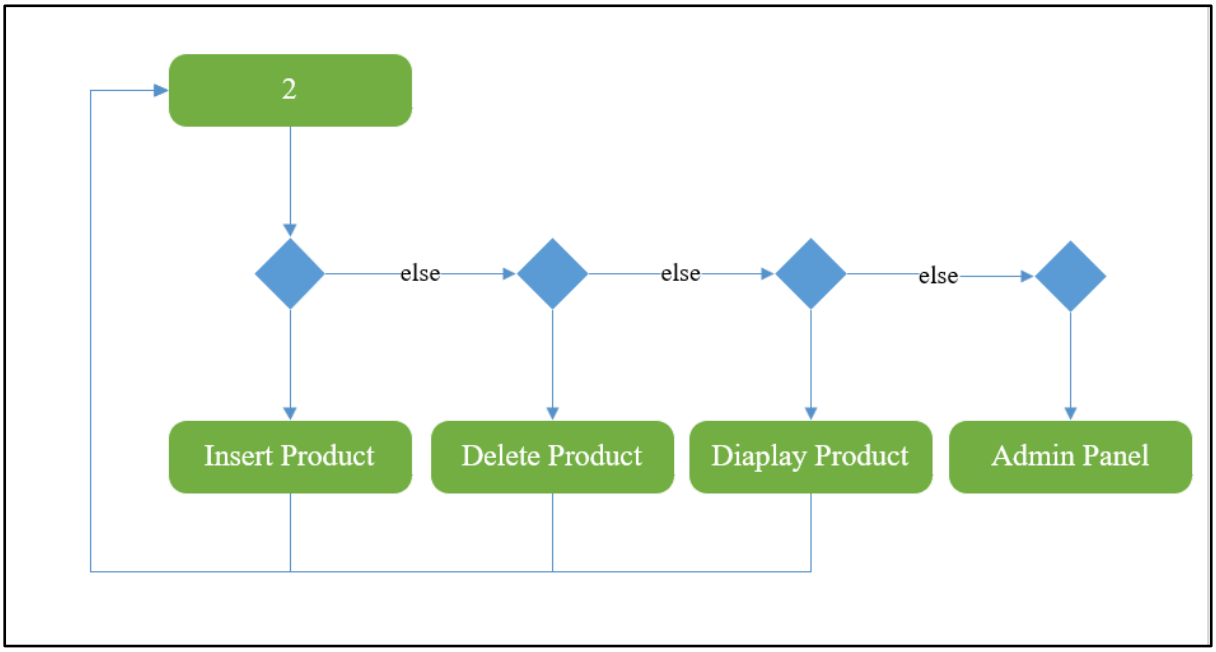
5.3.1.1 Admin Login:



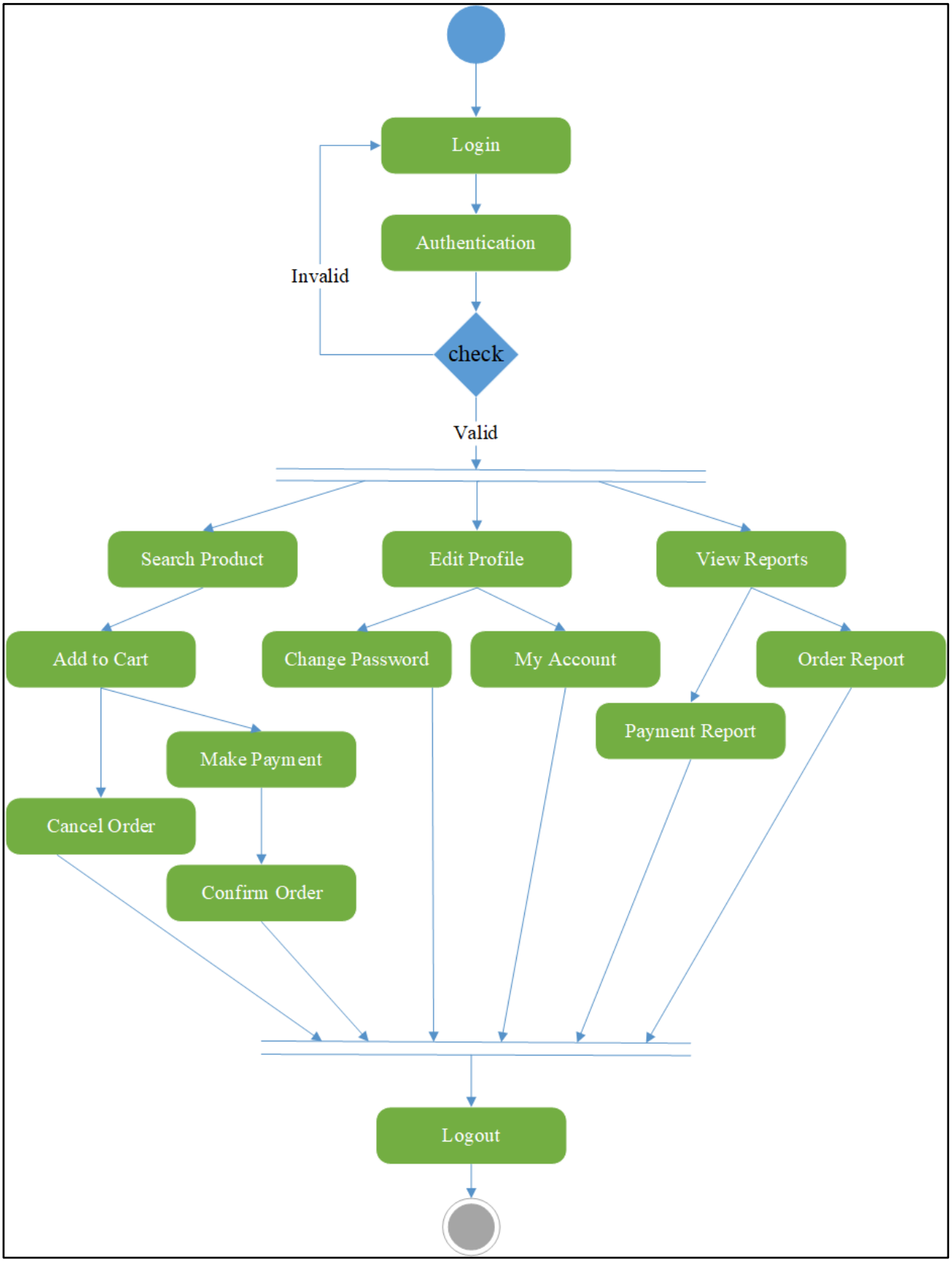
5.3.1.2 Manage User:



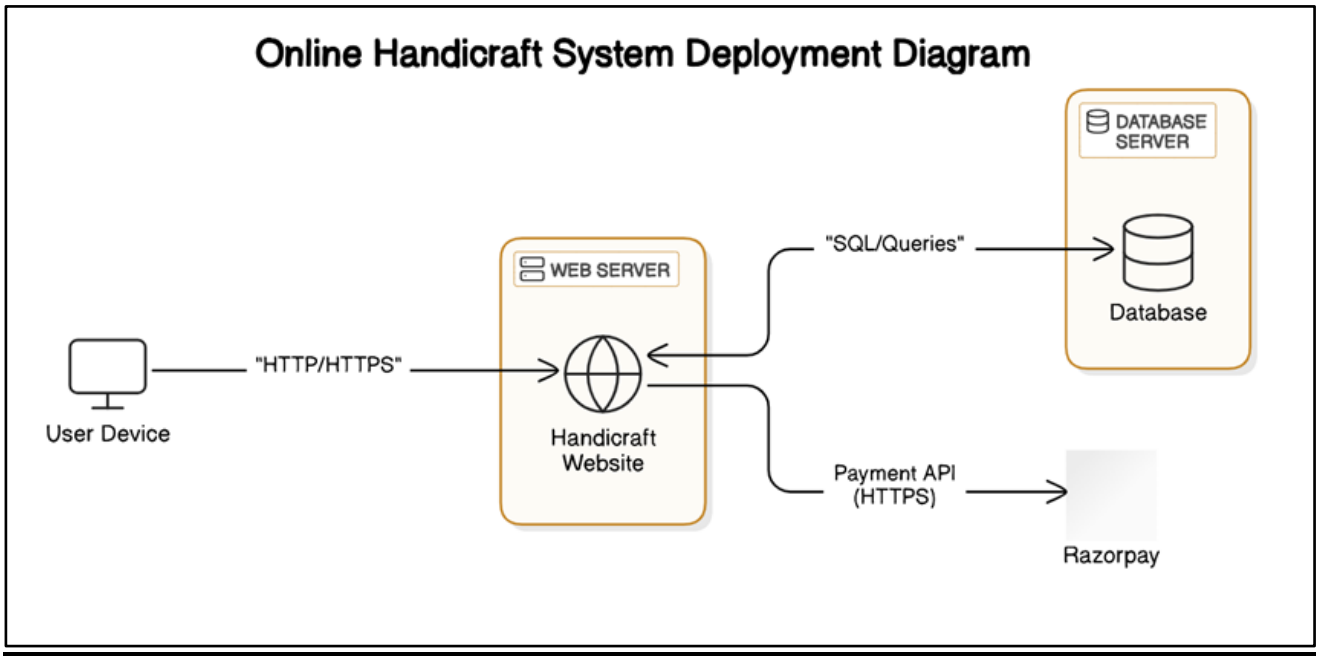
5.3.1.3 Manage Product:



5.3.2 User Module



5.4 DEPLOYMENT DIAGRAM



5.5 DATABASE DESIGN

1) Table User

Primary key: id

Name	Data Type	Size	Extra
id	int	11	Auto_Increment
username	varchar	255	Not null
email	varchar	255	Not null
password	varchar	255	Not null
Reset_token	varchar	255	Not null

2) Table Product

Primary key: Product_id

Name	Data Type	Size	Extra
id	int	11	Auto_Increment
name	varchar	50	Not null
description	varchar	100	Not null
category	varchar	50	Not null
qty	varchar	50	Not null
price	varchar	50	Not null
photo	varchar	50	Not null
location	varchar	200	Not null
upi	varchar	100	Not null

3) Table Cart

Primary key: id

Name	Data Type	Size	Extra
id	int	11	Auto_Increment
user_id	int	11	Not null
product_id	int	11	Not null
quantity	int	11	Not null

4) Table Cart

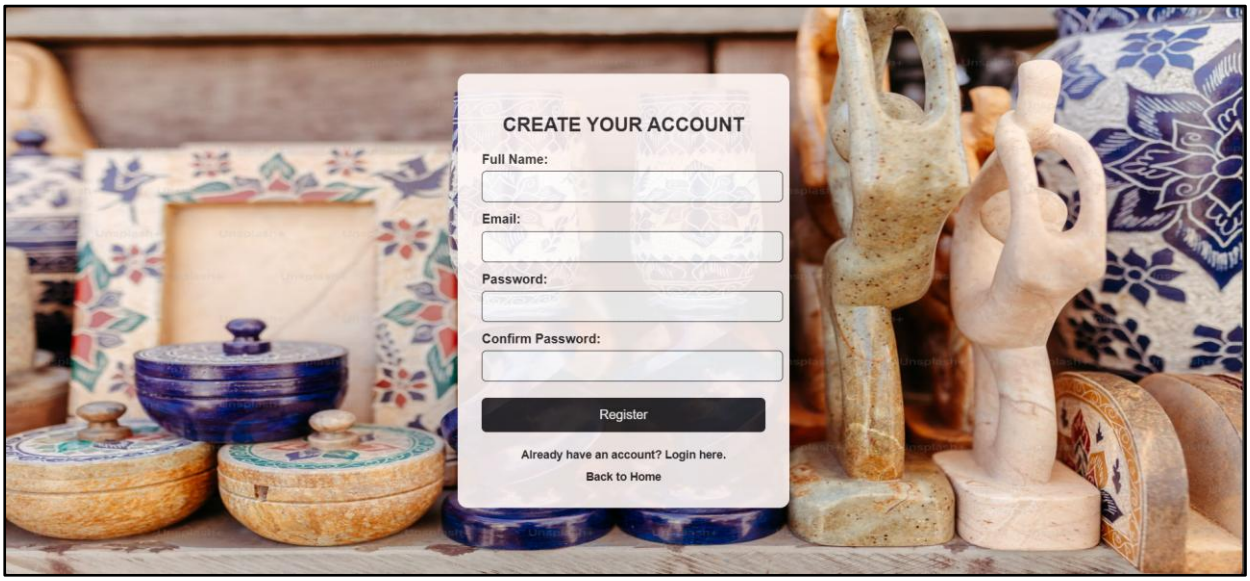
Primary key: id

Name	Data Type	Size	Extra
id	int	11	Auto_Increment
username	varchar	50	Not null
product_id	int	11	Not null
quantity	int	11	Not null
price	decimal	10,2	Not null
payment_id	varchar	100	Not null
order_date	timetamp	-	Not null
status	varchar	20	yes

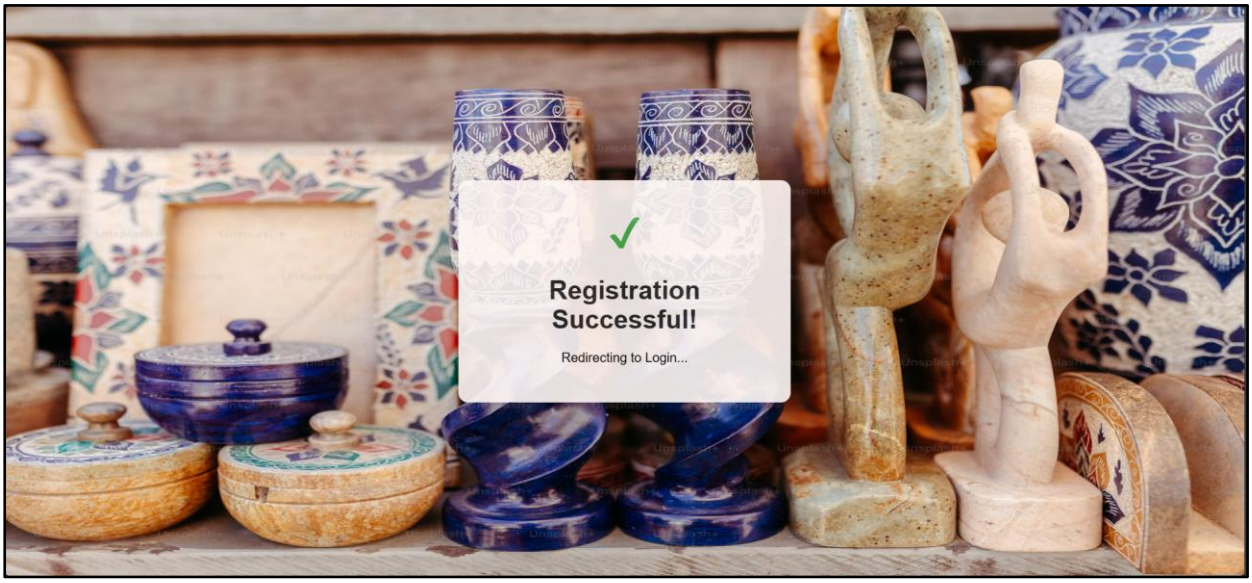
6. SYSYEM IMPLEMENTATION

6.1 SCREEN SHORT

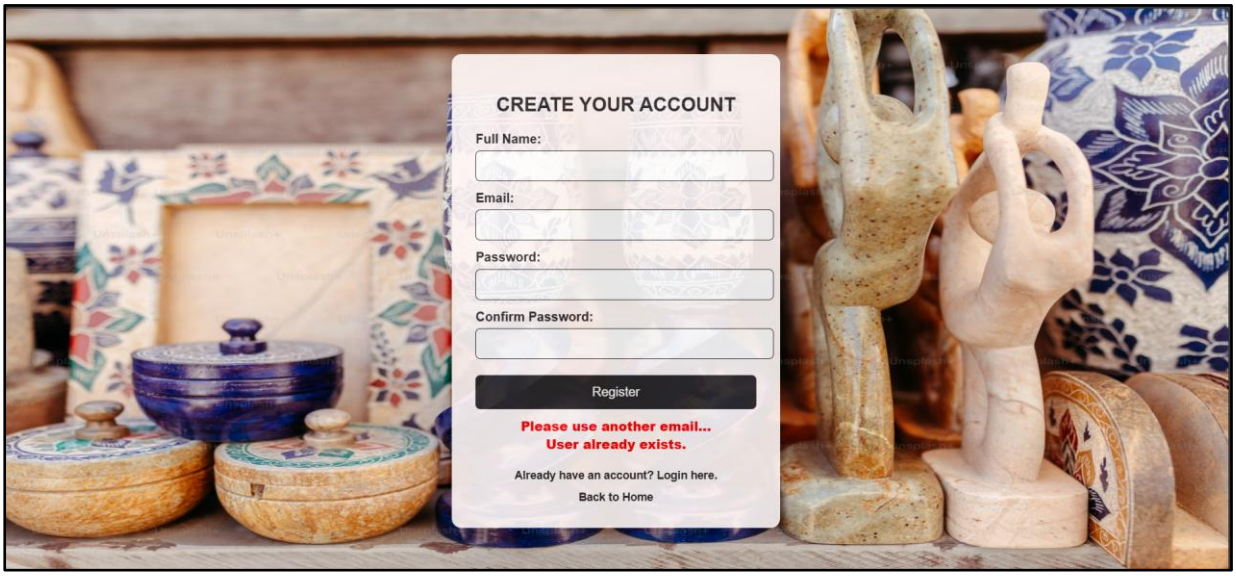
MODULE NAME – USER



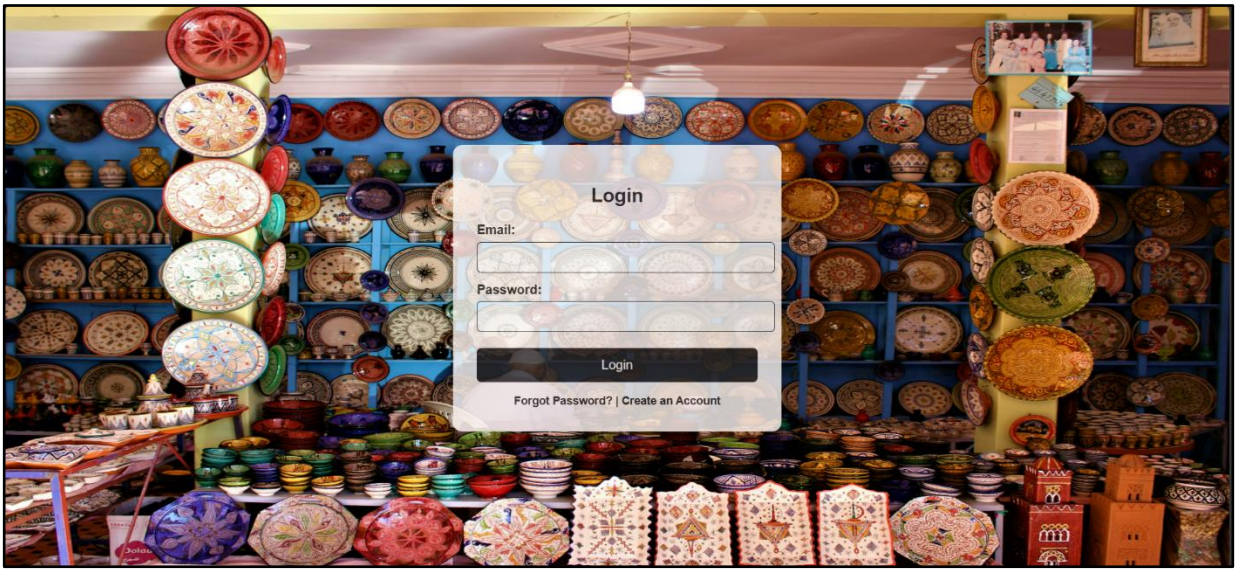
SUCCESS REGISTER



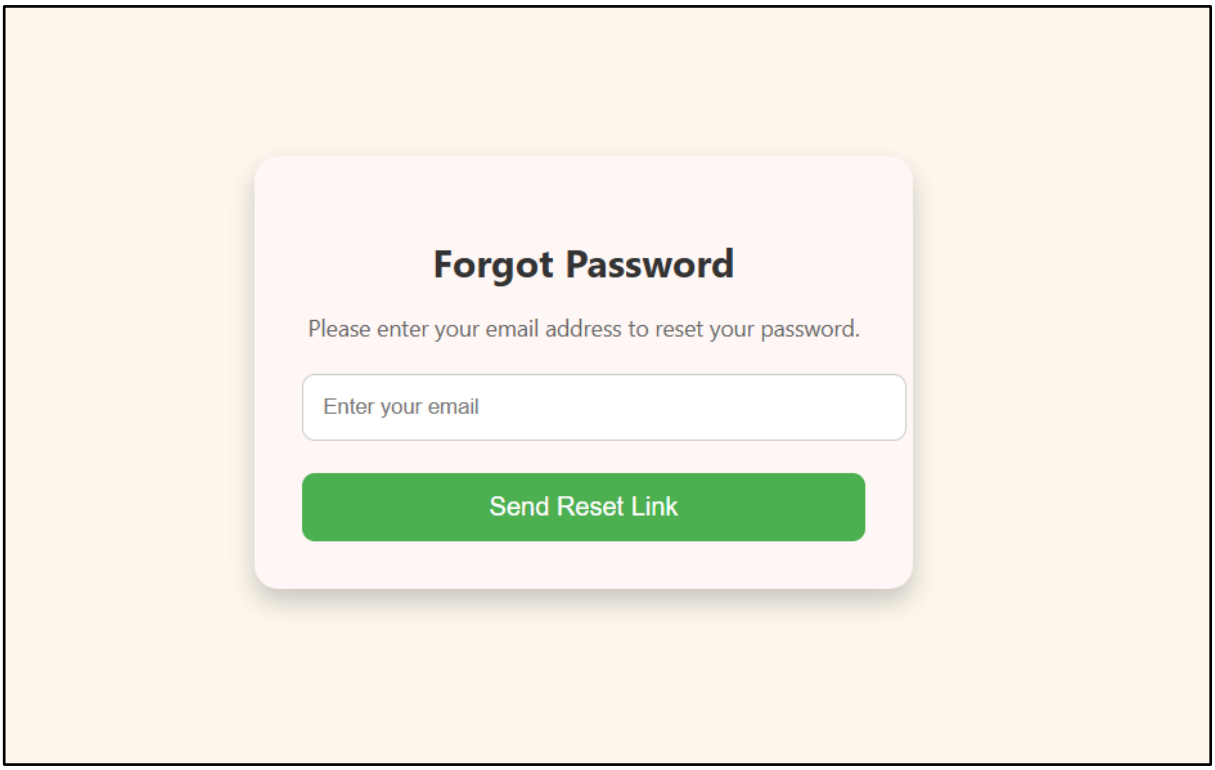
USER ALREADY EXISTS



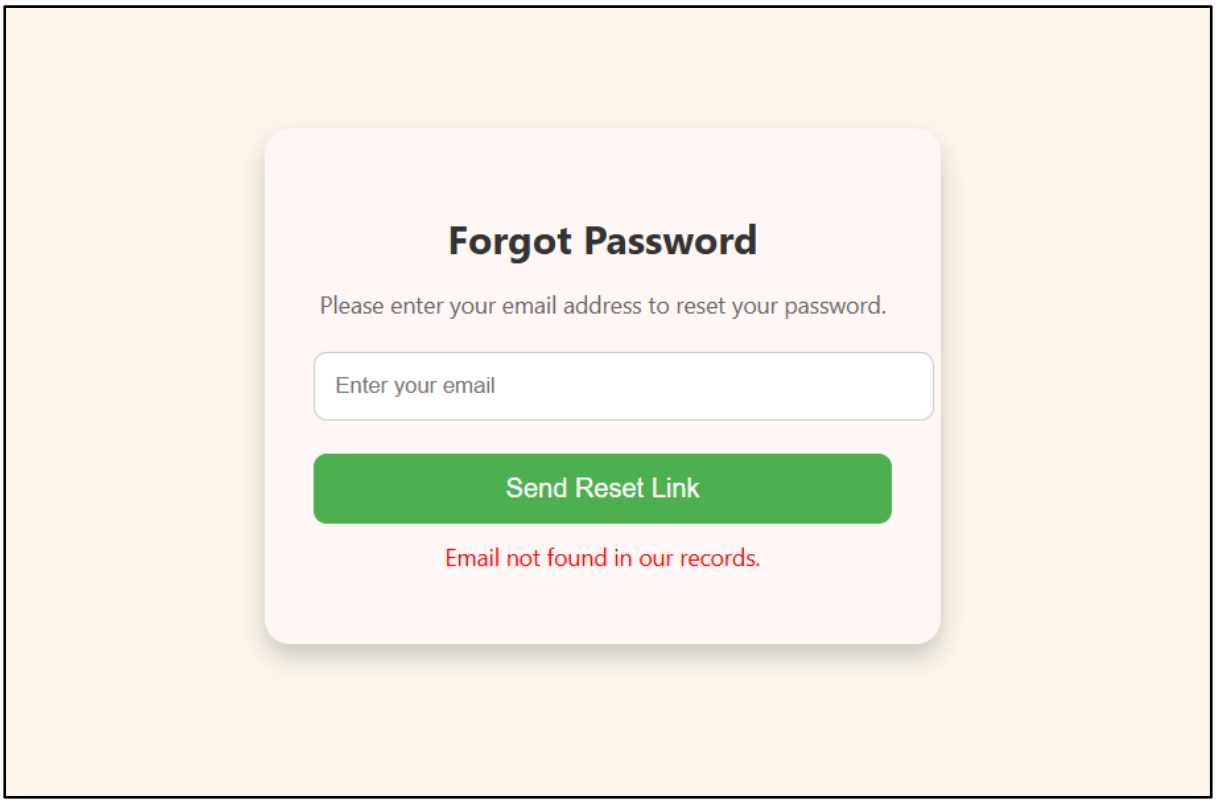
LOG IN



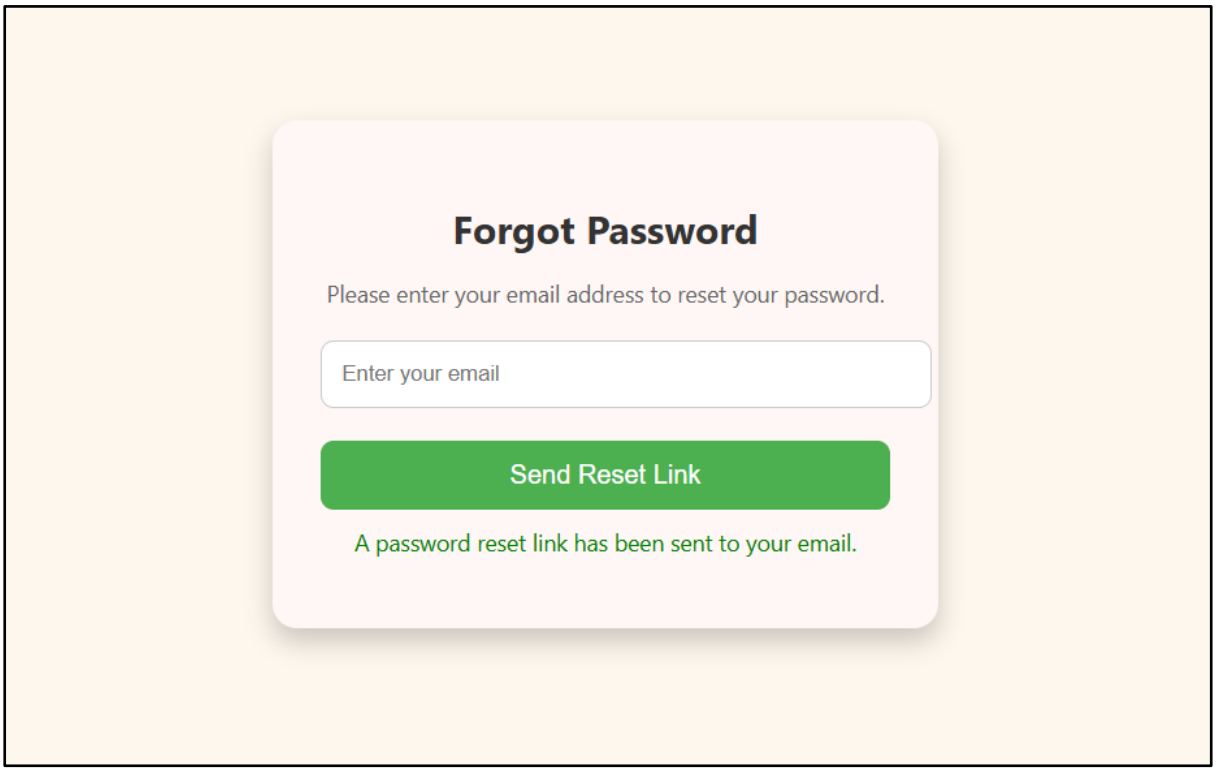
FORGET PASSWORD



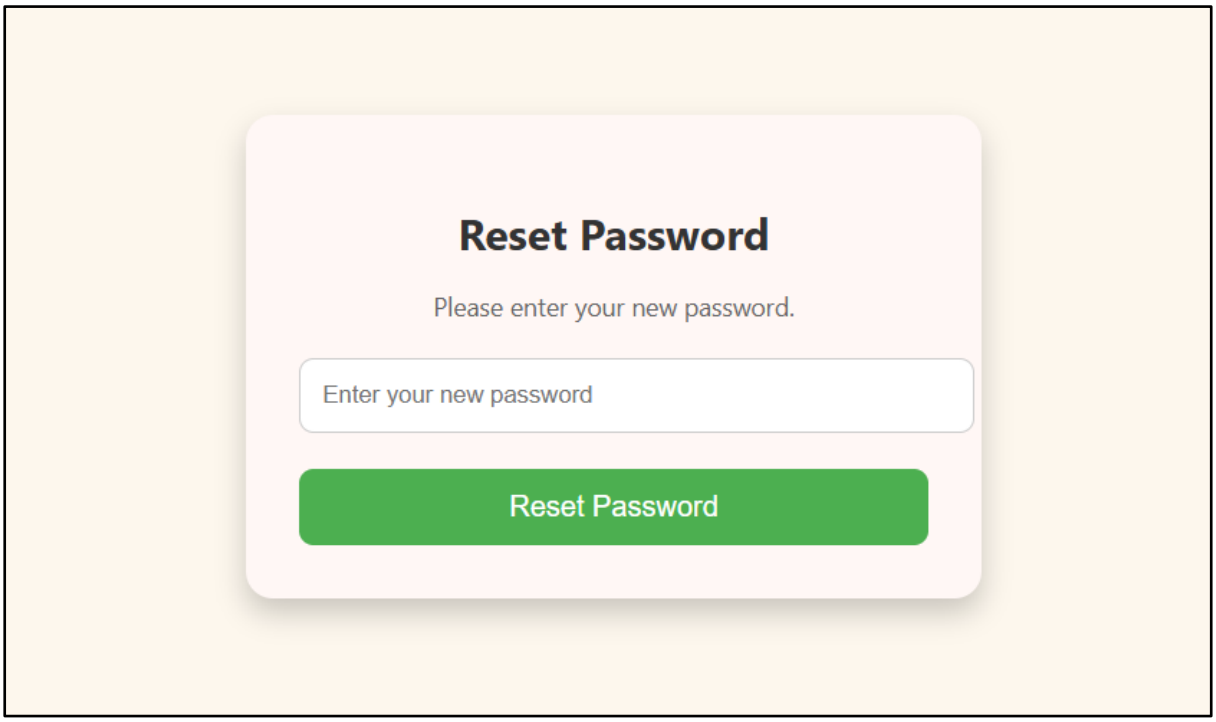
EMAIL NOT FOUND



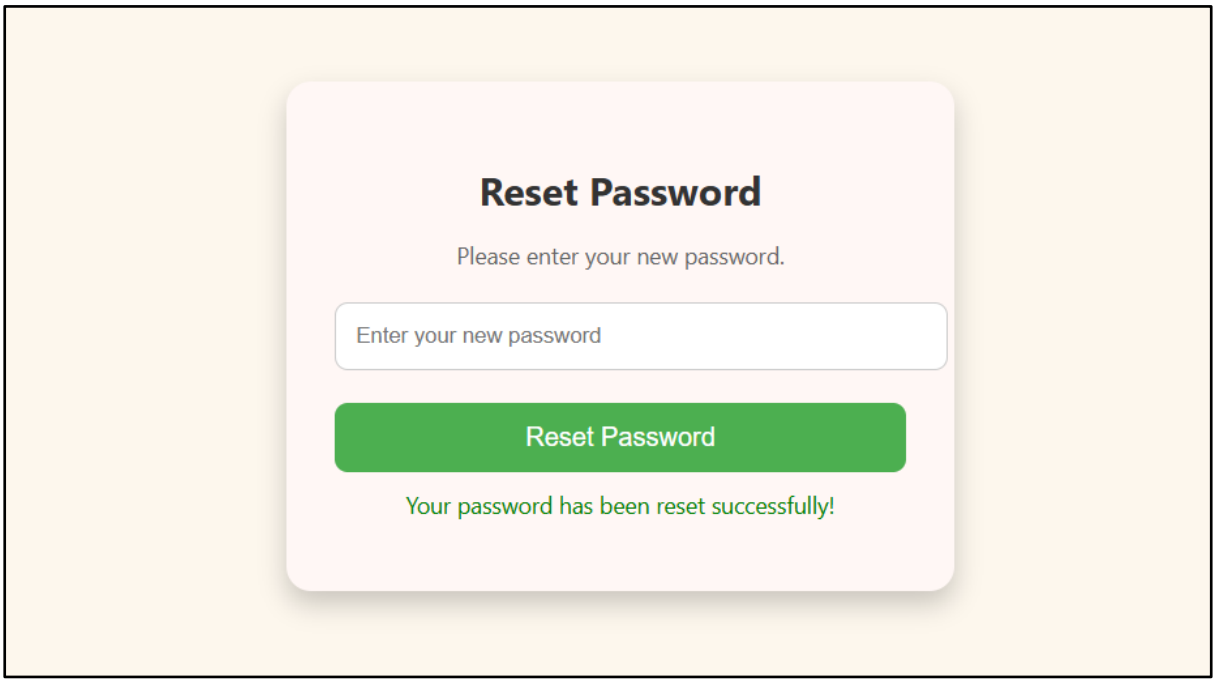
SEND RESET LINK ON EMAIL



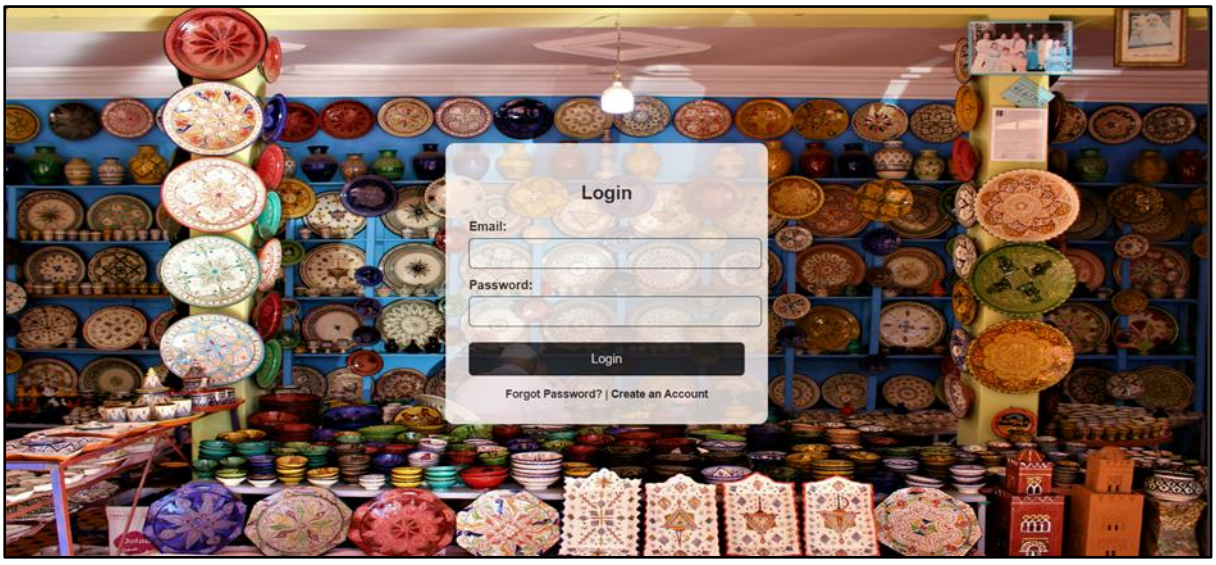
RESET PASSWORD



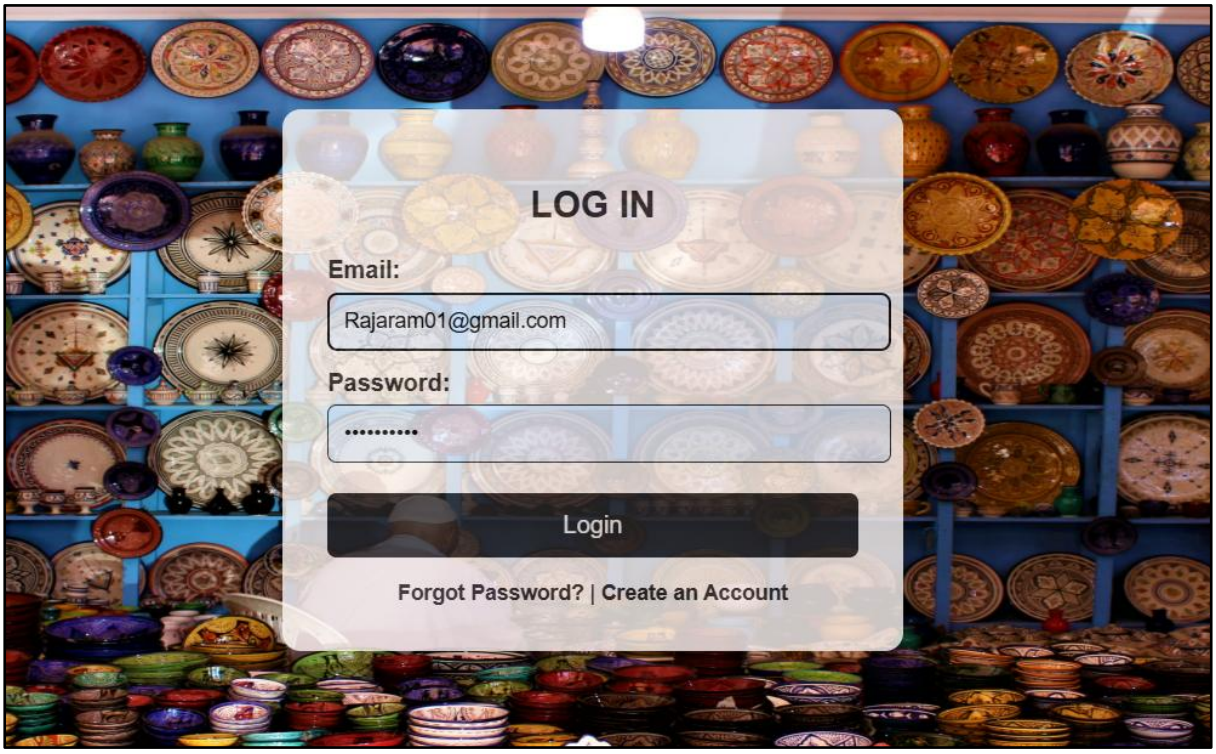
SUCCESSFULLY UPDATE PASSWORD

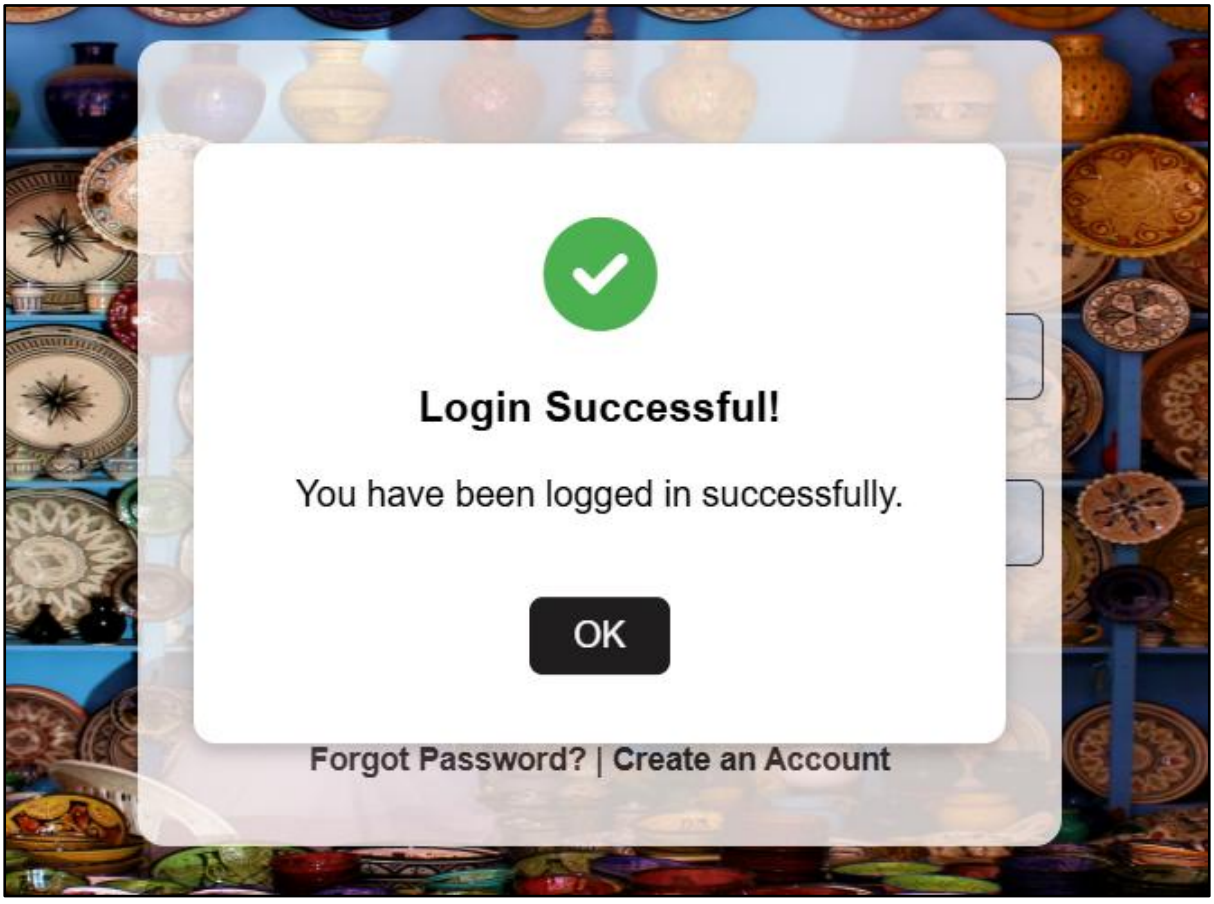


CHECK LOG IN



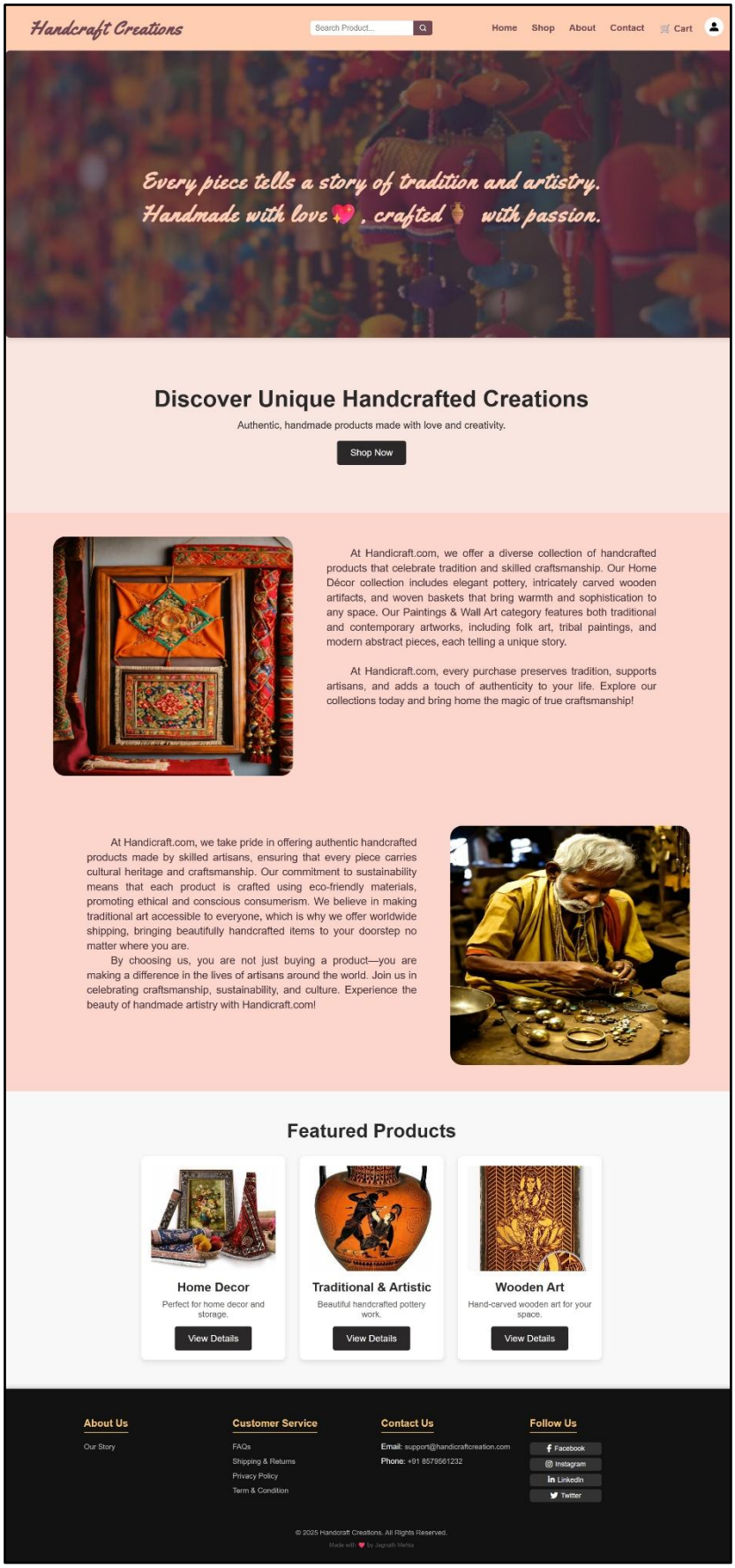
LOG IN SUCCESSFULLY





Online Handicraft System

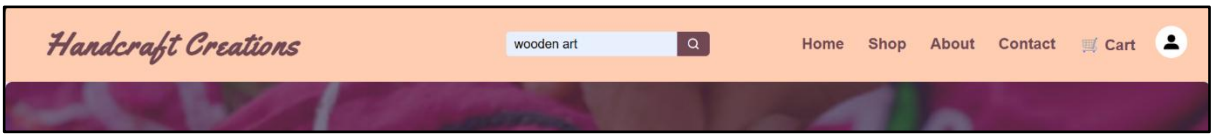
HOME PAGE

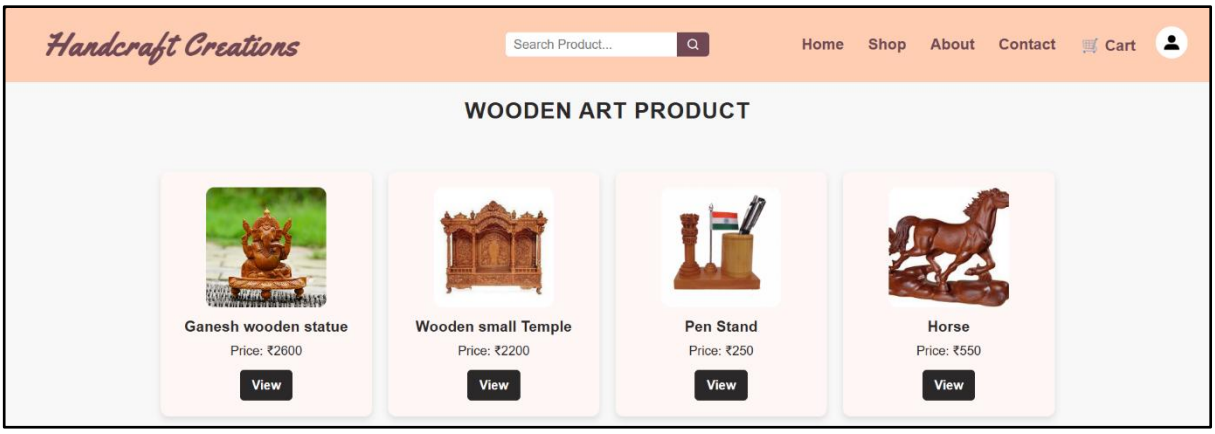


SEARCH OPTION



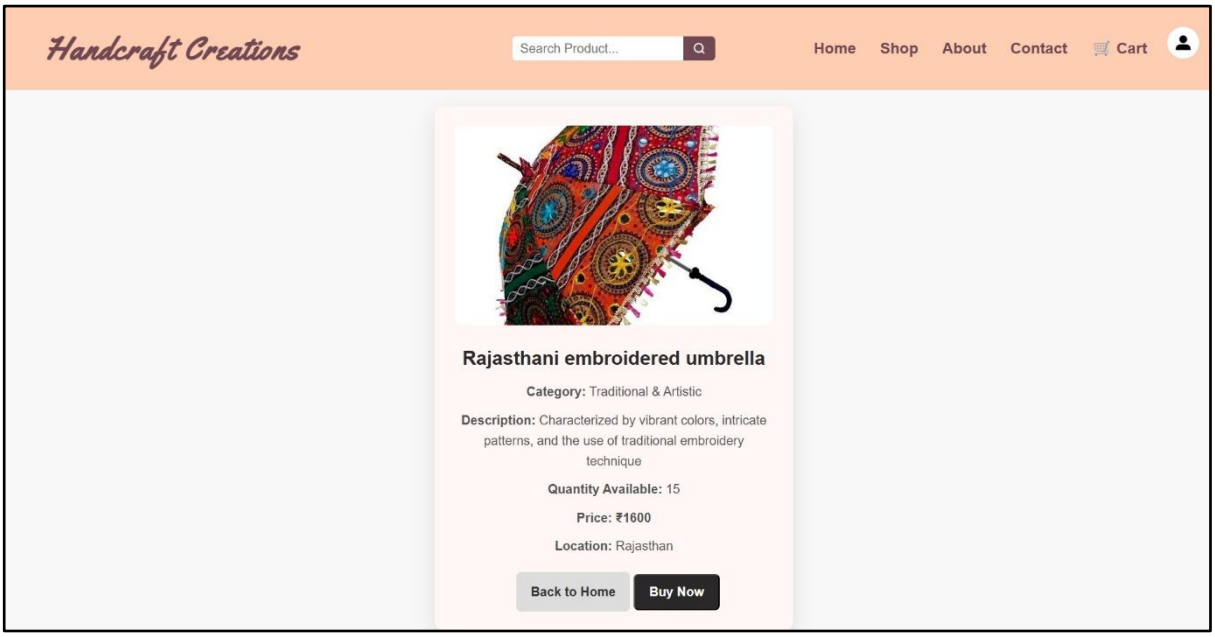
SEARCH BY CATEGORY



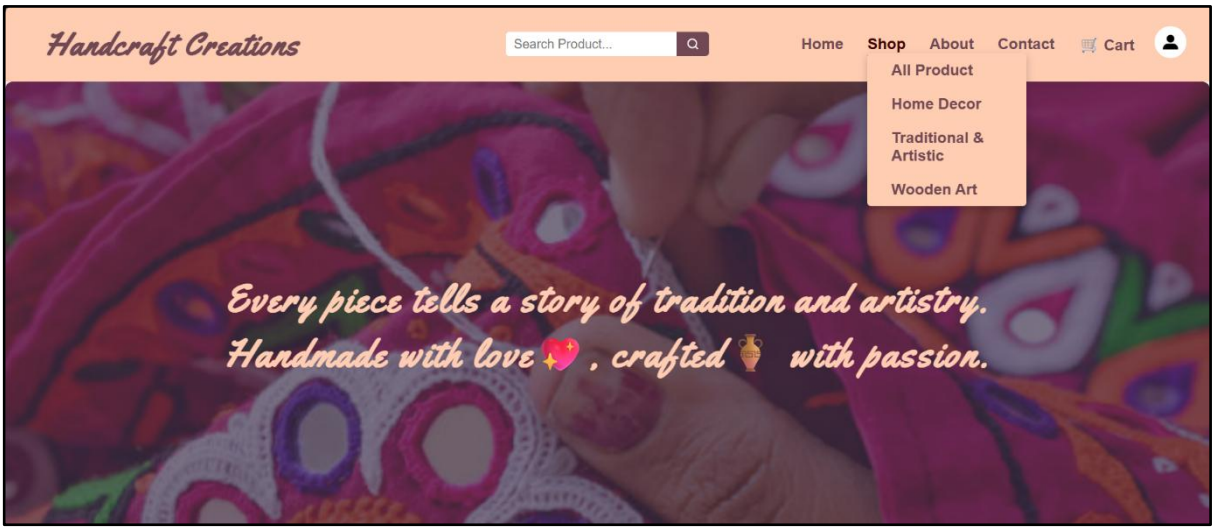


SEARCH BY PRODUCT

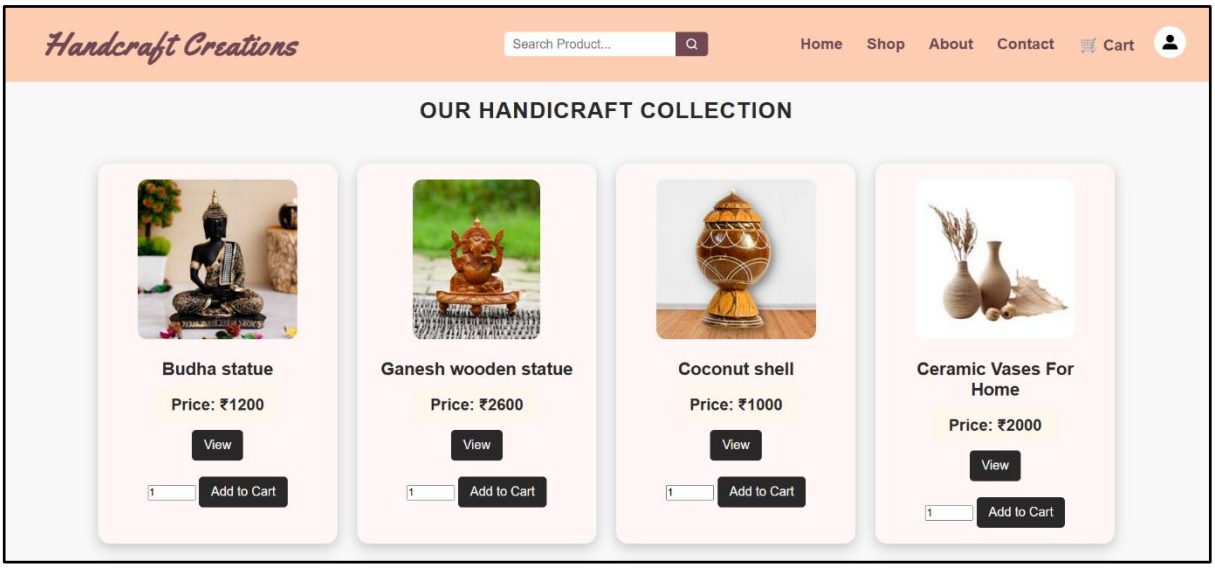




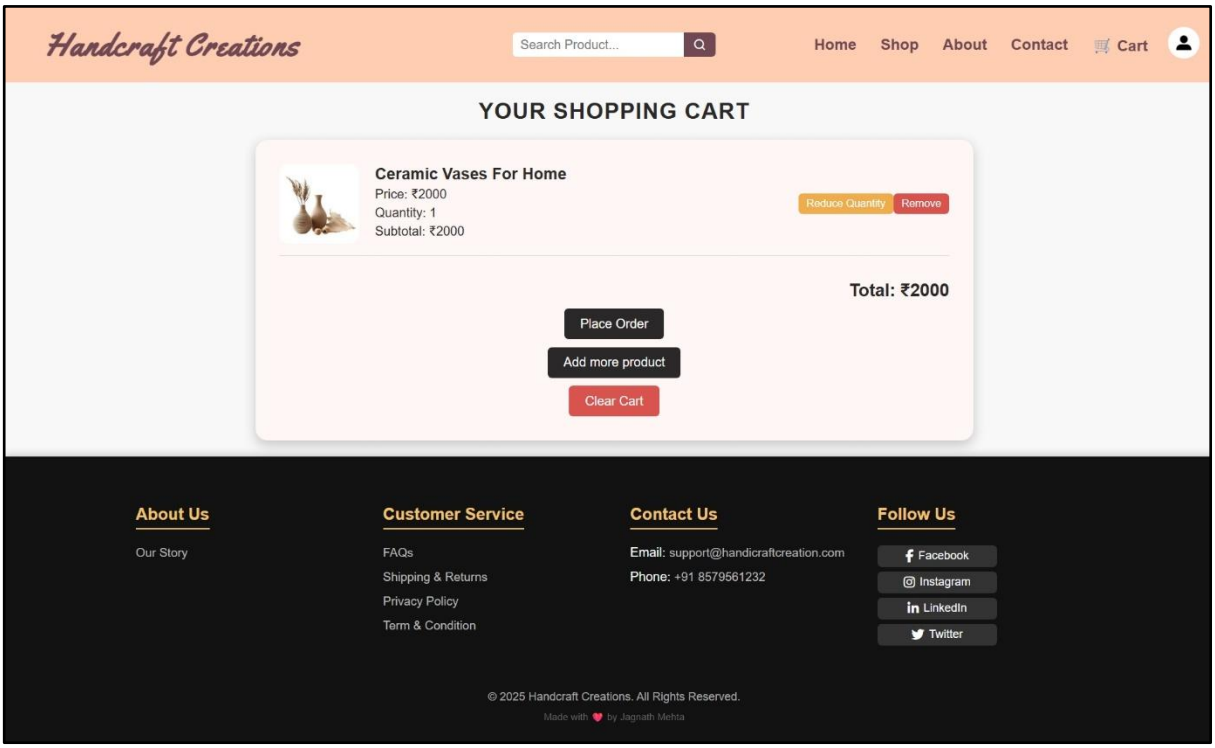
SHOP MENU



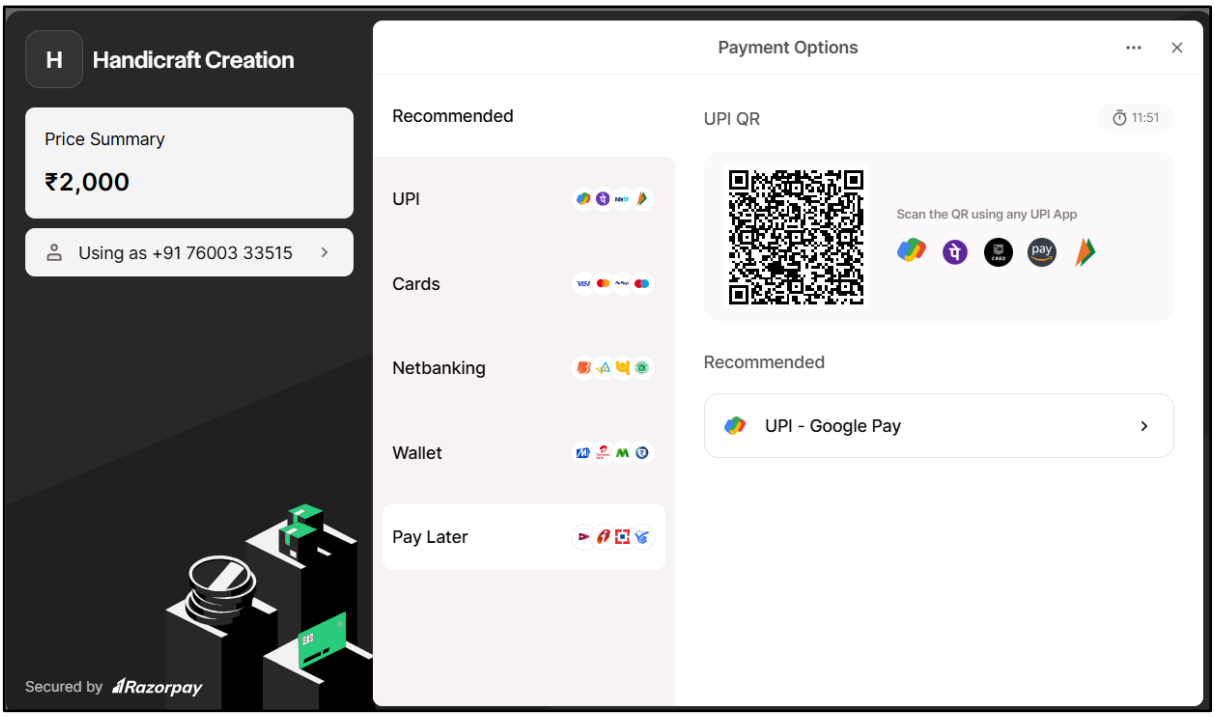
ALL PRODUCT



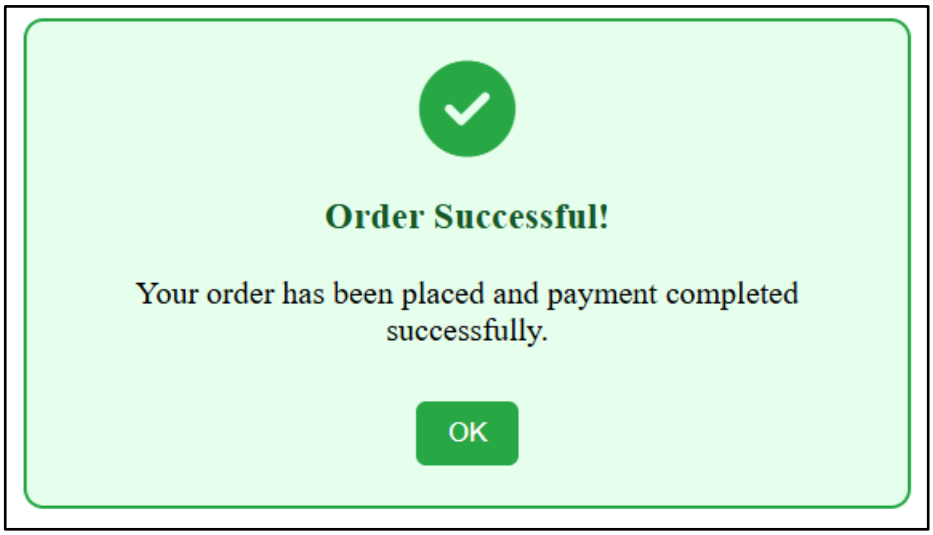
USING ADD TO CART FEATURE BUY CERAMIC VASES FOR HOME

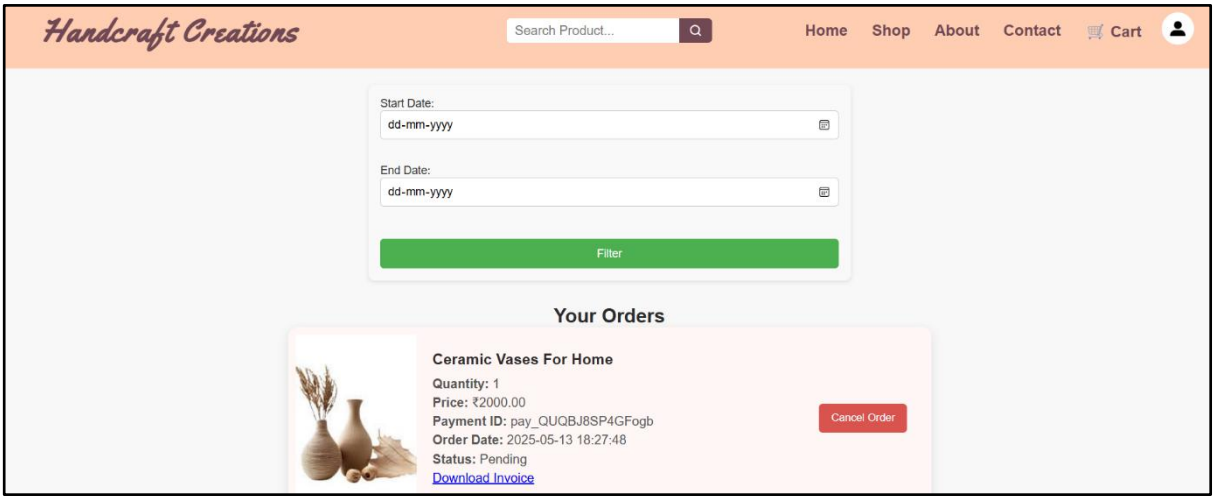


PAYMENT FOR CERAMIC VASES



CONFIRMATION OF ORDER COMPLETED SUCCESSFULLY





DOWNLOAD INVOICE OF CERAMIC VASES



Handcraft Creations - Invoice

Payment ID: pay_QUQBJ8SP4GFogb

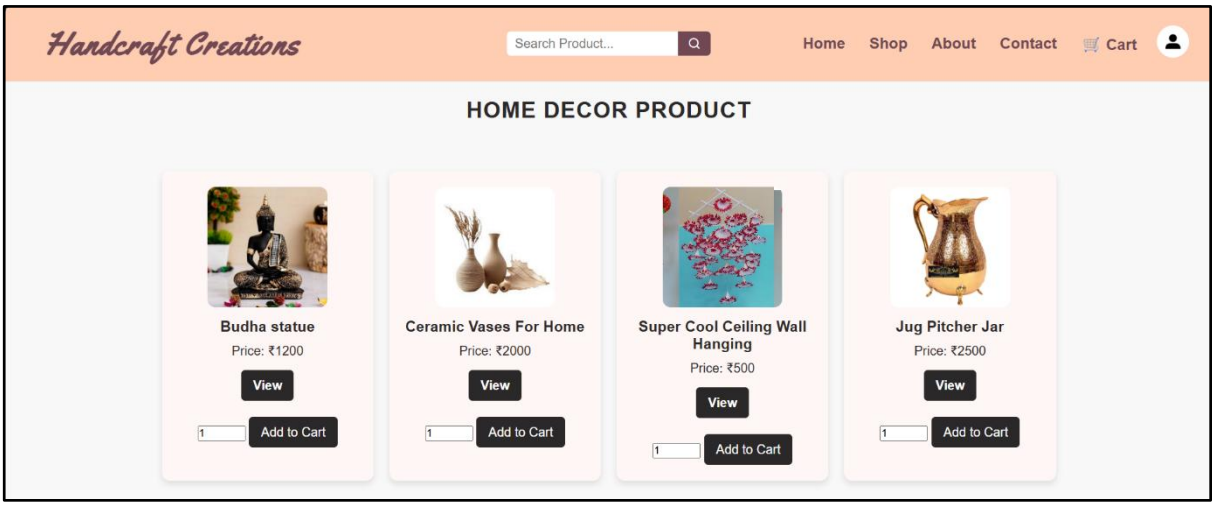
Username: Jagnath Mehta

Order Date: 2025-05-13 18:27:48

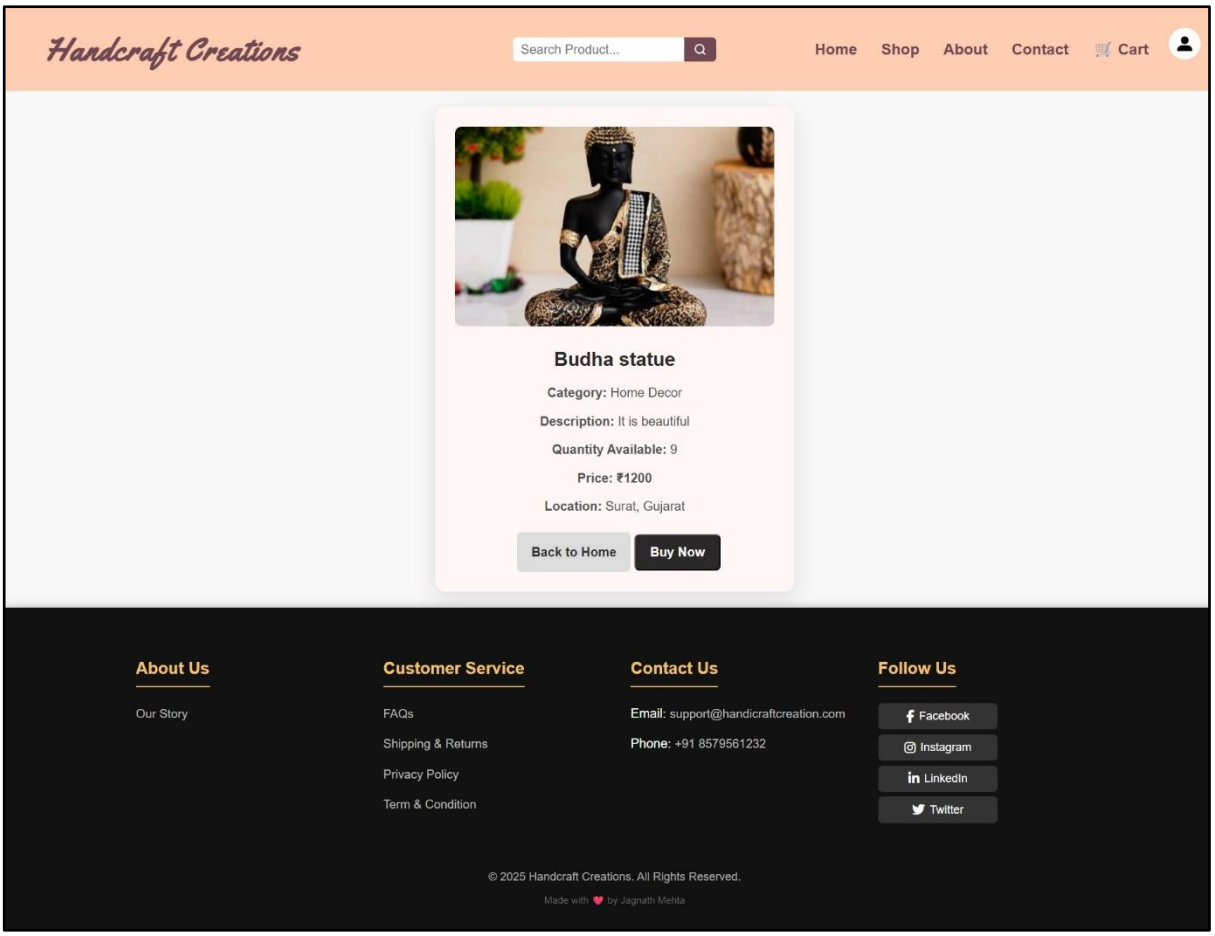
Product	Qty	Price	Subtotal
Ceramic Vases For Home	1	2,000.00	2,000.00

Total: 2,000.00

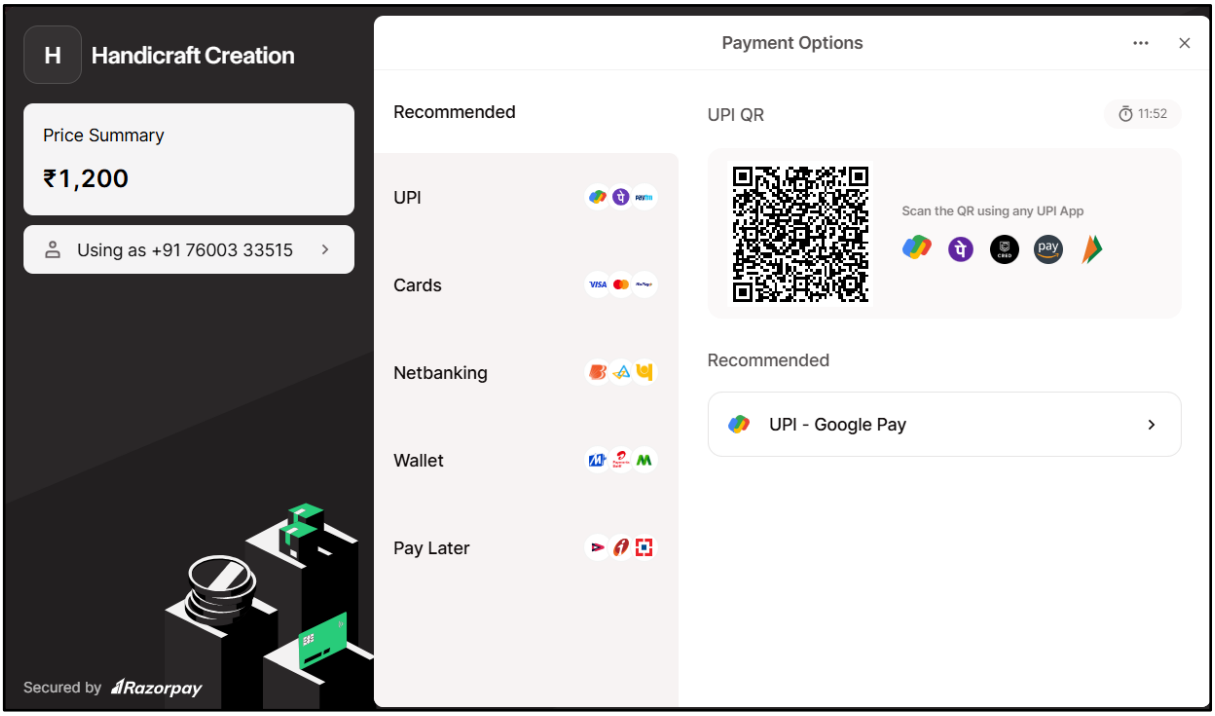
HOME DECOR



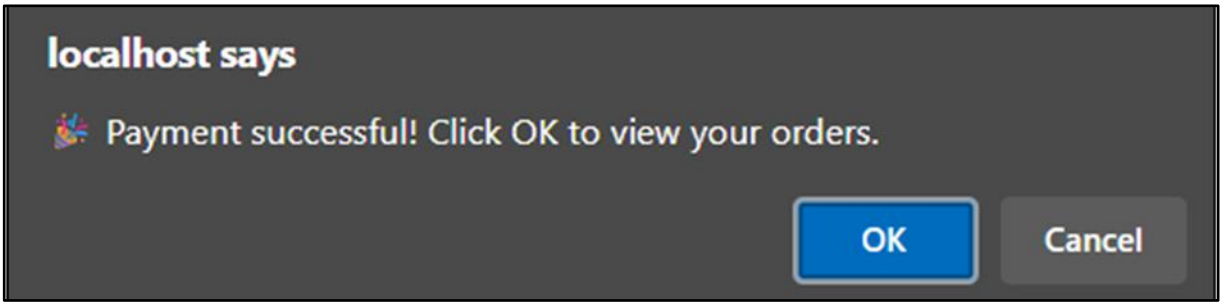
BUY BUDHA STATUE

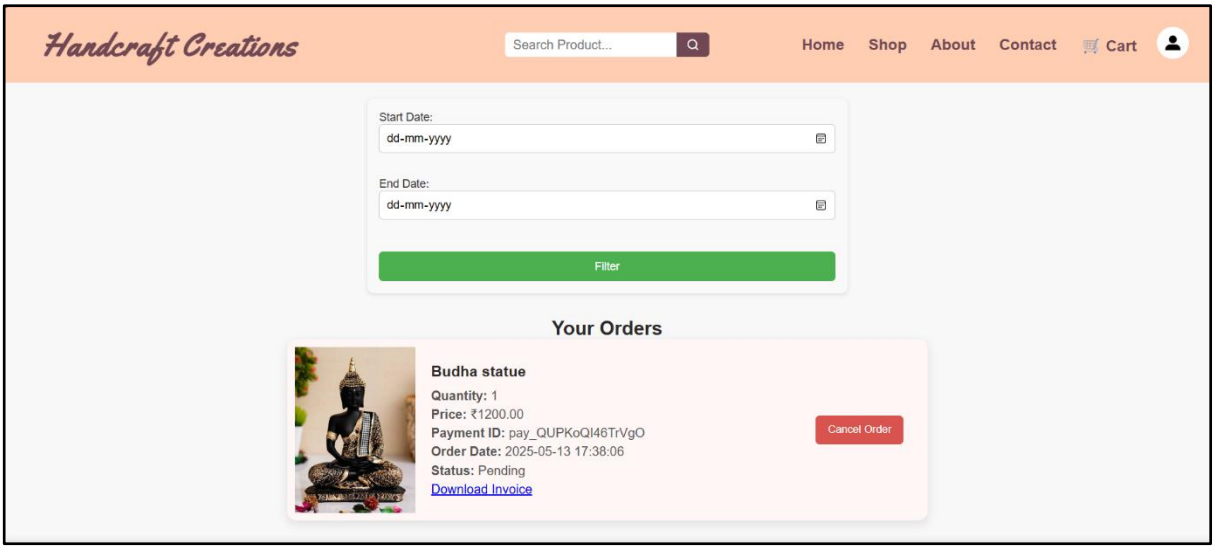


PAYMENT FOR BUDHA STATUE



COMPLETE PAYMENT PROCCESS FOR BUDHA STATUE ORDER





DOWNLOAD INVOICE OF BUDHA STATUE



Handcraft Creations - Invoice

Payment ID: pay_QUPKoQI46TrVgO

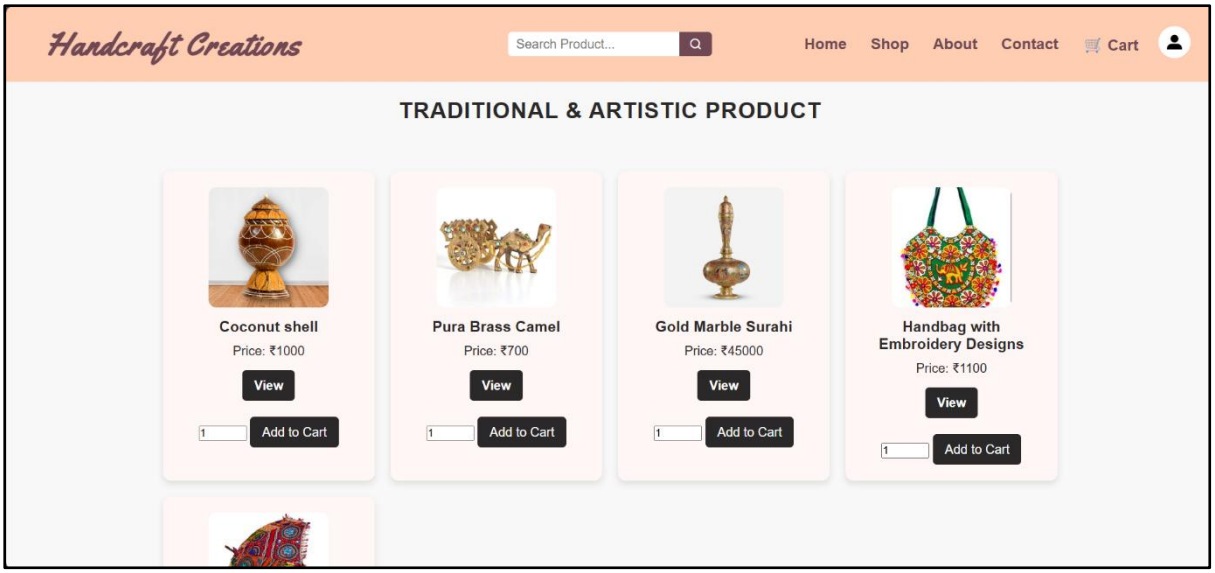
Username: Jagnath Mehta

Order Date: 2025-05-13 17:38:06

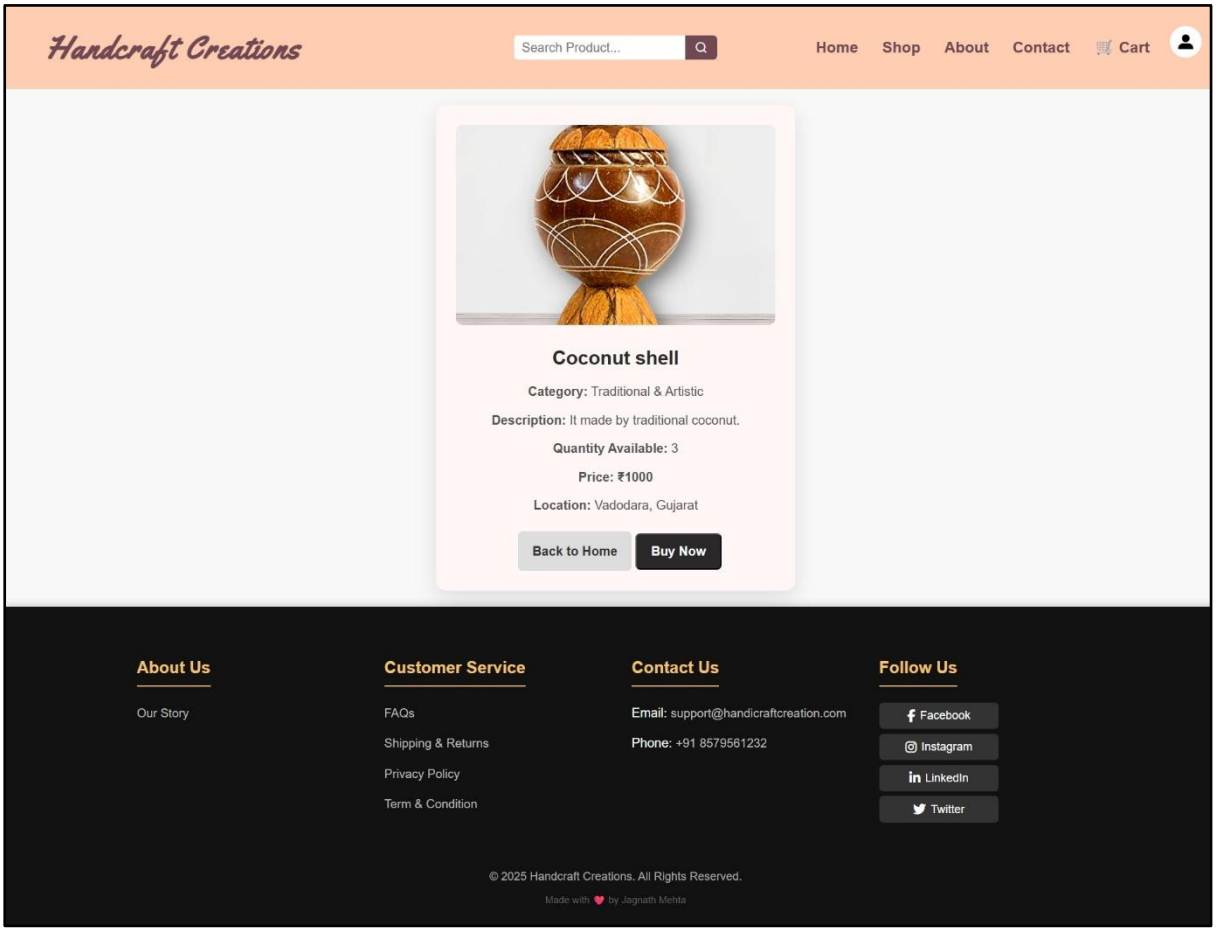
Product	Qty	Price	Subtotal
Budha statue	1	1,200.00	1,200.00

Total: 1,200.00

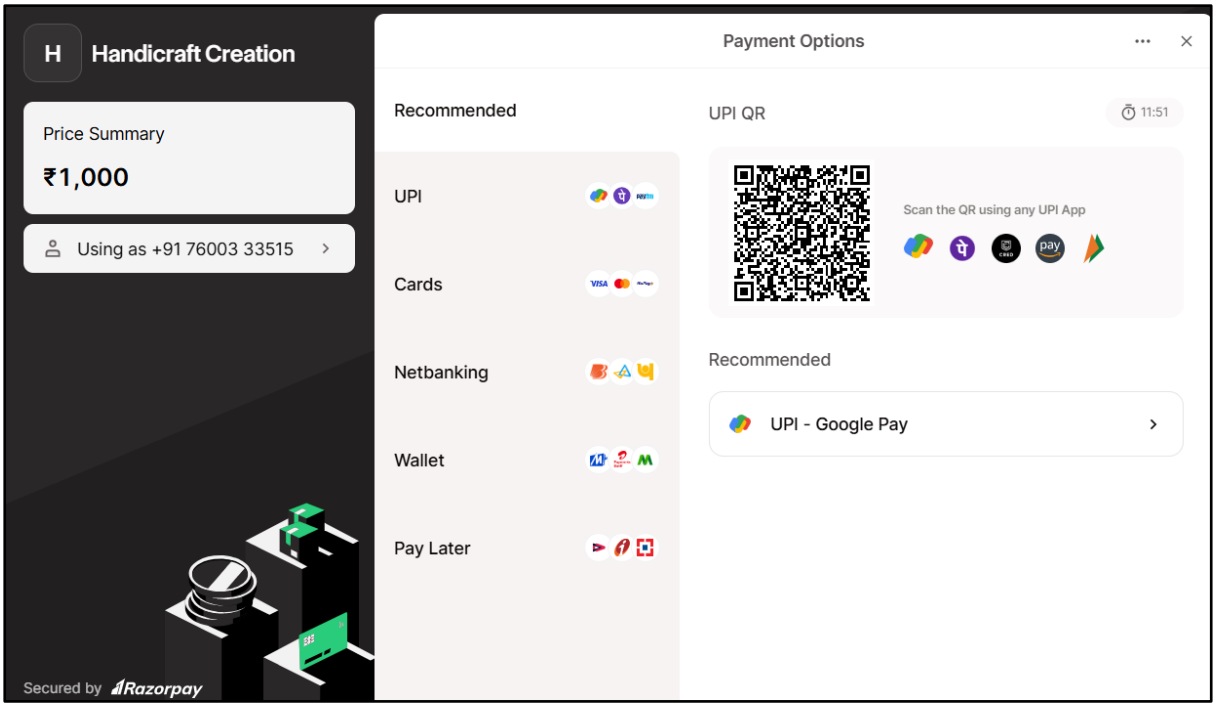
TRADITIONAL & ARTSTIC PRODUCT



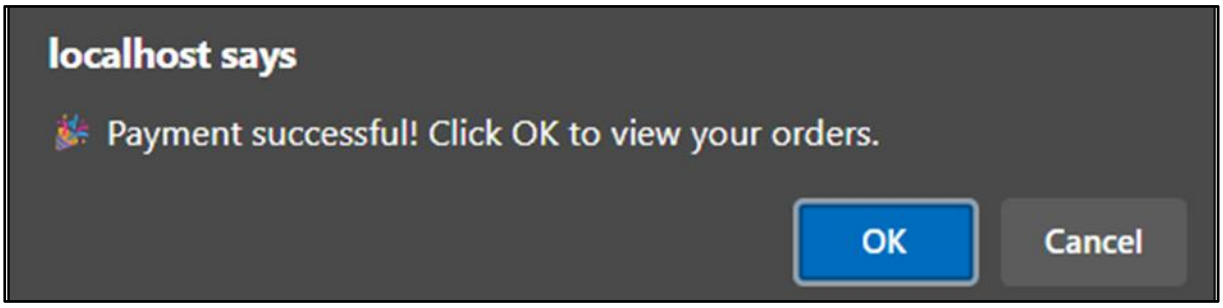
BUY COCONUT SHELL

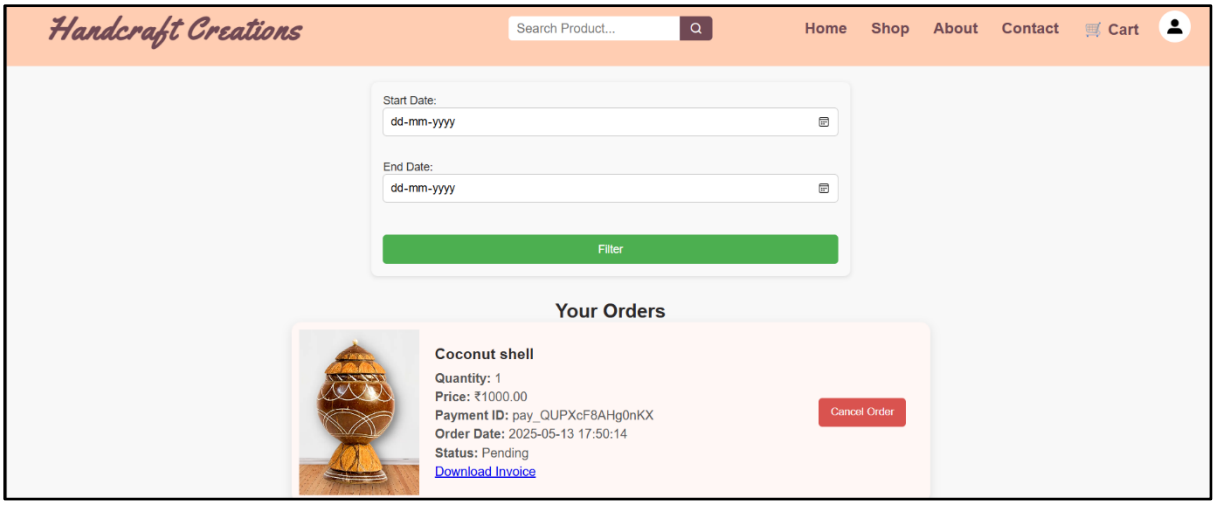


PAYMENT FOR COCONUT SHELL



COMPLETE PAYMENT PROCCESS FOR COCONUT SHELL ORDER

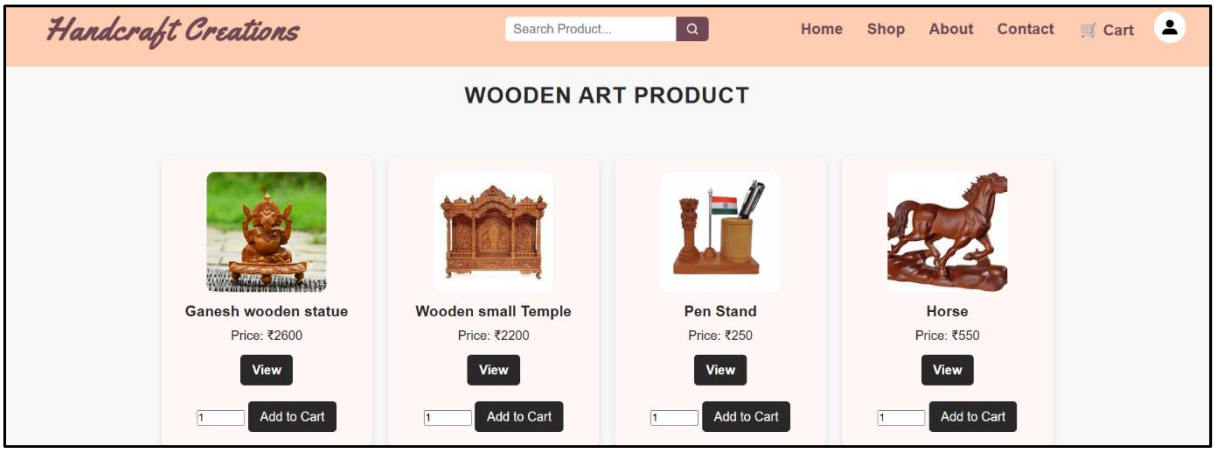




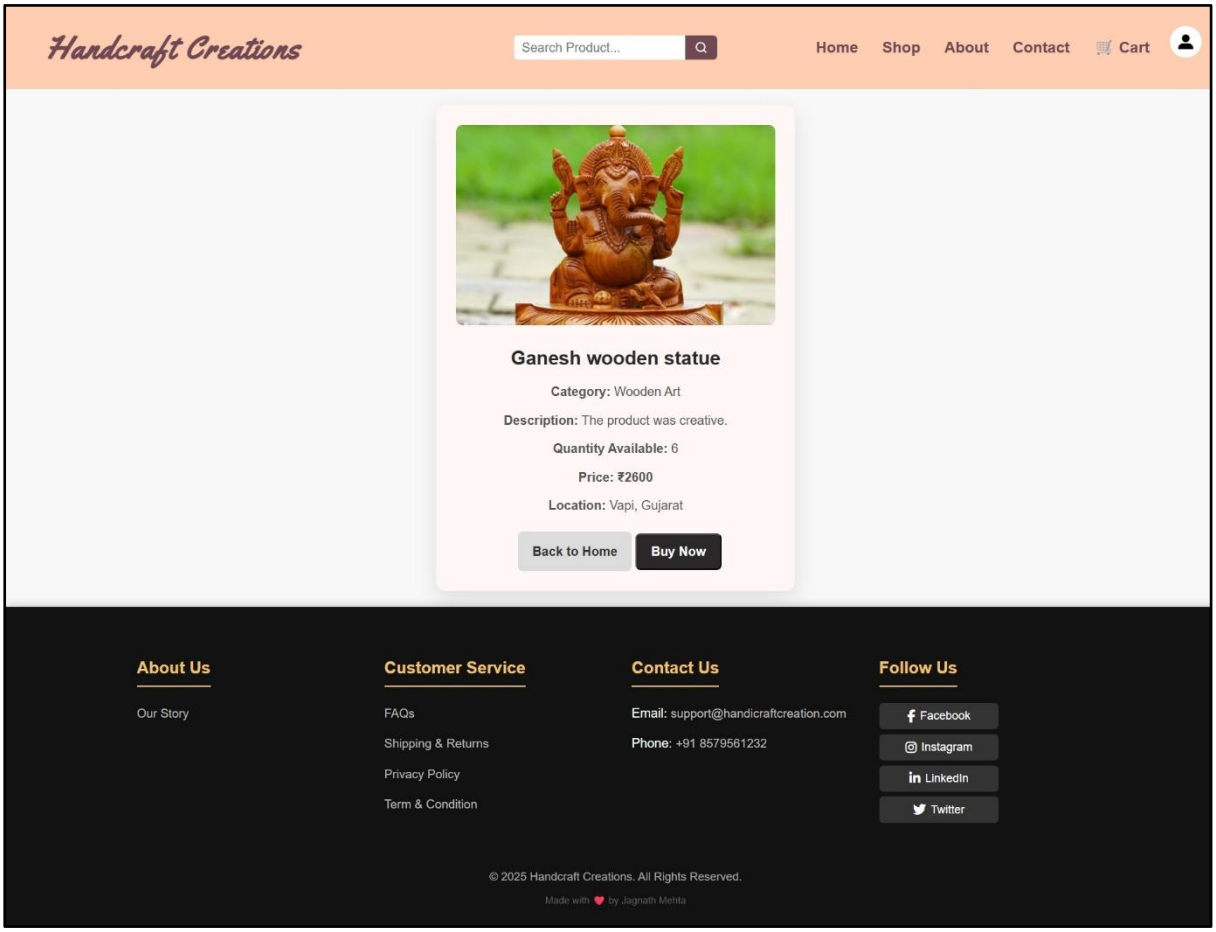
DOWNLOAD INVOICE OF COCONUT SHELL



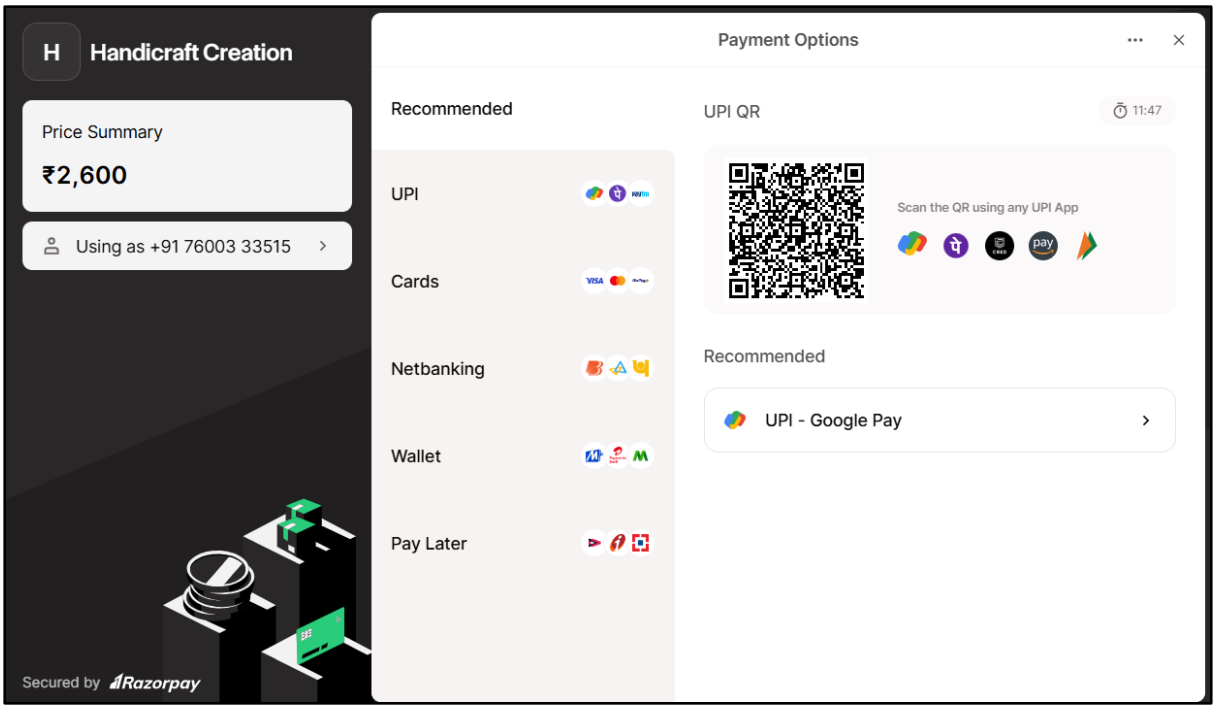
WOODEN ART PRODUCT



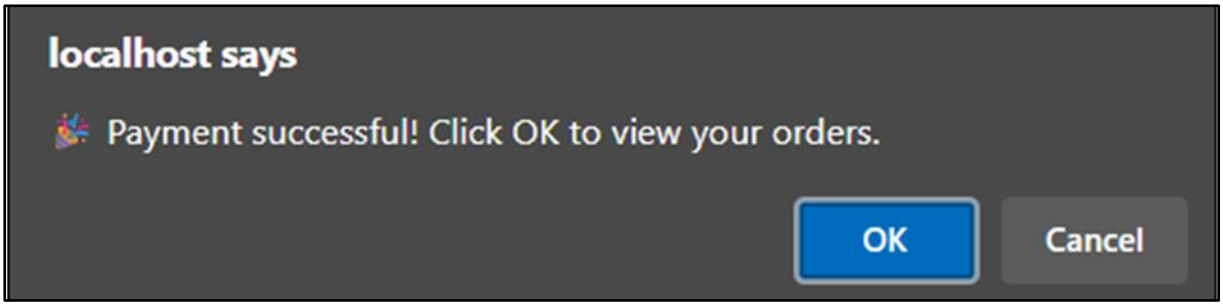
BUY GANESH WOODEN STATUE

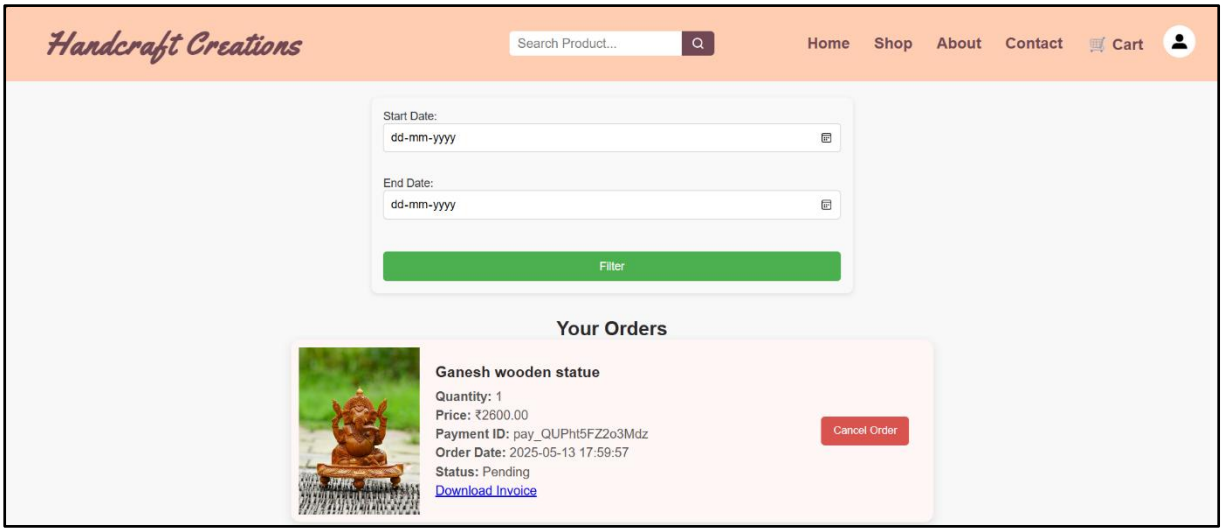


PAYMENT FOR GANESH WOODEN STATUE



COMPLETE PAYMENT PROCCESS FOR GANESH WOODEN STATUE ORDER





DOWNLOAD INVOICE OF GANESH WOODEN STATUE



Handcraft Creations - Invoice

Payment ID: pay_QUPht5FZ2o3Mdz

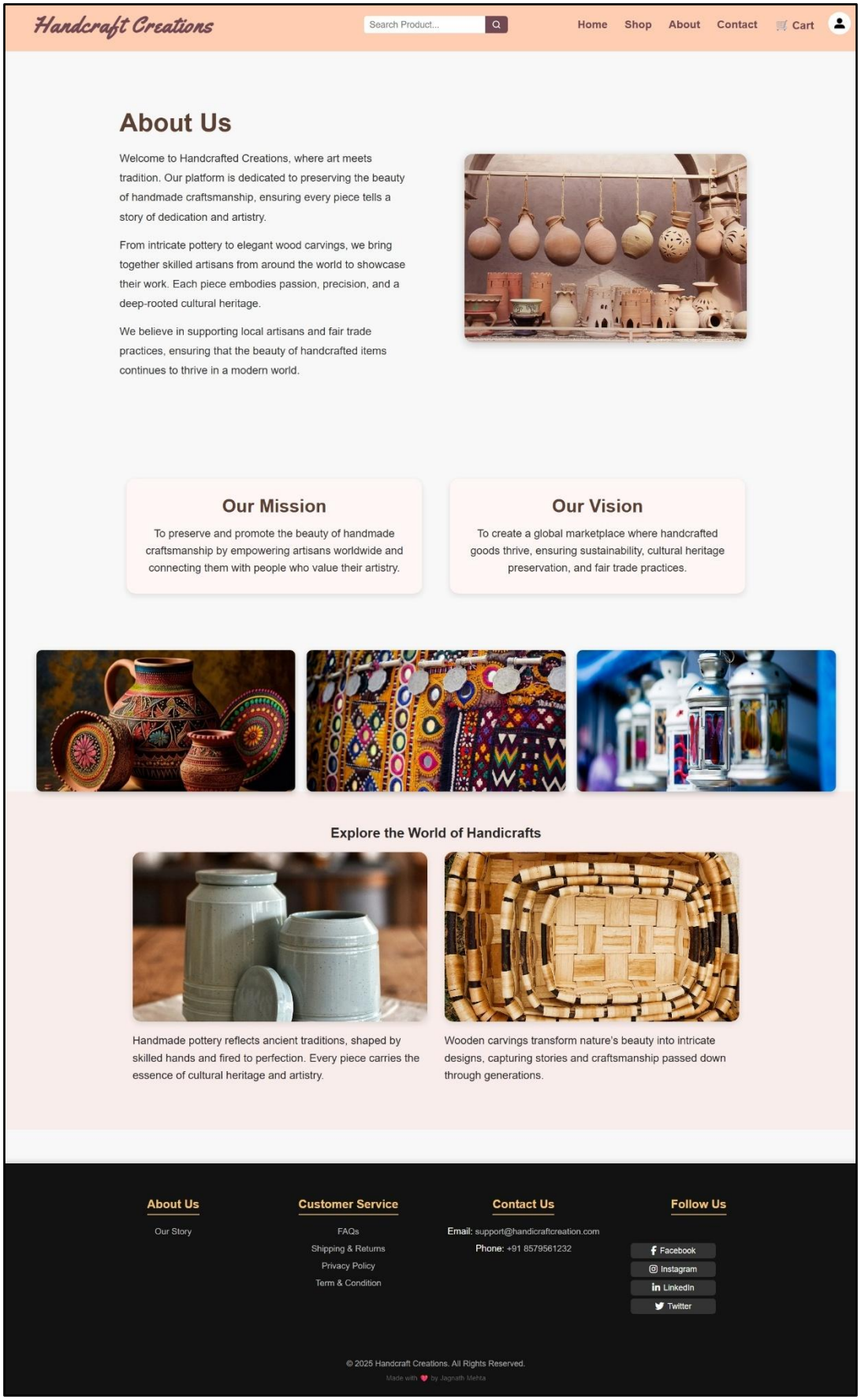
Username: Jagnath Mehta

Order Date: 2025-05-13 17:59:57

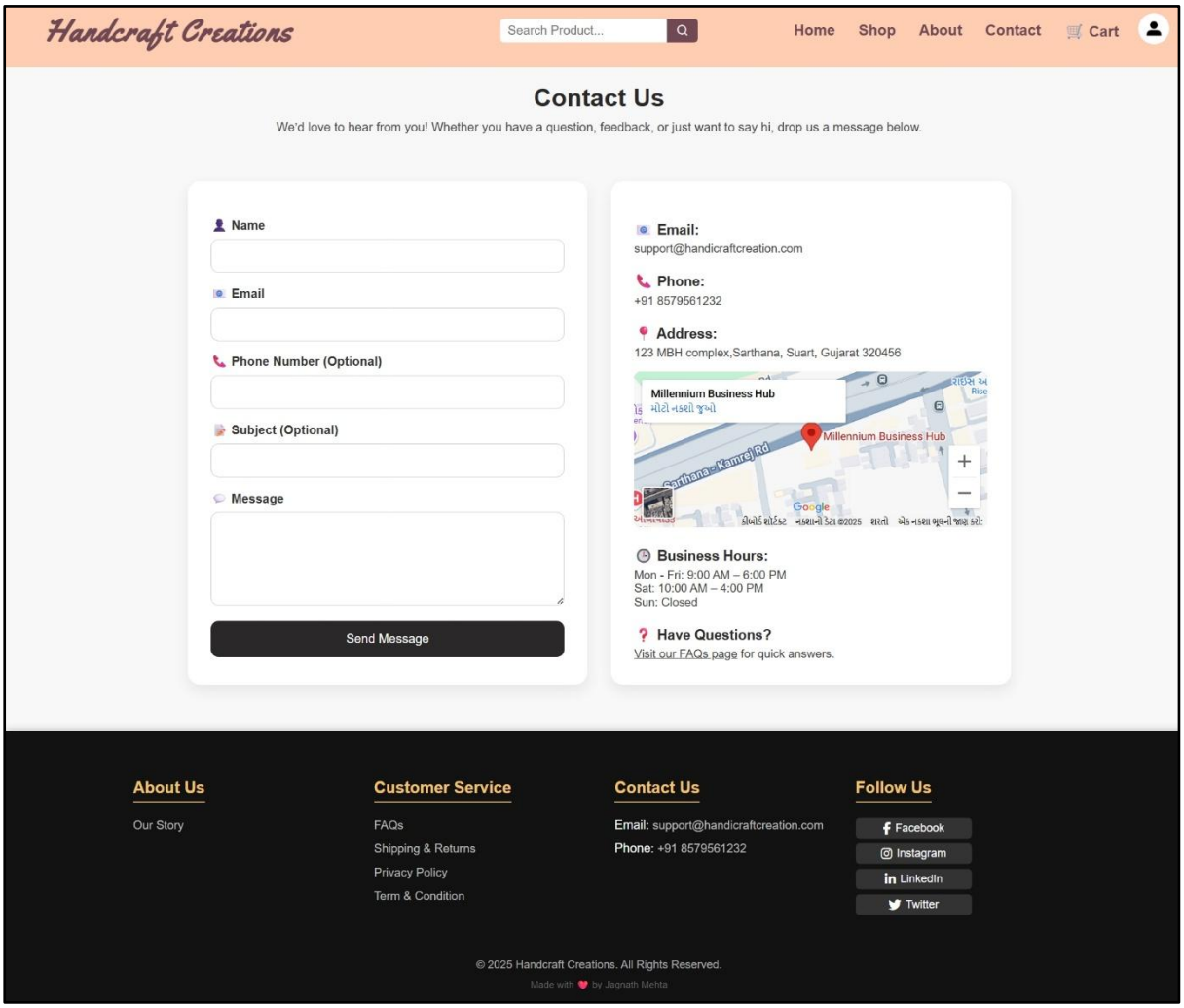
Product	Qty	Price	Subtotal
Ganesh wooden statue	1	2,600.00	2,600.00

Total: 2,600.00

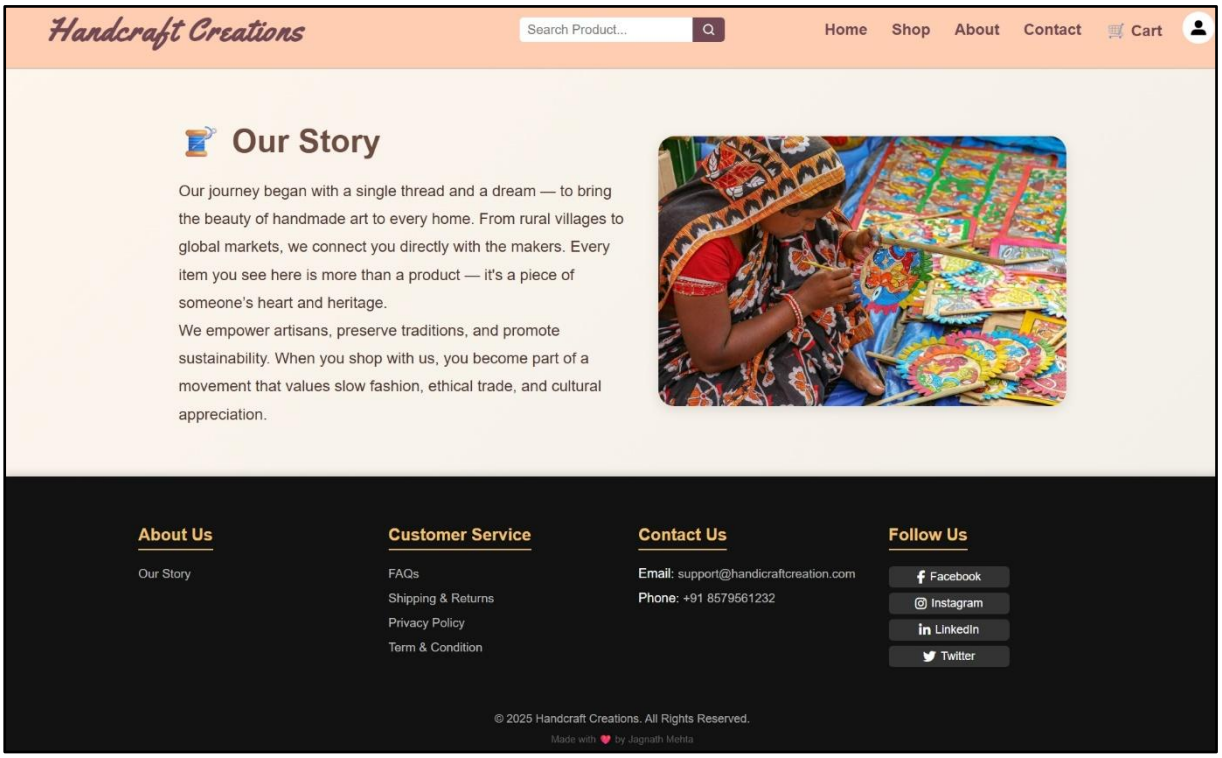
ABOUT US


Handcraft Creations
Search Product...
Home
Shop
About
Contact
Cart
About Us
Welcome to Handcrafted Creations, where art meets tradition. Our platform is dedicated to preserving the beauty of handmade craftsmanship, ensuring every piece tells a story of dedication and artistry.
From intricate pottery to elegant wood carvings, we bring together skilled artisans from around the world to showcase their work. Each piece embodies passion, precision, and a deep-rooted cultural heritage.
We believe in supporting local artisans and fair trade practices, ensuring that the beauty of handcrafted items continues to thrive in a modern world.
Our Mission
To preserve and promote the beauty of handmade craftsmanship by empowering artisans worldwide and connecting them with people who value their artistry.
Our Vision
To create a global marketplace where handcrafted goods thrive, ensuring sustainability, cultural heritage preservation, and fair trade practices.
Explore the World of Handicrafts
Handmade pottery reflects ancient traditions, shaped by skilled hands and fired to perfection. Every piece carries the essence of cultural heritage and artistry.
Wooden carvings transform nature's beauty into intricate designs, capturing stories and craftsmanship passed down through generations.
About Us
Our Story
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FAQs
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Email: support@handicraftcreation.com
Phone: +91 8579561232
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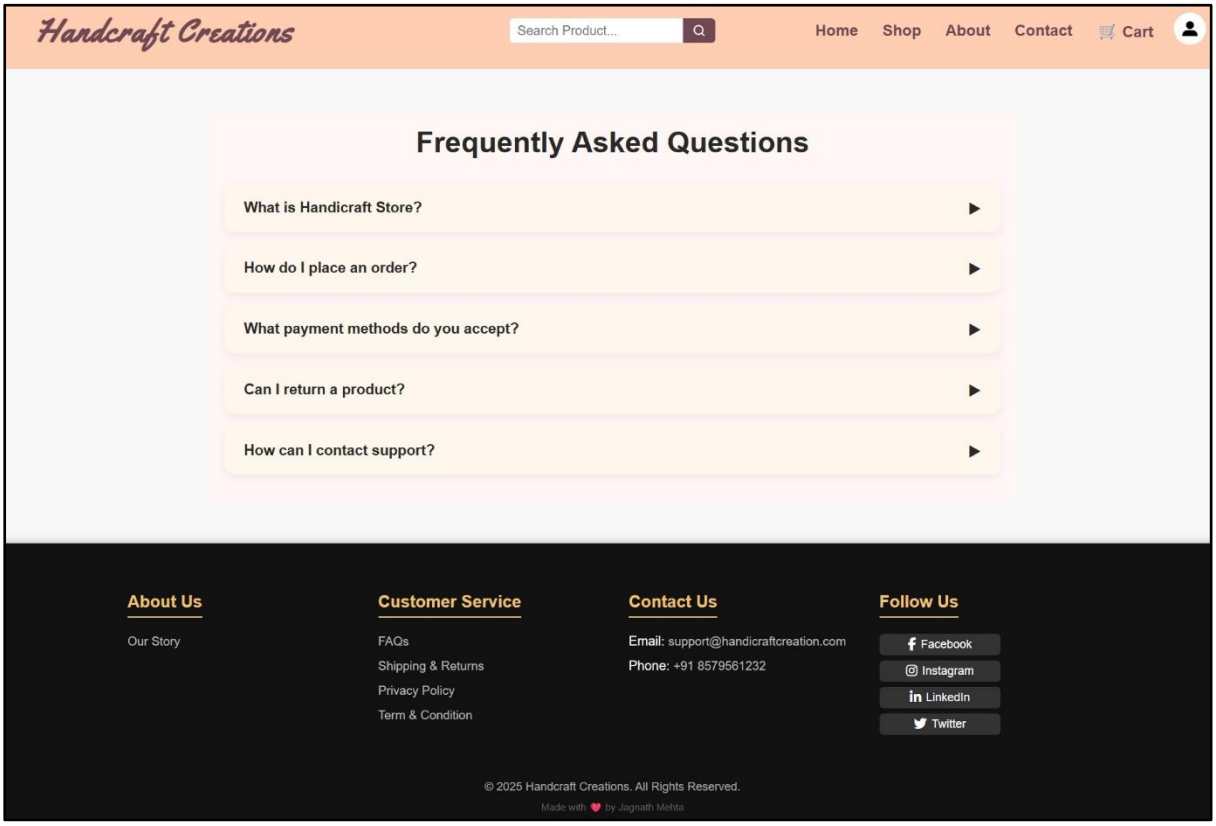
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We'd love to hear from you! Whether you have a question, feedback, or just want to say hi, drop us a message below.
Name
Email
Phone Number (Optional)
Subject (Optional)
Message
Send Message
Email:
support@handicraftcreation.com
Phone:
+91 8579561232
Address:
123 MBH complex,Sarthana, Suart, Gujarat 320456
Millennium Business Hub
Business Hours:
Mon - Fri: 9:00 AM – 6:00 PM
Sat: 10:00 AM – 4:00 PM
Sun: Closed
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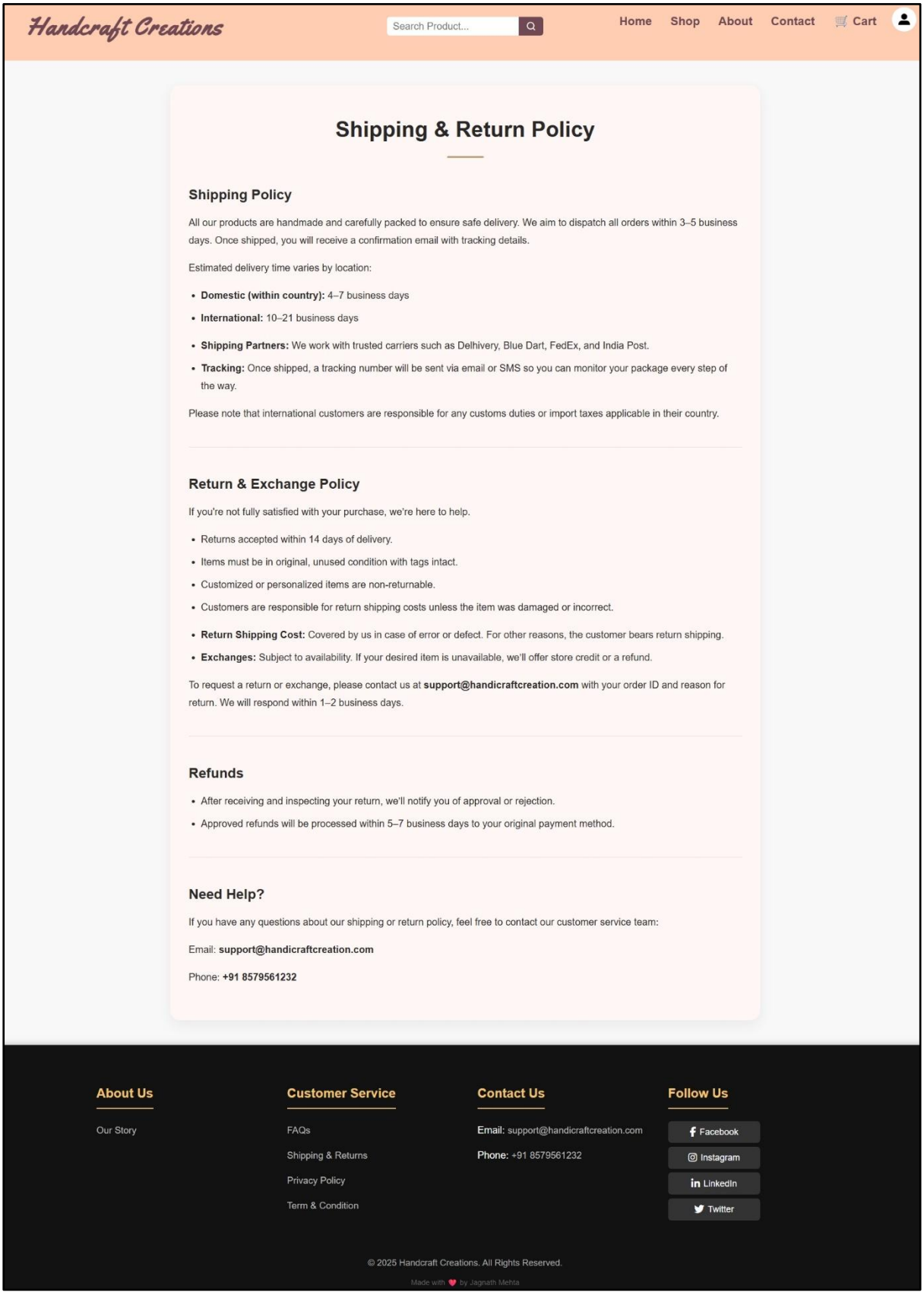
ABOUT US (OUR STORY)



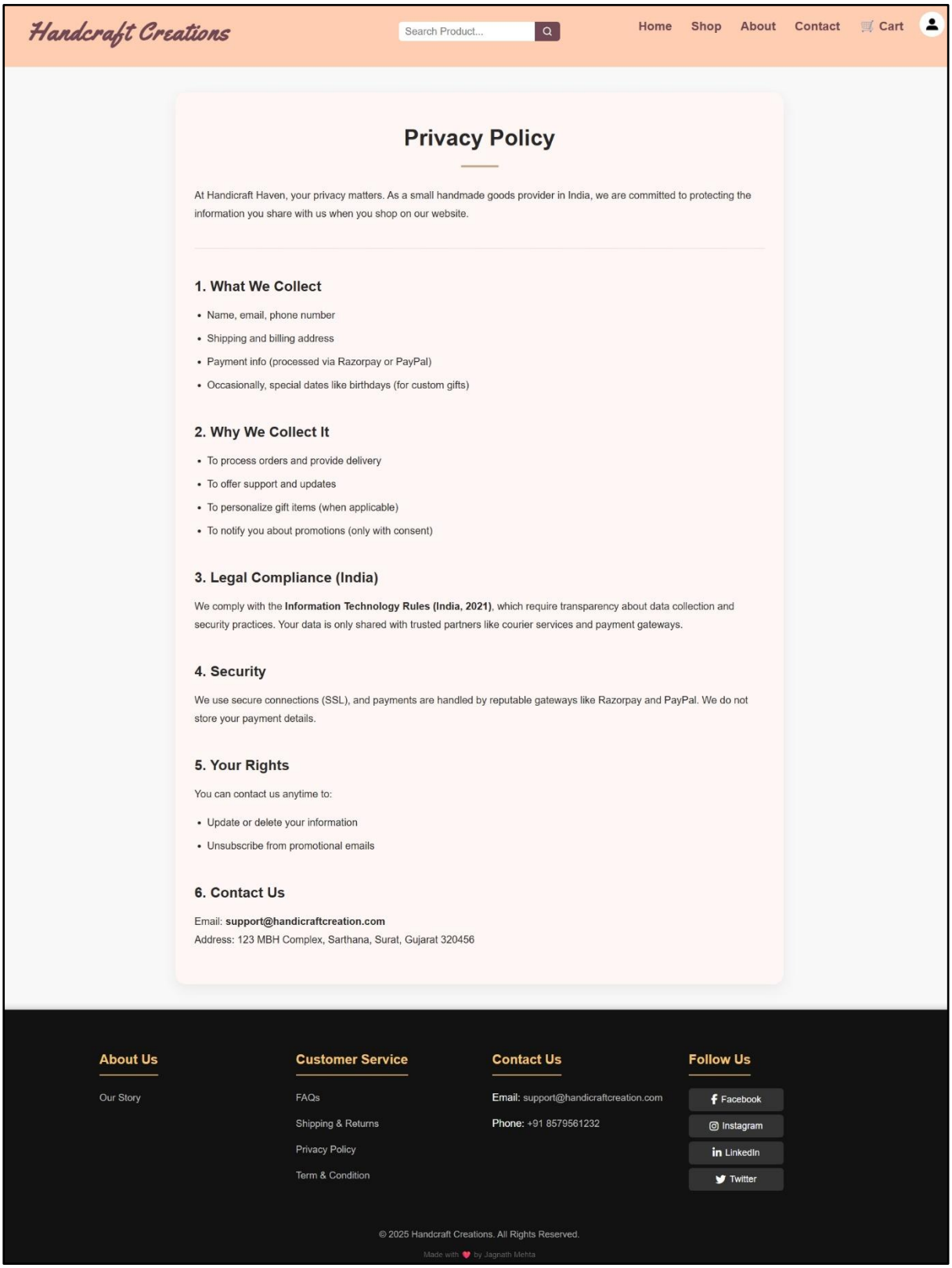
CUSTOMER SERVICE
FAQs



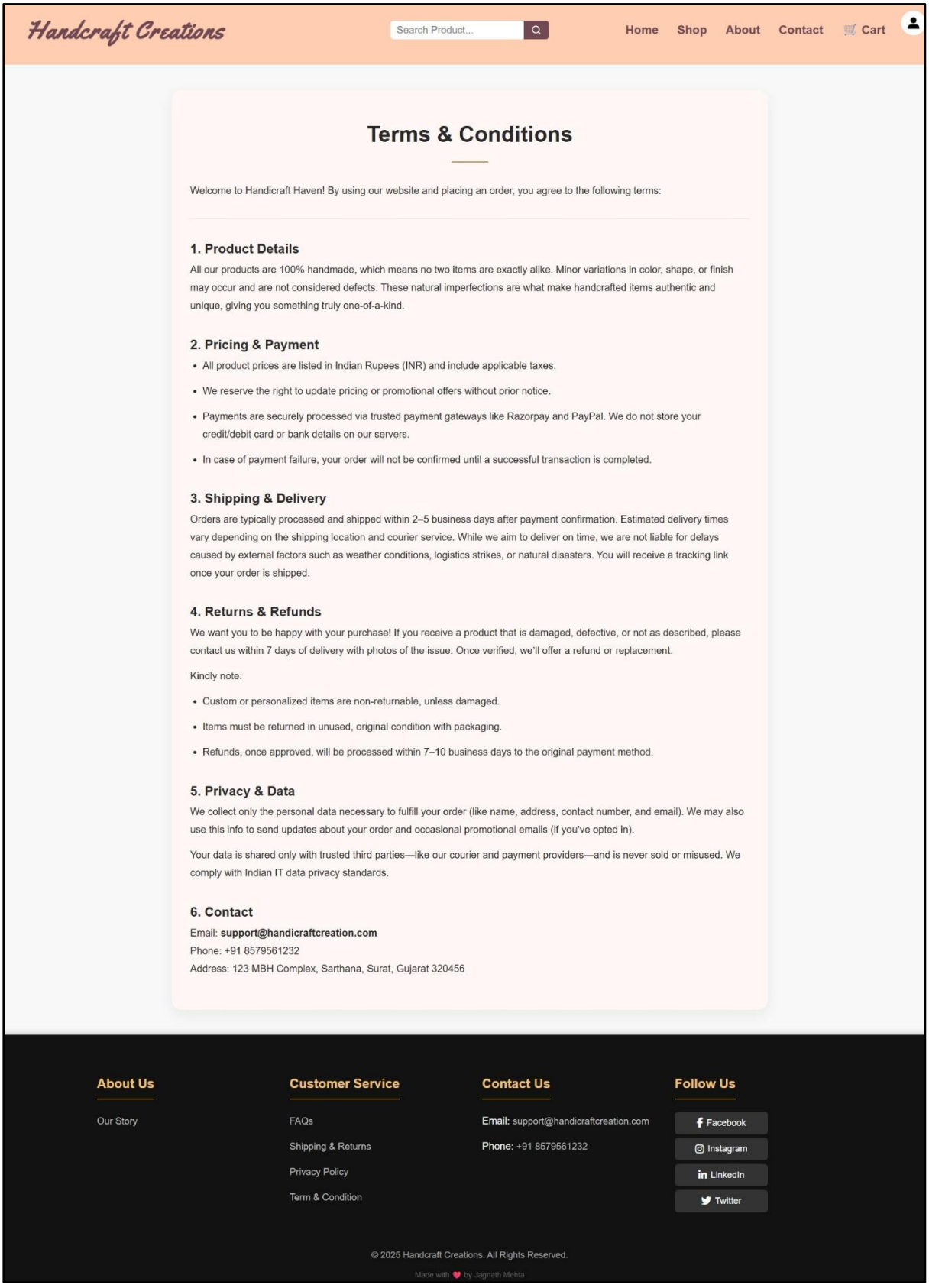
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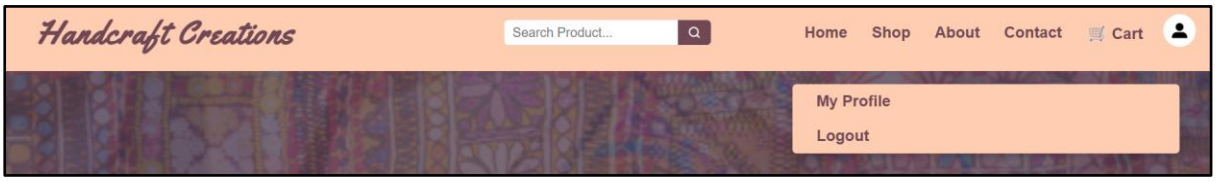
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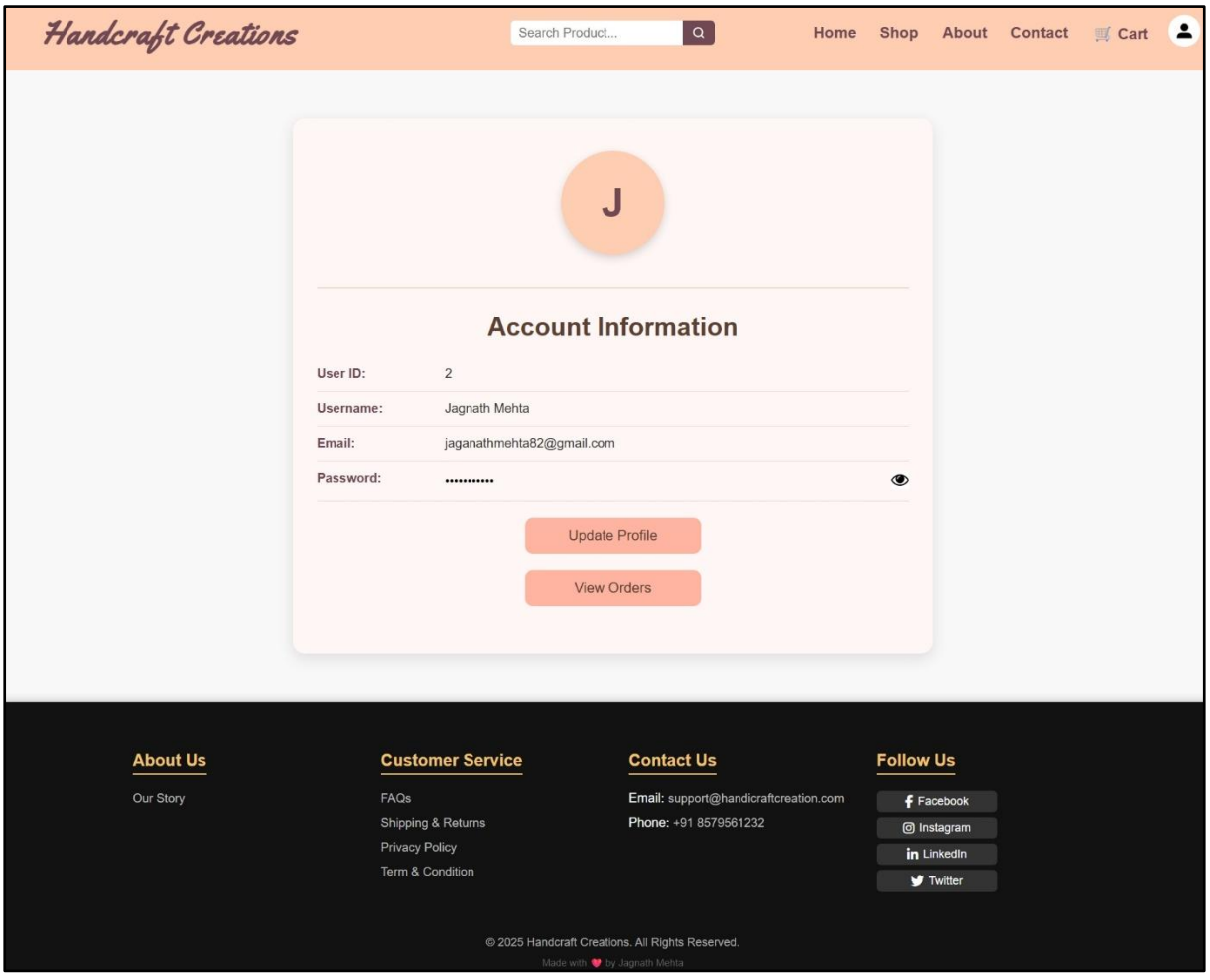
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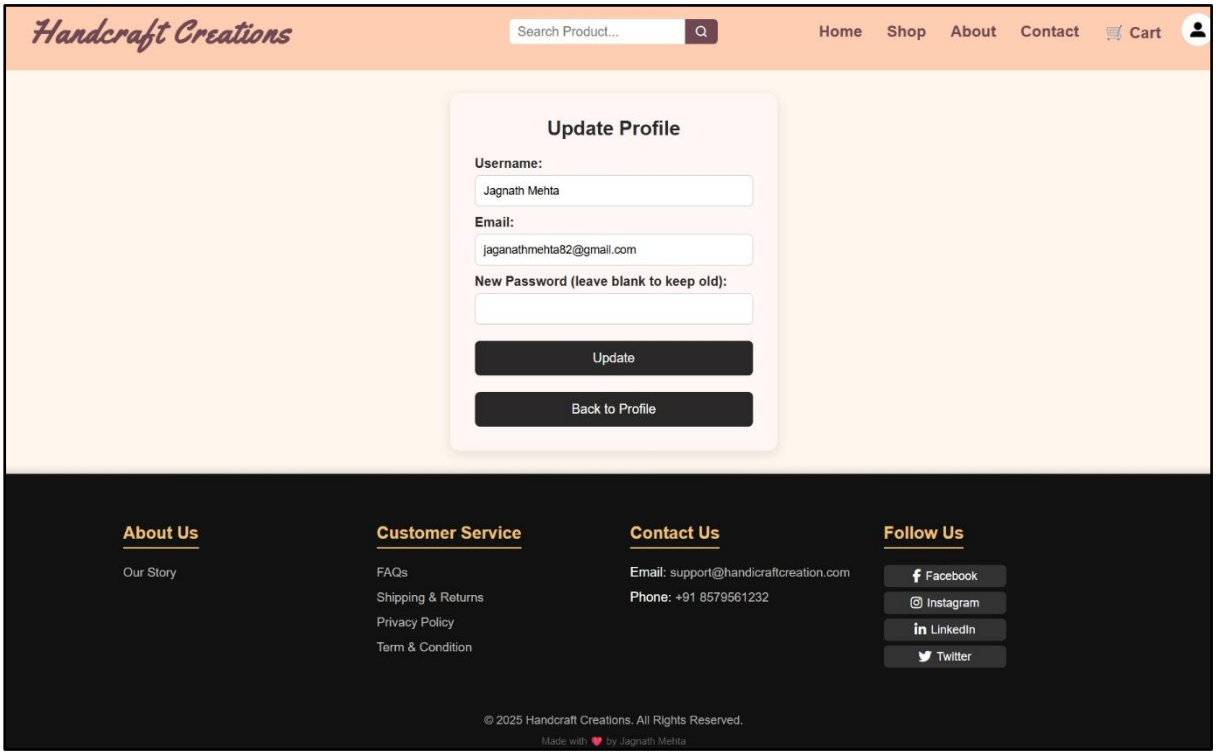
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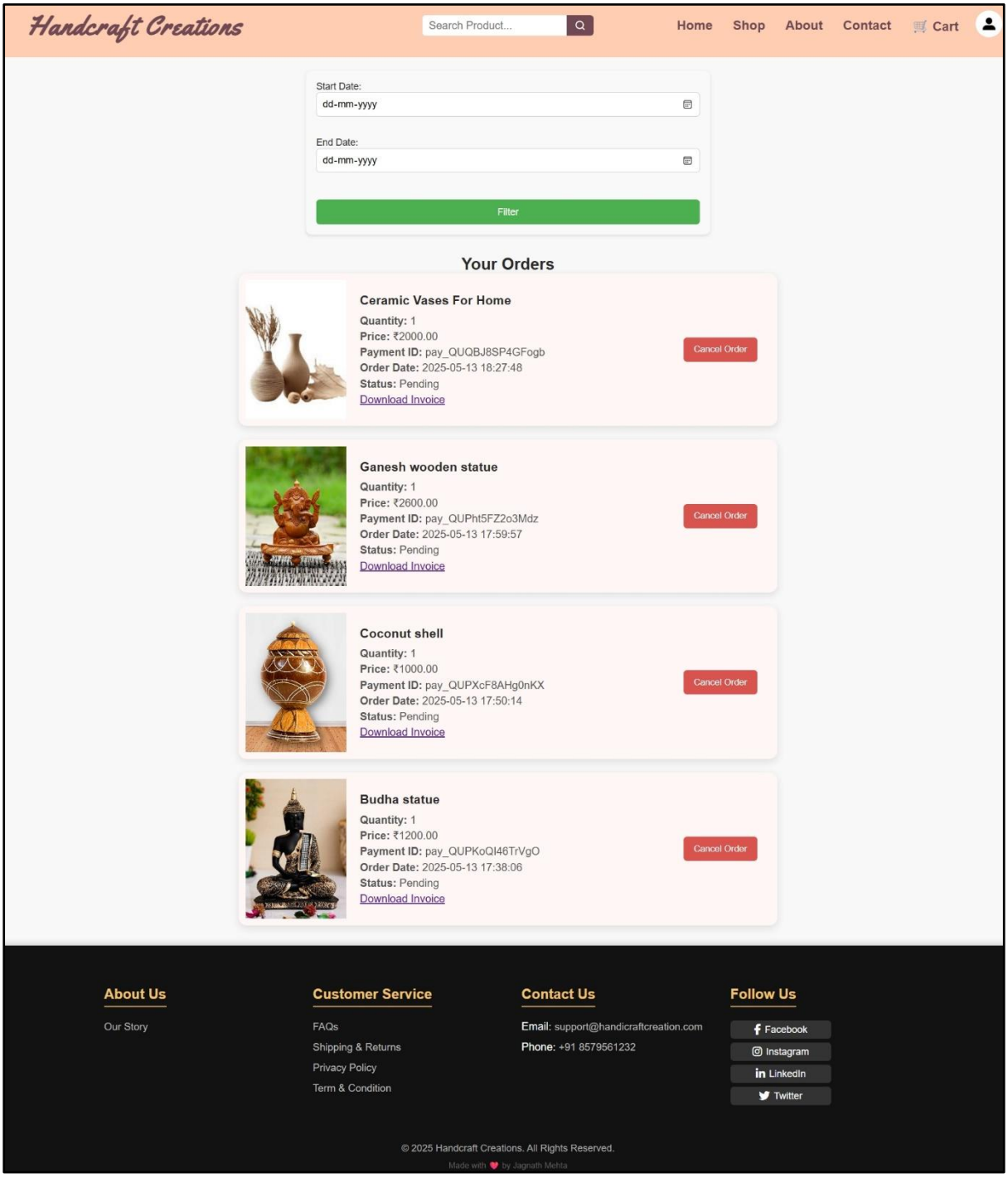
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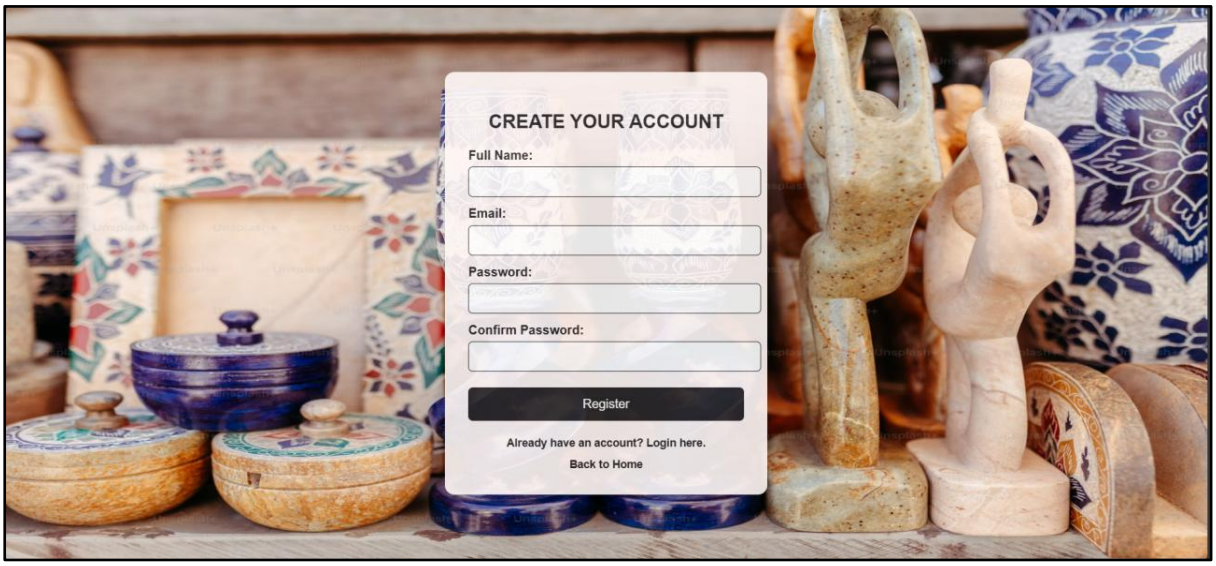
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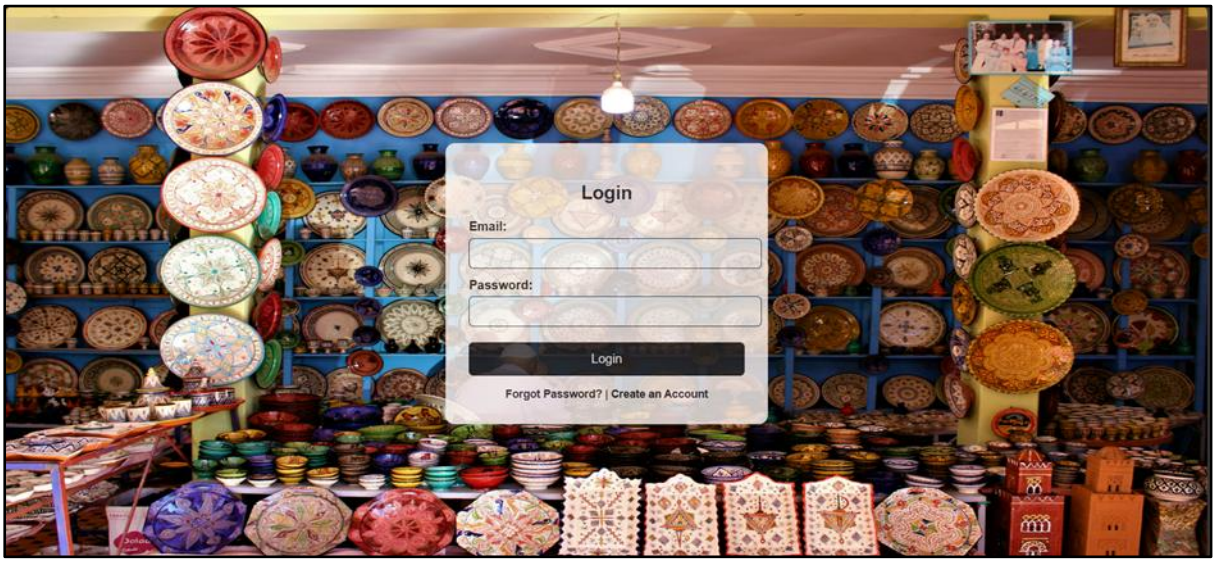


VIEW ORDER

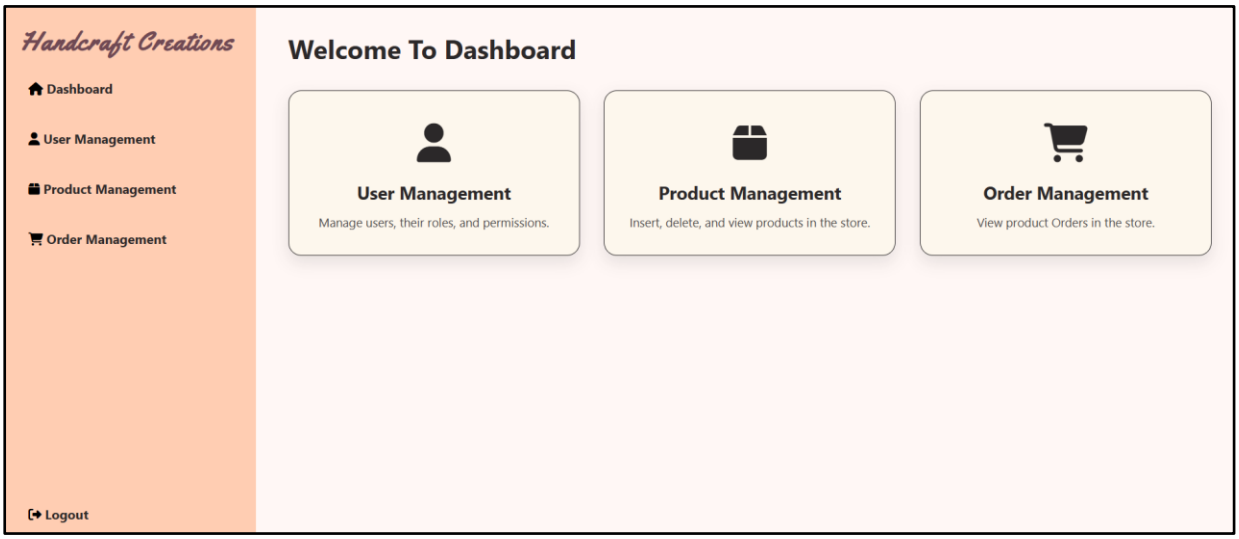

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Search Product...
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Start Date:
dd-mm-yyyy
End Date:
dd-mm-yyyy
Filter
Your Orders
Ceramic Vases For Home
Quantity: 1
Price: ₹2000.00
Payment ID: pay_QUQBJ8SP4GFogb
Order Date: 2025-05-13 18:27:48
Status: Pending
Download Invoice
Cancel Order
Ganesh wooden statue
Quantity: 1
Price: ₹2600.00
Payment ID: pay_QUPht5FZ2o3Mdz
Order Date: 2025-05-13 17:59:57
Status: Pending
Download Invoice
Cancel Order
Coconut shell
Quantity: 1
Price: ₹1000.00
Payment ID: pay_QUPXcF8AHg0nKX
Order Date: 2025-05-13 17:50:14
Status: Pending
Download Invoice
Cancel Order
Budha statue
Quantity: 1
Price: ₹1200.00
Payment ID: pay_QUPKoQI46TrVgO
Order Date: 2025-05-13 17:38:06
Status: Pending
Download Invoice
Cancel Order
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Made with by Jagnath Mehta

MODULE NAME – ADMIN

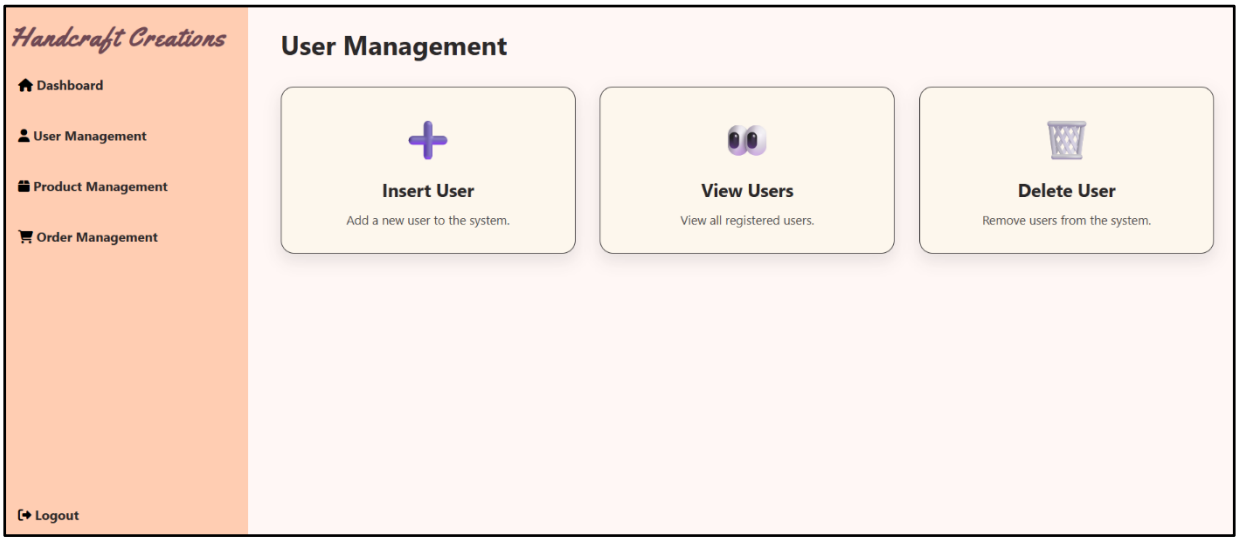




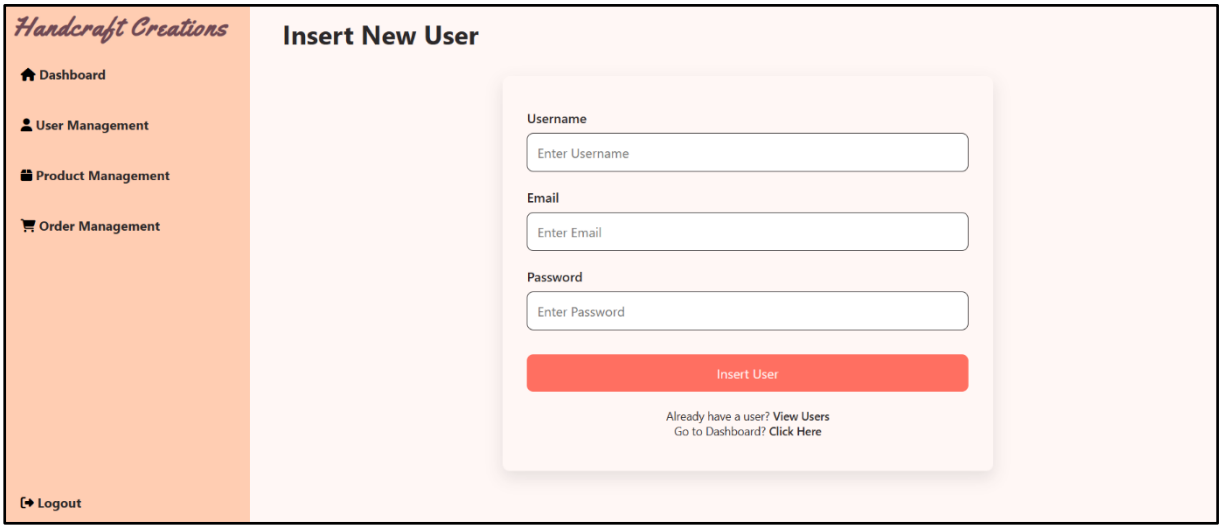
DASHBOARD



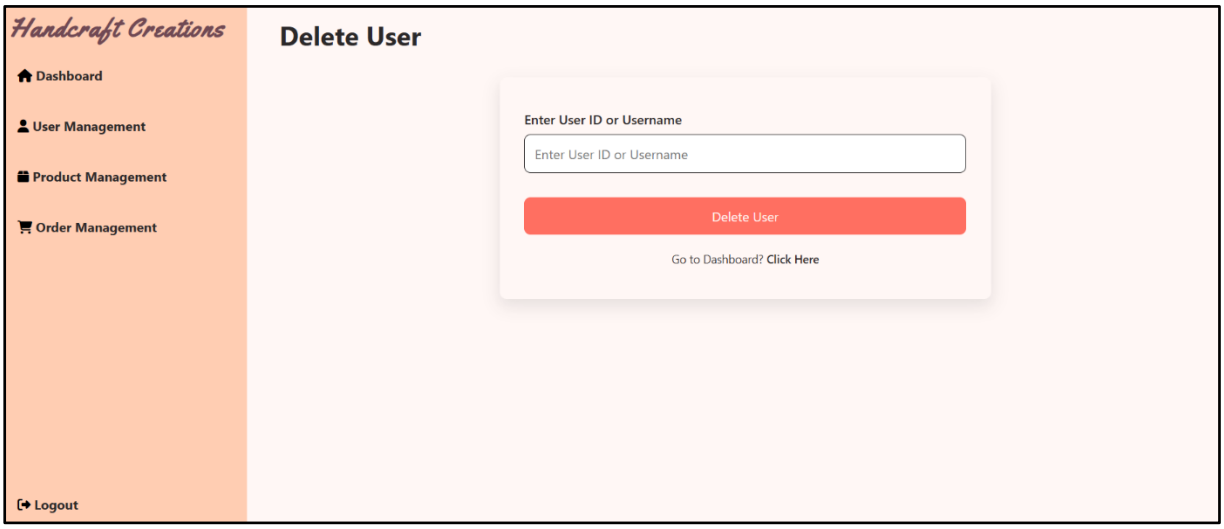
USER MANAGEMENT



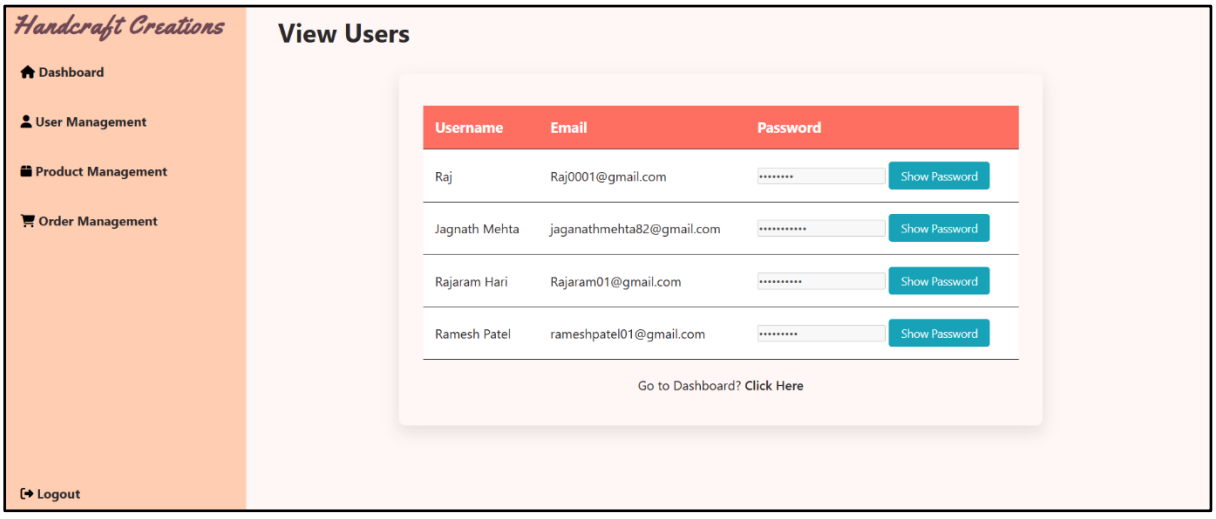
USER INSERT



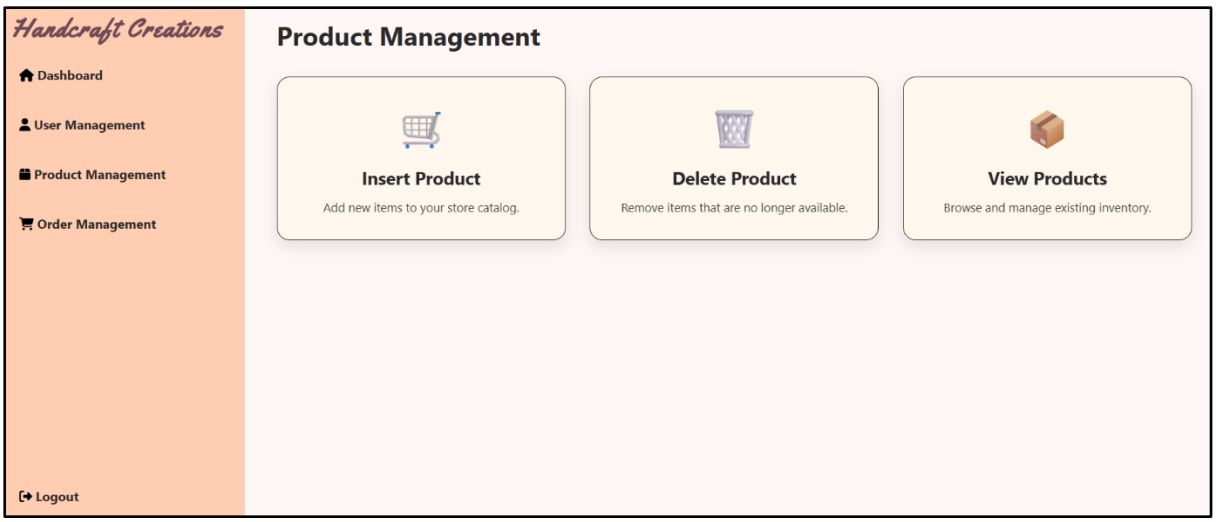
USER DELETE



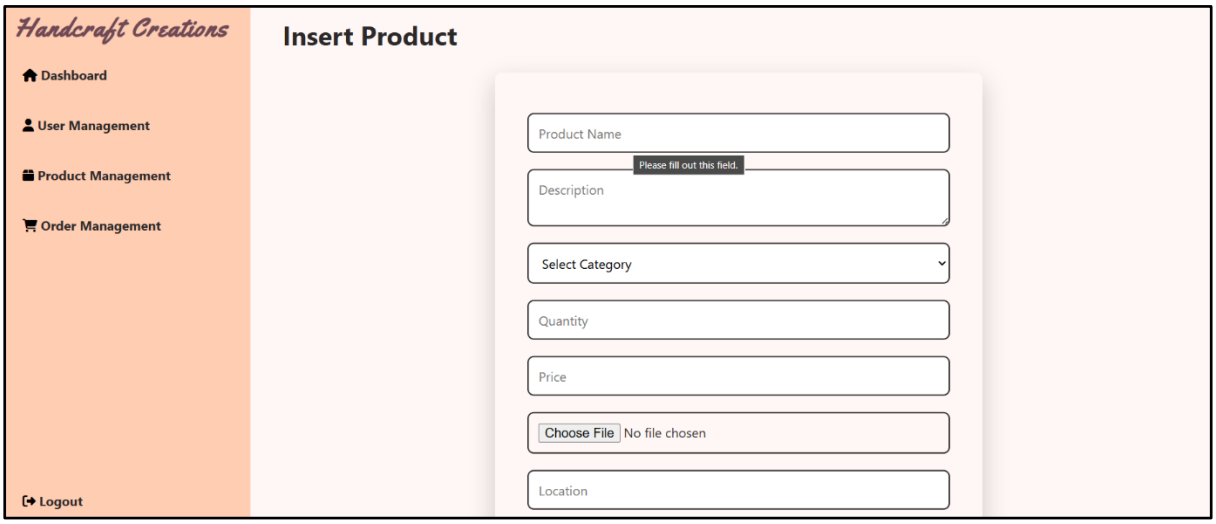
USER DISPLAY



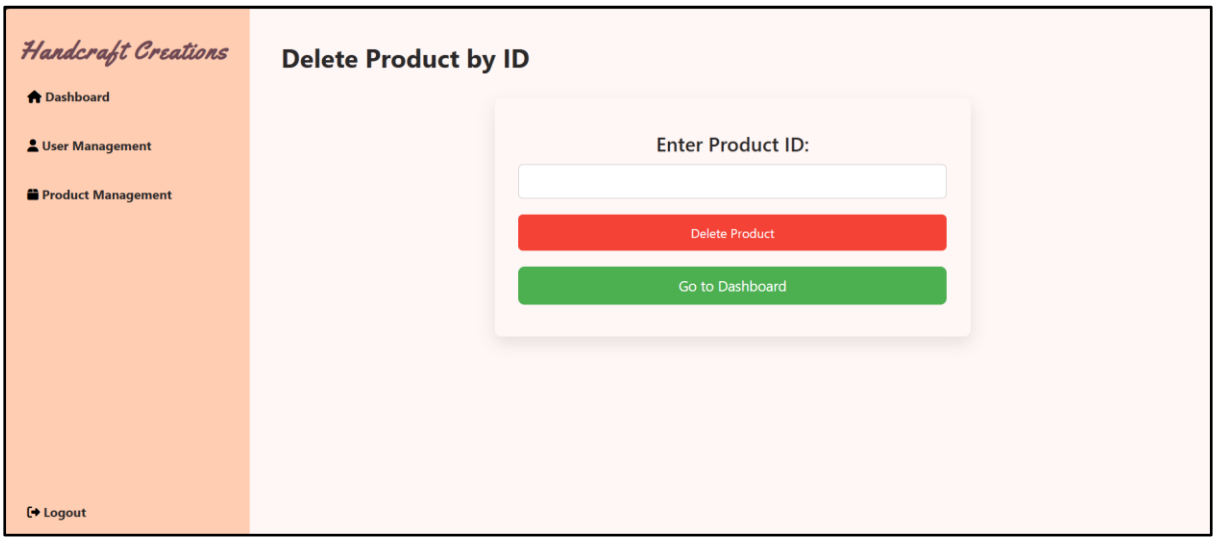
PRODUCT MANAGEMENT



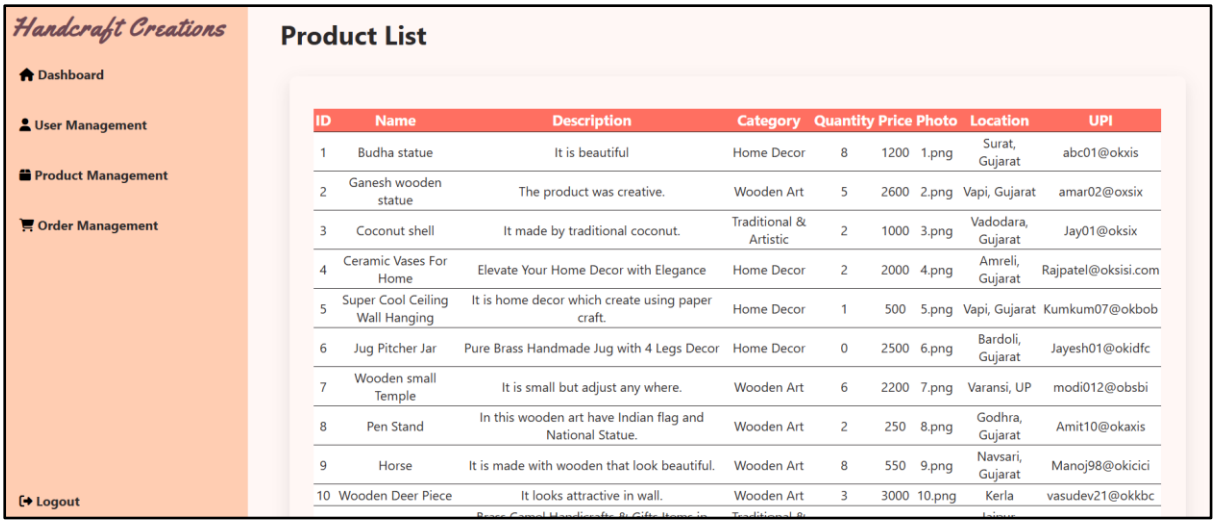
PRODUCT INSERT



PRODUCT DELETE

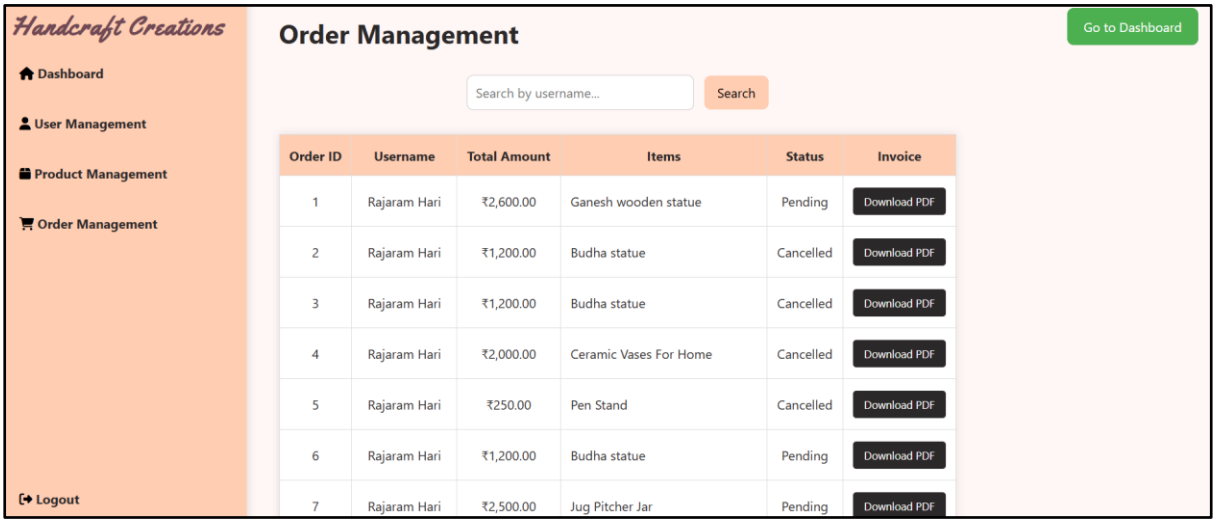


PRODUCT DISPLAY



ID	Name	Description	Category	Quantity	Price	Photo	Location	UPI
1	Budha statue	It is beautiful	Home Decor	8	1200	1.png	Surat, Gujarat	abc01@okxis
2	Ganesh wooden statue	The product was creative.	Wooden Art	5	2600	2.png	Vapi, Gujarat	amar02@oxsix
3	Coconut shell	It made by traditional coconut.	Traditional & Artistic	2	1000	3.png	Vadodara, Gujarat	Jay01@oksix
4	Ceramic Vases For Home	Elevate Your Home Decor with Elegance	Home Decor	2	2000	4.png	Amreli, Gujarat	Rajpatel@oksisi.com
5	Super Cool Ceiling Wall Hanging	It is home decor which create using paper craft.	Home Decor	1	500	5.png	Vapi, Gujarat	Kumkum07@okbob
6	Jug Pitcher Jar	Pure Brass Handmade Jug with 4 Legs Decor	Home Decor	0	2500	6.png	Bardoli, Gujarat	Jayesh01@okidfc
7	Wooden small Temple	It is small but adjust any where.	Wooden Art	6	2200	7.png	Varansi, UP	modi012@obsbi
8	Pen Stand	In this wooden art have Indian flag and National Statue.	Wooden Art	2	250	8.png	Godhra, Gujarat	Amit10@okaxis
9	Horse	It is made with wooden that look beautiful.	Wooden Art	8	550	9.png	Navsari, Gujarat	Manoj98@okicici
10	Wooden Deer Piece	It looks attractive in wall.	Wooden Art	3	3000	10.png	Kerla	vasudev21@okkbc

ORDER MANAGEMENT



7. SYSYEM TESTING

7.1 TESTING STRATEGIES

System testing is a critical phase in the software development lifecycle where the complete and integrated system is evaluated against its specified requirements. For the Online Handicraft System, system testing ensures that the platform functions correctly, is secure, performs efficiently, and provides a seamless experience to end-users. Various testing methodologies are employed to validate both functional and non-functional aspects of the application.

Below is a comprehensive overview of the system testing approaches applied to the Online Handicraft System:

1. White-Box Testing:

White-box testing, also known as glass-box testing, is a test design method that examines the internal structure of the Online Handicraft System. This technique is used to verify the internal logic and correctness of the code.

White-box testing attempts to find errors in the following categories:

- Guarantee that all independent paths within a module have been exercised at least once
- Exercise all logical decisions on their true and false sides
- Execute all loops at their boundaries and within their operational bounds
- Exercise internal data structures to ensure their validity

2. Black-Box Testing:

Black-box testing, also called behavioral testing, focuses on the functional requirements of the Online Handicraft System. This testing technique ensures that the system meets its intended functionalities by providing inputs and examining outputs without considering the internal code structure.

Black-box testing attempts to find errors in the following categories:

- Incorrect or missing functions
- Interface errors
- Errors in data structures or external database access
- Behavior or performance errors
- Initialization and termination errors

3. Unit Testing:

Unit testing focuses on verifying the smallest unit of software design – the software component or module. Using the component-level design description as a guide, important control paths are tested to uncover errors within the boundary of the module.

- User Registration & Login Module: Testing correct registration, login authentication for customers and sellers.
- Product Listing Module: Ensuring sellers can add, update, and delete handicraft items properly.
- Shopping Cart Module: Verifying item addition, quantity updates, and cart total calculations.
- Order Management Module: Checking order placement, order status tracking, and cancellation processes.
- Payment Module: Testing secure payment handling and transaction success/failure scenarios.
- Review & Rating Module: Verifying customer ability to submit reviews and sellers' responses.

4. Integration Testing:

Integration testing verifies the interaction between different modules of the Online Handicraft System to ensure they work together correctly. It identifies issues in data flow, communication, and integration.

Key Areas of Integration Testing:

1. **User Authentication & Product Browsing:**
    - Verify that only logged-in users can add items to the cart or place orders.
    - Ensure that guests can browse items but have limited functionality.
2. **Shopping Cart & Payment Gateway:**
    - Check if cart details are passed correctly to the payment gateway.
    - Ensure successful payment updates the order status and reduces stock accordingly.
3. **Database Management & Notification System:**
    - Verify if users receive order confirmation emails/SMS after successful purchase.
    - Ensure sellers are notified of new orders promptly.
4. **Review System & Product Listings:**
    - Ensure that submitted reviews appear correctly under the respective products.
    - Verify that reviews affect the average product rating.

5. Alpha & Beta Testing:

Alpha and Beta Testing are conducted internally by the project team and faculty before releasing the Online Handicraft System for external use. Developers test all key features like user registration, product listing, shopping cart functionality, and payment processing, while faculty members review usability, performance, and technical issues.

- **Alpha Testing**:

  Alpha testing is conducted internally by the project team and faculty members. All features such as browsing handicrafts, placing orders, payments, and user feedback are tested in a controlled environment. Bugs, UI/UX issues, and performance bottlenecks are identified and fixed before progressing.

- **Beta Testing**:

  Beta testing is conducted by selected students or external volunteers acting as end users. The system is deployed in a test environment where users interact with the platform by browsing products, adding items to their carts, making purchases, and leaving reviews. Feedback is collected for further improvements before final deployment.

8. CONCLUSION AND FUTURE SCOPE

Conclusion:

The **Online Handicraft System** with **search functionality** and **buy functionality** provides a seamless, user-friendly platform that connects consumers with artisans who produce unique handmade items. By implementing an efficient search system, users can quickly find products based on various categories, keywords, or preferences, making the shopping experience more intuitive and enjoyable. The buying functionality ensures a secure transaction process, enhancing customer trust and satisfaction. This system promotes the growth of the handicraft industry by reaching a broader audience, offering artisans a platform to showcase their talent, and empowering consumers to purchase quality handmade goods from the comfort of their homes.

Key benefits of the system:

1. **Ease of Access**: Users can easily find products via a robust search system.
2. **Convenient Purchasing**: The buying process is streamlined and secure.
3. **Support for Artisans**: This platform offers artisans an online marketplace to sell their handmade products.
4. **Cultural Promotion**: The system promotes and preserves traditional crafts through modern technology.

Future Scope:

1. **Artificial Intelligence (AI) for Personalized Recommendations**: The integration of AI can enhance the user experience by offering personalized recommendations. By analyzing users' past purchases, browsing history, and preferences, AI can suggest products that the customer is more likely to buy, improving customer satisfaction and increasing conversion rates.
2. **Augmented Reality (AR) Integration**: AR technology can be used to allow customers to virtually "try" or "see" the handicraft products in their homes. For example, they could use an app to see how a handmade painting or sculpture would look in their living room, helping customers make more confident purchasing decisions.
3. **Mobile Application Development**: In the future, the online handicraft platform could expand by creating dedicated mobile apps for iOS and Android. Mobile apps could offer push notifications, a more personalized shopping experience, and easier navigation for customers. Apps also increase accessibility, allowing users to shop anytime and anywhere.
4. **Global Expansion and Multi-Language Support**: The platform can expand internationally to cater to a broader market. Adding multi-language support would help reach customers from different countries and cultures. This would provide artisans with a global marketplace and increase the visibility of their products worldwide.

5. **Advanced Search Filters and Sorting Options**: The future version of the website could have more advanced search filters. Customers would be able to filter products by material, size, price, color, craftsmanship type, etc. This would enhance the search functionality, making it easier for users to find exactly what they are looking for quickly.
6. **Integration of Social Media and User-Generated Content**: Social media integration can be a great addition to the platform. Customers could share their purchases, reviews, and photos of how they've used the products in their homes. This can help build an online community and promote the platform to a wider audience through organic user-generated content.
7. **Customer Loyalty Programs**: Implementing a customer loyalty program could help retain customers by offering them rewards like discounts, early access to new products, or special offers. This can increase customer retention and enhance overall customer satisfaction.
8. **Improved Payment Options**: Future enhancements could include integrating multiple payment gateways, including digital wallets, cryptocurrency, and international payment systems, to make it easier for customers from different regions to make secure transactions.
9. **Cross-Platform Integration with E-Commerce Platforms**: The online handicraft system could integrate with major e-commerce platforms (like Etsy, Amazon, or eBay) to broaden the market reach for artisans. This would allow them to cross-sell their products on multiple platforms while managing inventory from a single centralized system.
10. **Automated Inventory Management**: As the platform grows, integrating automated inventory management systems can help artisans keep track of their stock, reduce human error, and streamline the process of managing orders.
11. **Customer Feedback and Review System Enhancements**: The review system could be enhanced to include video reviews, ratings for different product attributes (e.g., quality, craftsmanship, durability), and a more robust Q&A section where customers can ask artisans about product details.
12. **Integration of Virtual Assistants**: A virtual assistant powered by AI can be implemented to assist customers in real-time with product inquiries, order tracking, and suggestions. This can improve customer service and reduce response time for common questions.

9. BIBLIOGRAPHY

Category	Details
Books	1. E-Commerce 2021: Business, Technology, Society by Kenneth C. Laudon and Carol Guercio Traver 2. Web Development and Design Foundations with HTML5 by Terry Felke-Morris 3. Designing Web Usability: The Practice of Simplicity by Jakob Nielsen and Hoa Loranger 4. The Art of Scalability: Scalable Web Architecture, Processes, and Organizations for the Modern Enterprise by Martin L. Abbott and Michael T. Fisher 5. PHP and MySQL for Dynamic Web Sites: Visual QuickPro Guide by Larry Ullman
Study Material	1. E-commerce Systems and Applications - Online Course notes by Udemy or Coursera 2. User Experience (UX) Design Principles for Websites and E-commerce - Faculty Lecture Notes 3. Web Application Security for E-commerce Platforms - Research papers on cybersecurity for online stores 4. System Design for E-commerce Platforms - Faculty notes or online tutorials from websites like Codecademy, freeCodeCamp
Websites	1. www.geeksforgeeks.org – System development concepts.
	2. www.w3schools.com – Frontend development tutorials.
	3. www.tutorialspoint.com – SDLC, backend, and databases.
	4. https://chatgpt.com – Assistance with content generation, documentation, and explanation of technical concepts.
Research Papers	1. "Design and Implementation of an E-commerce System" – Journal of E-commerce Research. 2. "Online Marketplaces and Handicrafts: Trends and Challenges" – IEEE Transactions on Technology and Society.
College Materials	1. SDLC notes and testing models from faculty. 2. Lecture notes on requirement gathering and system design.
Tools & Technologies	Frontend: HTML, CSS, JavaScript. Backend: PHP. Database: MySQL. Design Tools: Figma. Local Server: XAMPP / WAMP.